

**DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE
COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE
STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY**

A Design Project
Submitted to the Faculty of the
Department of Engineering
Cavite State University-CCAT Campus
Rosario, Cavite

In partial fulfillment
Of the requirements for the degree
Bachelor of Science in Computer Engineering

**JONEL B. ANDAMON
JOVAN P. NADALA
GEROME Q. VOLUNTAD**
September 2026



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU CCAT Campus
Rosario, Cavite
(046) 437-9507 / (046) 437-6659
cvsurosarlio@cvsu.edu.ph
www.cvsu-rosario.edu.ph

DEPARTMENT OF ENGINEERING

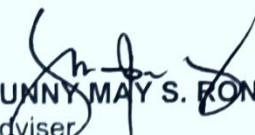
MANUSCRIPT APPROVAL FORM

Title : DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY

Name of Authors(s) : JONEL B. ANDAMON
JOVAN P. NADALA
GEROME Q. VOLUNTAD

Degree Program : BS IN COMPUTER ENGINEERING

A P P R O V E D :


SUNNY MAY S. RONAS
Adviser

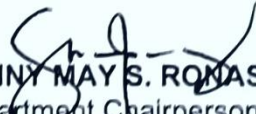
6/15/26
Date


PRINCESS YNA D. JUYAD
Technical Critic

6/15/26
Date


EMJAY A. CAMPO
Department Research
Coordinator, BSCPE

6/15/26
Date


SUNNY MAY S. RONAS, MEng
Department Chairperson

6/15/26
Date


ARIEL G. SANTOS, EdD
Director for Instructions

Date


KARLO JOSE E. NABABLIT, PhD
Director for Research and
Extension

6-16
Date


LAURO B. PASCUA, EdD
Campus Administrator

6-16-27
Date

BIOGRAPHICAL DATA

Mr. Jonel B. Andamon was born on August 14, 2003, in Nabua, Camarines Sur. He is the third child of Mrs. Nilda B. Andamon and Mr. Joy P. Andamon. They currently reside in Banana Island, Daang Amaya 2, Tanza, Cavite.

He completed his elementary education at Felipe G. Calderon Elementary School in 2016. He then continued his secondary education at Tanza National Comprehensive High School, where he received academic distinction for three consecutive years, until finally completing junior high school education and graduated with Honors. In 2022, he completed his senior high school education under the Science, Technology, Engineering, and Mathematics (STEM) strand at Cavite Science Integrated School, where he graduated with High Honors.

In the same year, he enrolled under the Bachelor of Science in Computer Engineering (BSCPE) program at Cavite State University-CCAT Campus. During his third year in college, he achieved the distinction of being a President's Lister. He also successfully passed the Civil Service Examination-Professional Level, demonstrating his academic excellence and commitment to public service.

He earned his degree in September 2026.

BIOGRAPHICAL DATA

Mr. Jovan P. Nadala was born on March 17, 2004, in Sta. Margarita, Samar. He is the youngest of the family raised by Mrs. Elena P. Nadala and Mr. Benjamin C. Nadala. They currently reside in Pabuhay Housing, Santa Rosa 1, Noveleta, Cavite.

He completed his elementary education as a salutatorian at Inoraguiao Elementary School in 2016. He transferred here in Noveleta and continued his journey at Noveleta National High School, where he graduated with Honors after four years. In 2022, he completed his senior high school education under the STEM strand at Holy Cross Catholic School of Noveleta, where he graduated with Honors.

In the same year, he enrolled under the BSCPE program at Cavite State University-CCAT Campus. During his time at the campus, he became a public relation officer (P.R.O) of the Association of Computer Engineering Students (ACES) in the same campus. He became a Vice President's Lister in his third year. He also passed the Civil Service Examination-Professional Level in August 2025. Additionally, he participated in extracurricular activities including the street dance competition during the *Juana* and Anti-VAWC events during the celebration of women's month and anti-violence against women and children campaign, respectively. In both events, their group won as third placer.

He earned his degree in September 2026.

BIOGRAPHICAL DATA

Mr. Gerome Q. Voluntad was born on September 20, 2003, in Cavite City, Cavite. He is the eldest child in the family and has two younger sisters. He was raised by his parents, Mrs. Roselle Q. Voluntad and Mr. George T. Voluntad. He currently resides in B. Toledo Street, Ligtong IV, Rosario, Cavite.

He completed his elementary education at Mayor Calixto D. Enriquez Elementary School in 2015. He then finished his junior high school education at Bagbag National High School in 2019, where he graduated with Honors. He pursued his senior high school education at Rosario Institute, graduating with Honors in 2021.

In September 2021, he enrolled at Cavite State University-CCAT Campus, where he pursued a bachelor's degree in computer engineering.

He successfully earned his degree in September 2026.

ACKNOWLEDGMENT

The authors sincerely acknowledge the guidance and support of different groups and individuals who made the completion of this design project possible:

Engr. Sunny May S. Ronas, design project adviser and chairperson of the Department of Engineering, for her unwavering guidance, patience, and constructive feedback throughout the entire research process which were essential in overcoming the challenges encountered in this study;

Engr. Princess Yna D. Juyad, technical critic, for her timely assistance and guidance in ensuring compliance with the requirements of the research;

Engr. Emjay A. Campo, unit research coordinator for BSCPE, for his oversight in ensuring that the study met the appropriate academic standards and for his valuable recommendations;

Dr. Karlo Jose E. Nabablit, director for Research and Extension, for encouraging research initiatives and supporting student-led studies;

Dr. Ariel G. Santos, director for Instructions, for his commitment to quality education and for providing students with opportunities for academic growth;

Dr. Lauro B. Pascua, campus administrator, for his exemplary leadership and for cultivating an environment conducive to academic achievement;

Family, for their unconditional love, understanding, and continuous emotional and financial support throughout this project; and

Friends and classmates, for their cooperation, encouragement, and contributions during the preparation and completion of this study.

THE AUTHORS

ABSTRACT

ANDAMON, JONEL B., NADALA, JOVAN P., and VOLUNTAD, GEROME Q. Development of LibLocker: An Automated Library Baggage Counter with Dual Frequency RFID Authentication for Cavite State University-CCAT Campus e-Library. Undergraduate Design Project. Bachelor of Science in Computer Engineering. Cavite State University-CCAT Campus. Rosario, Cavite. September 2026. Adviser: Engr. Sunny May S. Ronas. Technical Critic: Engr. Princess Yna D. Juyad.

This study aimed to design, develop, and evaluate LibLocker, an automated library baggage counter system with dual-frequency RFID authentication for the Cavite State University–CCAT Campus e-Library. The system was developed to address the limitations of manual baggage handling, including slow processing, weak security, and inefficient monitoring.

The system integrated high frequency (HF) RFID at 13.56 MHz for kiosk-based authentication and ultra-high frequency (UHF) RFID at 915 MHz for entrance monitoring. A Raspberry Pi 4 Model B served as the central controller managing locker assignment, identity verification, transaction logging, and HF-UHF identity matching. The system also featured a user registration module with reusable dual-frequency RFID cards, solenoid-based locker control with magnetic reed switches, USB camera-assisted verification, and a web-based dashboard for real-time monitoring. Offline–online synchronization was implemented to ensure continuous operation during network interruptions.

The system was evaluated using ISO/IEC 25010 quality standards and the Technology Acceptance Model (TAM) with 30 respondents from various academic programs. Performance testing yielded a 100 percent RFID pairing accuracy, 100 percent locker retrieval accuracy, an average transaction time of 6.8 s (within the 10-s target), and 95 percent system uptime during the testing period. The ISO/IEC 25010-based evaluation resulted in an overall mean of 4.63, while the TAM-based evaluation produced an overall mean of 4.63, both interpreted as "Strongly Agree".

The findings confirm that LibLocker is a reliable and efficient solution for automated baggage management. The system demonstrates strong potential to support the modernization of library services and the implementation of smart campus technologies in academic institutions.

TABLE OF CONTENTS

	Page
APPROVAL SHEET	ii
BIOGRAPHICAL DATA	iii
ACKNOWLEDGMENT	vi
ABSTRACT	vii
LIST OF TABLES	xi
LIST OF FIGURES	xiii
LIST OF APPENDIX TABLES	xv
LIST OF APPENDIX FIGURES	xvi
LIST OF APPENDICES	xvii
INTRODUCTION	1
Objectives of the Study	2
Significance of the Study	3
Time and Place of the Study	4
Scope and Limitations of the Study.....	4
Definition of Terms.....	4
Conceptual Framework	6
REVIEW OF RELATED LITERATURE	8
METHODOLOGY	14
Description of the Project/Technology	14
Respondents and Sampling Technique	15
Research Instrument	15
Statistical Treatment of Data	15
Ethical Considerations	16
Components	16

Design and Fabrication Procedures	21
Principles of Operation	30
Testing and Evaluation	39
Data Analysis	40
RESULT AND DISCUSSION	42
SUMMARY, CONCLUSION, AND RECOMMENDATIONS	60
Summary	60
Conclusion	61
Recommendations	61
REFERENCES	63
APPENDICES.....	65

LIST OF TABLES

Table		Page
1	Five-point Likert scale used in interpreting the evaluation based on ISO/IEC 25010 and TAM	41
2	System design validation	43
3	Documentation completeness quality	43
4	User verification through the RFID modules	45
5	Locker mechanism	45
6	Database	46
7	Synchronization logic	47
8	Camera snapshot function	47
9	Kiosk interface	48
10	Monitoring dashboard	48
11	Student information entry	49
12	Data validation	49
13	HF RFID assignment	50
14	UHF RFID assignment	50
15	Dual-frequency matching	51
16	Secured record storage	51
17	RFID card encoding	52
18	RFID card linking	53
19	Reusability of RFID cards	53
20	Data overwriting.....	54
21	Unauthorized access detection	54
22	Encoding accuracy	55
23	RFID pairing and authentication accuracy	56

24	Locker unlocking mechanism retrieval transactions	56
25	Transaction time and measurement	57
26	System reliability	58
27	Summary of evaluation based on ISO/IEC 25010 evaluation	58
28	Summary of evaluation based on TAM	59

LIST OF FIGURES

Figure		Page
1	Conceptual framework of the study.....	6
2	Raspberry Pi 4 model 4	16
3	I/O expander (MCP23017)	17
4	HF RFID reader (RC522)	17
5	UHF RFID reader (915 MHz)	18
6	Solenoid lock (12v)	18
7	Magnetic door sensor (Reed Switch)	19
8	USB webcam	19
9	Capacitive touchscreen monitor	20
10	Dual-frequency RFID card	20
11	Three-dimensional design of the LibLocker	22
12	Side view of the design of the LibLocker	22
13	Front view of the design of the LibLocker.....	23
14	Rear view of the design of the LibLocker	23
15	Kiosk main menu interface	24
16	User registration interface	24
17	Locker selection interface	25
18	RFID verification prompt (Deposit)	25
19	RFID verification prompt (Retrieval)	26
20	Deposit success screen	26
21	Retrieval success screen	27
22	Dashboard overview	27
23	Locker status grid	28
24	Activity log view	28

25	Violation page	29
26	System network diagram of Liblocker	30
27	System operation flowchart	32
28	RFID authentication sequence	33
29	Locker access mechanism	34
30	Camera-assisted verification	35
31	HF-UHF identity matching	36
30	Error handling logic	37
33	System integration flow	38
34	Process block diagram	39

LIST OF APPENDIX TABLES

Appendix Table		Page
1	Summary of responses for evaluation based on ISO/IEC 25010 in terms of functional suitability and performance efficiency.....	67
3	Summary of responses for evaluation based on ISO/IEC 25010 in terms of reliability and security	68
2	Summary of responses for evaluation based on ISO/IEC 25010 in terms of compatibility and usability	69
4	Summary of responses for evaluation based on TAM in terms of perceived usefulness and perceived ease of use.....	70
5	Summary of responses for evaluation based on TAM in terms of behavioral intention and user satisfaction	71
6	Cost analysis	72

LIST OF APPENDIX FIGURES

Appendix Figure		Page
1	Admin log interface	74
2	Admin main interface	74
3	User transaction saved in admin dashboard	75
4	Landing page of kiosk interface	75
5	Kiosk main interface	76
6	Components wirings of kiosk	76
7	Wirings of solenoid, reed switch, and locker	77
8	Components testing	77
9	Kiosk interface testing	78
10	Kiosk and locker integration	78
11	Kiosk casing (front view) with UHF RFID reader (915 MHz)	79
12	Kiosk casing (side view)	79
13	Final setup of prototype	80
14	Deployed at e-library	80
15	Prototype testing	81

LIST OF APPENDICES

Appendix		Page
A	Appendix Tables	66
B	Appendix Figures	73
C	Communication Letter with Library	82
D	Source Code	85
E	Research Instrument (Survey Questionnaire).....	107
F	Sample Data Sheet	113
G	Gantt Chart	119
H	Curriculum Vitae	121
I	Summary of Recommendations	125
J	Compliance to Recommendations	127
K	Routing Slip (For Undergraduate)	129
L	Certification of English Critique	131
M	Mentee Consent/Attestation Form for Utilization of Intellectual Property-Related Work and Output	133

DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY

**Jonel B. Andamon
Jovan P. Nadala
Gerome Q. Voluntad**

An undergraduate design project submitted to the faculty of the Department of Engineering, Cavite State University-CCAT Campus, in Rosario, Cavite, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Engineering with Contribution No. CPE-2026-02-21. Prepared under the supervision of Engr. Sunny May S. Ronas, adviser, and Engr. Princess Yna D. Juyad, technical critic

INTRODUCTION

Libraries in academic institutions serve as essential spaces for study and information access. To complement this function, baggage counters are used to secure students' personal belongings before library entry. However, many institutions, including the CCAT e-Library, still rely on manual systems involving handwritten logs and claim stubs. These conventional approaches contribute to slower processing times, misplaced slips, and increased error rates, particularly during peak hours, negatively affecting user experience and operational efficiency (Sureshkumar *et al.*, 2025).

At the CCAT e-Library, students currently use open cabinets without proper tracking or verification mechanisms. As library usage grows, the absence of an identity-based system makes ownership verification and item security increasingly difficult. Studies confirm that radio-frequency identification (RFID)-based automation addresses these limitations by reducing manual labor, accelerating transactions, and improving service reliability in high-traffic library environments (Yusuf *et al.*, 2023). Furthermore, identity-based RFID authentication has been shown to minimize security

risks and human error by ensuring that only authorized individuals can access stored items (Wang *et al.*, 2022).

To address these challenges, LibLocker integrates a dual-frequency RFID system, where one frequency handles close-range locker authentication while the other manages library doorway verification. This approach follows best practices recommended in literature for simultaneously managing authentication and secure access (Jain, Kumar, & Mehta, 2024). Prior to system utilization, students register their information and are issued a reusable high-frequency (HF) plus ultra-high-frequency (UHF) RFID card, eliminating anonymous access and preventing identity mismatches (Nirudi, 2025). By linking every locker transaction to a verified, registered user, LibLocker ensures accountability and traceability throughout the deposit and retrieval process.

Overall, LibLocker offers a secure, organized, and efficient alternative to manual baggage counters. By shifting toward dual-frequency RFID authentication, the CCAT e-Library can reduce congestion, improve user experience, and advance its transition to a smarter academic environment.

Objectives of the Study

The main objective of this study was to design, develop, and implement LibLocker, an automated library baggage counter system that integrates user registration and RFID card assignment with dual-frequency RFID authentication, camera-assisted verification, secured database, and offline–online operation to enhance security, efficiency, and accountability at the CCAT e-Library.

Specifically, the study aimed to:

1. identify the required hardware and software components, and to implement a user registration module with RFID card encoding for the dual-frequency RFID-based baggage counter system, and to deploy the resulting LibLocker prototype at the CCAT e-Library for testing and evaluation;

2. design the system architecture and workflow, including RFID authentication flow, locker mechanism, camera integration, dashboard interface, and HF-UHF identity-matching logic;
3. develop and integrate all hardware and software components of the LibLocker system;
4. develop a user registration module that records student's information and assigns unique dual-frequency RFID identifiers;
5. implement an RFID card encoding process that assigns reusable physical credentials to registered LibLocker users; and
6. test and evaluate the system's performance in terms of locker security, retrieval accuracy, transaction efficiency, reliability, and user acceptability based on ISO/IEC 25010 and the Technology Acceptance Model (TAM).

Significance of the Study

LibLocker addresses the limitations of the CCAT e-Library's manual baggage handling by introducing a secure, automated RFID-based solution. Through user registration, RFID card encoding and assignment, dual-frequency authentication, camera-assisted verification, and real-time monitoring, the system ensures that only authorized students can access their deposited belongings, effectively reducing the risks of loss, theft, and misidentification.

For students, LibLocker provides a faster, tap-based deposit and retrieval process without the need for claim stubs, offering convenience and improved user experience.

For library personnel, the automated system minimizes manual workload, improves transaction accuracy, and enhances monitoring through digital logs, snapshots, and unclaimed-deposit notifications.

At the institutional level, the system supports CCAT e-Library's move toward smart campus technologies by integrating secure RFID processes, digital dashboards,

and dual offline–online operation. This contributes to improved service delivery, operational efficiency, and strengthened security protocols within the campus environment.

Time and Place of the Study

The study was conducted at the e-Library of Cavite State University-CCAT Campus, where the LibLocker system was installed, tested, and evaluated under actual library operating conditions. The duration of the study was from September 2025 to June 2026, following the approved research timeline.

Scope and Limitations of the Study

The scope of the study includes the design, development, and implementation of the LibLocker system, which integrates user registration and RFID card encoding, dual-frequency RFID authentication, automated locker control, camera-assisted transaction logging, and a digital dashboard for real-time monitoring. The system supported both offline and online operation modes and is tested for security, retrieval accuracy, efficiency, and user acceptability among CCAT e-Library students and staff.

The study was limited to the baggage counter operations within the CCAT e-Library and does not cover other library services or campus-wide access systems. The RFID authentication was tested using system-issued reusable RFID cards assigned via the kiosk's registration interface, and UHF detection was restricted to the designated library entrance area. The evaluation did not include long-term durability testing or integration with external campus databases beyond what was necessary for system operation. Additionally, the handling of unclaimed deposits follows the library's existing policies.

Definition of Terms

This section presents significant terms, along with their definition as how they are used in this study.

Access control refers to a security technique that regulates who can view or use resources in a computing or physical environment.

Automated system refers to a system that operates automatically using programmed instructions and electronic components with minimal human intervention.

Authentication refers to the process of verifying the identity of a user before granting access to a system or device.

Baggage counter refers to a designated area where users temporarily store their personal belongings before entering a restricted area such as a library.

Dual frequency RFID refers to a radio frequency identification technology that operates using two different frequency ranges to improve compatibility, security, and detection accuracy.

Embedded system refers to a combination of hardware and software designed to perform a specific function within a larger system.

E-Library refers to a digital or electronically managed library system that provides access to library resources and services through technology.

HF-UHF identity matching refers to the logical process of verifying that the HF RFID card used at the kiosk and the UHF tag detected at the library entrance belong to the same registered user.

LibLocker refers to the automated locker system developed in this study that uses RFID authentication to manage and secure users' baggage in the library.

Locker system refers to a storage mechanism consisting of secured compartments where users can temporarily store their personal belongings.

Radio Frequency Identification (RFID) refers to a wireless technology that uses radio waves to automatically identify and track tags attached to objects.

RFID card refers to a small electronic device containing microchip and antenna that stores identification information and communicates with an RFID reader.

RFID reader refers to an electronic device that communicates with RFID card to read or retrieve stored data.

Security system refers to a set of technologies and procedures used to protect property, data, or facilities from unauthorized access.

User identification refers to the process of recognizing and verifying a users' identity before allowing access to a system.

Conceptual Framework of the Study

The conceptual framework of this study follows the Input-Process-Output (IPO) model, which presents the overall development of the LibLocker system. It shows how the hardware and software requirements (input) undergo planning, development, testing, and implementation (process) to produce an automated library baggage counter with dual-frequency RFID authentication (output). The framework guided the researchers in developing a secure, efficient, and reliable system for the CCAT Campus e-Library.

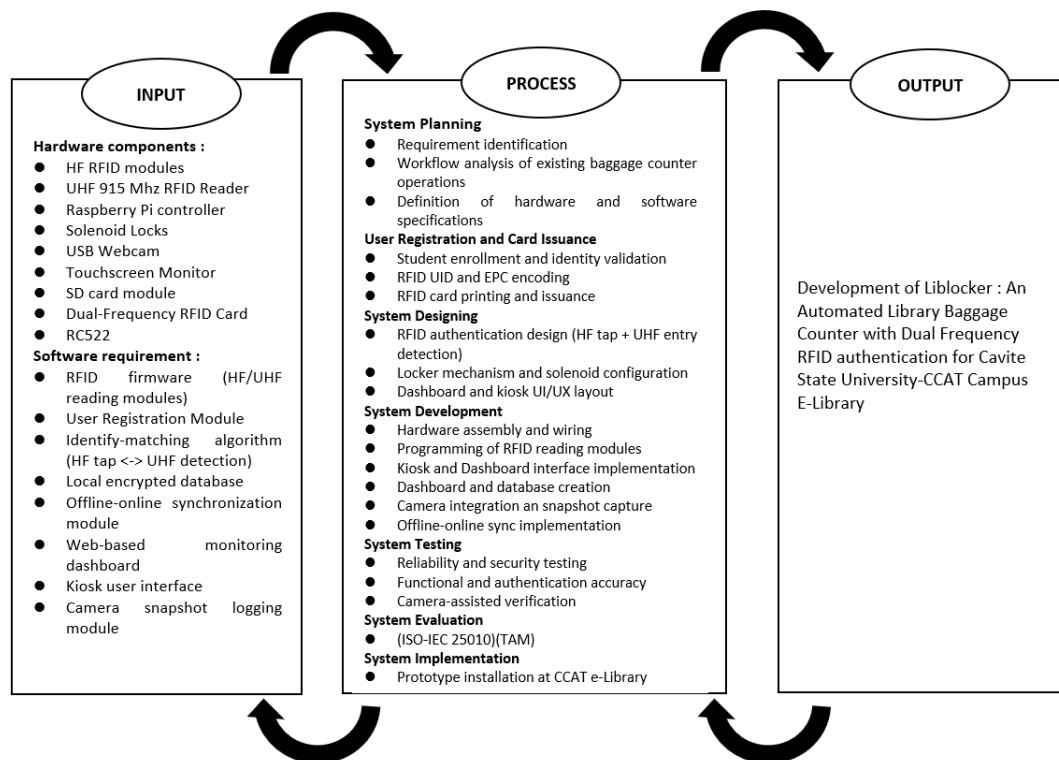


Figure 1. Conceptual framework of the study

LibLocker follows an Input–Process–Output (IPO) model. Hardware inputs include HF RFID modules, a UHF 915-MHz reader, Raspberry Pi controller, solenoid locks, webcam, touchscreen monitor, SD card module, and reusable RFID cards. Software inputs consist of RFID firmware, a user registration module, local user records, a monitoring dashboard, offline–online synchronization, and HF-UHF identity-matching algorithms. These undergo system planning, user registration with RFID card encoding, system design and development, and evaluation based on ISO/IEC 25010 and TAM. The resulting output is LibLocker, an automated baggage counter system integrating dual-frequency RFID authentication, secure locker assignment and retrieval, real-time monitoring, and transaction logging, delivering efficient and accountable baggage management for the CCAT e-Library.

REVIEW OF RELATED LITERATURE

The digital transformation of libraries necessitates technologies that enhance efficiency, security, and user experience. RFID has emerged as a foundational technology in this shift, moving beyond basic inventory to enable comprehensive automation of library services. This review explores the application of RFID in core library functions, its expansion into asset and locker management, the integration of advanced dual-frequency and sensor-fusion technologies, and the critical security and user registration frameworks required for reliable systems. Together, this literature provides the technological and conceptual justification for developing integrated solutions like LibLocker, which aims to automate ancillary services such as baggage handling.

RFID in Library Automation

The increasing need for efficient, accurate, and secure library services has led educational institutions to shift from manual processes toward automated systems. RFID has emerged as one of the most widely adopted technologies for modernizing library operations due to its ability to support fast identification, tracking, and inventory management. Aguila and Balba (2024) developed SmartLibrary, an RFID-based system that automates attendance tracking and book inventory, demonstrating that RFID significantly reduces manual workload and enhances monitoring accuracy. Similarly, the study of Dwivedi, Singh, and Tiwari (2012) highlighted that RFID systems improve user satisfaction by minimizing queue times, reducing handling errors, and supporting efficient circulation workflows. These findings reinforce the growing need for RFID-integrated solutions such as LibLocker, which seeks to automate baggage handling at library entrances.

RFID and IoT Applications in Library and Asset Management

RFID's role in library automation has expanded beyond book borrowing to

include real-time asset tracking, anti-theft detection, and centralized data management. Arfan (2024) emphasized that RFID-based smart libraries create seamless interactions between users and library systems by automating identification and transaction logging. De La Salle University Manila's (2018) early research on RFID-based book monitoring systems further demonstrated the feasibility of real-time inventory management and automatic security detection through RFID gateways.

Vedanth *et al.* (2023) also contributed to this domain by developing an automated library management system that integrates UHF RFID readers for accurate multi-tag detection, allowing faster scanning and enhanced stock control. As RFID increasingly becomes a standard for library automation, these works establish strong justification for applying RFID to ancillary library services, such as baggage counters, where identity accuracy, security, and real-time logging are equally important.

RFID-Based Locker and Storage Systems

Smart lockers have been widely explored as secure storage solutions in schools, workplaces, and public facilities. Bharatiraja *et al.* (2023) presented an IoT-based centralized smart locker system using RFID technology, where each compartment can be monitored and controlled remotely through a cloud platform. Their system highlights the importance of centralized dashboards, user authentication logs, and automated access control key design principles adopted in LibLocker.

Bhat, Manohar, and Acharya (2025) extended the concept into dormitory environments by integrating RFID for identifying students and visitors, preventing unauthorized access, and maintaining digital logs of locker usage. These studies illustrate that RFID-enabled lockers not only improve security but also provide accountability by linking locker activity to verified user identities, which is an essential requirement for baggage counters that must prevent item loss and disputes.

Dual-Frequency RFID Technologies

Recent advancements in RFID technology have introduced more sophisticated

sensing capabilities and improved interaction with IoT systems. Costa *et al.* (2021) described RFID sensors as an emerging frontier in IoT, highlighting how passive tags can support not only identification but also environmental sensing. Duroc (2022) expanded on this by explaining the transition of RFID from simple identification tools to multifunctional devices within IoT environments, supporting real-time analytics, long-range detection, and automated event documentation.

These developments are relevant to LibLocker's use of dual-frequency RFID, HF for kiosk authentication and UHF for doorway detection, which leverages the strengths of both short- and long-range communication. The literature validates that combining RFID modes enhances precision, reduces false detection, and supports complex interactions such as verifying whether a user who deposited baggage is the same person entering the library.

RFID–Camera Fusion Technologies

Sensor fusion is increasingly applied to improve security and accountability in automated systems. Liu *et al.* (2021) demonstrated that combining RFID with camera systems enhances recognition of human-object interactions, providing an additional verification layer that can resolve ambiguous or disputed events. This is particularly important in environments where item tracking, identity confirmation, and behavioral monitoring intersect. LibLocker's design, which captures images during baggage deposit and retrieval, mirrors these findings by incorporating multi-sensor validation to ensure accurate logging and minimize unauthorized use or retrieval errors.

User Registration in RFID-Based Systems

User registration is a fundamental requirement in RFID-based systems that handle access control and personal property. Registration ensures that each system user is uniquely identified and that all transactions are associated with verified individuals. In secure environments such as libraries and storage facilities, registration prevents anonymous access and reduces errors related to ownership verification.

Bharatiraja *et al.* (2023) emphasized that RFID-enabled locker systems become more secure and reliable when access is limited to registered users. Their study highlighted that centralized user enrollment allows systems to enforce access policies, improve monitoring accuracy, and reduce disputes over stored items. This supports the design of LibLocker, where users must complete registration before being allowed to deposit or retrieve baggage.

Similarly, Arfan (2024) explained that smart library systems rely on structured user registration to maintain data integrity and ensure reliable automation. By formally enrolling users, RFID-based systems can accurately track usage patterns, maintain clean databases, and support real-time monitoring—features that are directly applied in the LibLocker system.

RFID Credential Encoding and Issuance for Secure Access

RFID credential encoding and issuance play a critical role in strengthening access control by ensuring that credentials are system-issued and uniquely bound to registered users. Unlike generic or externally issued RFID cards, system-assigned tokens allow tighter control over credential management and reduce the risk of unauthorized access.

Bhat *et al.* (2025) demonstrated that RFID credential issuance in locker and storage environments improves authentication accuracy and minimizes identity mismatches. Their findings indicate that issuing specific RFID cards during registration ensures consistency between user records and physical credentials, resulting in more reliable access control.

Dwivedi *et al.* (2012) also noted that RFID-based systems that integrate formal credential issuance achieve higher user trust and operational efficiency. When users are issued system-verified RFID cards, transaction errors decrease, queues are reduced, and system acceptability improves. These findings justify LibLocker's inclusion of an RFID card encoding and assignment process as a core system

component.

Registration-Based RFID Security and Accountability

Proper registration and controlled RFID card issuance are essential in mitigating security risks associated with RFID systems. Unmanaged RFID environments are vulnerable to issues such as tag cloning, unauthorized use, and anonymous transactions, which compromise system integrity.

Munoz-Ausecha *et al.* (2021) emphasized the importance of credential control, encryption, and identity binding in securing RFID applications. Their study recommends limiting RFID usage to registered users and implementing strong identity linkage between tags and system records. LibLocker addresses these concerns by issuing RFID cards only through its registration module and by linking each card to secured user records and transaction logs.

By integrating user registration, RFID card encoding and printing, and transaction logging, LibLocker establishes a secure and accountable baggage management system. The literature supports this approach, confirming that registration-based RFID systems provide improved security, traceability, and reliability in environments where personal belongings are involved.

RFID Security Issues and Mitigation

As RFID becomes deeply integrated into security-sensitive applications, concerns regarding tag cloning, unauthorized scanning, and data tampering have been widely documented. Munoz-Ausecha *et al.* (2021) provided a comprehensive review of RFID security risks, emphasizing the need for encryption, authentication checks, and controlled reader access. These challenges highlight the importance of implementing secure communication protocols and robust logging mechanisms in systems like LibLocker, where user belongings are stored and must be protected from unauthorized retrieval.

The literature stresses that while RFID improves speed and convenience,

system designers must counter vulnerabilities through multi-layer security, reinforcing LibLocker's adoption of dual-frequency verification, secured logs, camera-assisted validation, and a registration-based access model.

IoT Architectures for Real-Time Monitoring

Effective IoT systems rely on seamless communication between sensors, gateways, databases, and user interfaces. Studies such as those by Aguila and Balba (2025) and Arfan (2024) highlighted the importance of centralized dashboards and real-time data synchronization to support efficient library operations. Cloud-assisted RFID systems, as discussed across multiple works, underscore the value of high-availability data access, remote monitoring, and automated logging.

However, the literature also notes challenges including network dependency and the need for offline continuity, which are factors that LibLocker addresses by incorporating local data handling and integration with stable campus networks to ensure uninterrupted service even during connectivity fluctuations.

METHODOLOGY

The methodology of the study follows the major phases identified in the LibLocker conceptual framework: planning, designing, development, testing, evaluation, implementation, and data analysis. These phases guided the systematic procedures for designing, constructing, and validating the automated library baggage counter system.

Description of the Project/Technology

LibLocker is an automated library baggage counter system developed to improve security, accountability, and efficiency in managing personal belongings within the Cavite State University-CCAT Campus e-Library. The system employs dual-frequency RFID authentication, utilizing HF RFID for user authentication and locker, and UHF RFID for passive library entry verification at the doorway. The HF reader is the only mechanism that opens the lockers. On the other hand, the UHF reader passively confirms that the student who deposited a bag has entered the library and supports violation tracking for non-compliance.

The system includes a user registration module where students are enrolled and issued a dual-frequency RFID card linked to their identity. An HF RFID kiosk is used for tap-based authentication and locker assignment, while a UHF doorway gate detects movement during deposit and retrieval. A Raspberry Pi-based local gateway serves as the central controller, coordinating RFID validation, locker actuation, and image capture using a USB webcam. Secured transaction records and snapshots are stored locally and synchronized to a cloud server. A web-based dashboard allows librarians and administrators to monitor locker status, review transactions, and manage unclaimed items.

Through the integration of RFID technology, embedded systems, and web-based monitoring, LibLocker provides a secure, scalable, and practical solution for

library baggage management.

Respondents and Sampling Technique

The study involved 30 respondents composed of selected students currently enrolled at Cavite State University-CCAT Campus and library personnel of the CCAT e-Library. Respondents were selected through purposive sampling. The inclusion criteria required respondents to be either active CCAT students who regularly use the library, or library staff directly involved in baggage counter operations. Respondents who had no prior interaction with the LibLocker prototype were excluded. The sample size of 30 was considered sufficient for the prototype evaluation stage of this study, given the controlled deployment environment and the formative nature of the system testing.

Research Instrument

The study used a structured survey questionnaire as the main research instrument. The questionnaire was divided into two parts. The first part contained items based on the ISO/IEC 25010 software quality model, covering six characteristics: functional suitability, performance efficiency, reliability, security, usability, and compatibility. The second part contained items based on the Technology Acceptance Model (TAM), covering perceived usefulness, perceived ease of use, behavioral intention to use, and user satisfaction. All items used a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The questionnaire was reviewed by the design project adviser and the technical critic before distribution to confirm that the items were clear, complete, and aligned with the system's objectives.

Statistical Treatment of Data

Quantitative data from system testing and the survey were analyzed using descriptive statistics. For system performance, the researchers computed averages for transaction times, accuracy percentages for RFID pairing, retrieval, and encoding

trials, and frequencies for error or violation events. For user evaluation, weighted means were computed for each ISO/IEC 25010 criterion and each TAM construct. The resulting means were interpreted using the Likert scale interpretation shown in Table 1. Microsoft Excel was used to consolidate responses and compute the descriptive statistics.

Ethical Considerations

Respondents were informed of the purpose of the study before completing the survey, and participation was voluntary. No personally identifiable information was collected beyond what was needed for the study, and survey responses were treated as confidential and used only for academic purposes, in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173).

Components

The following materials discussed in this section were used in the development of the LibLocker.



Figure 2. Raspberry Pi 4 model B

The Raspberry Pi 4 Model B, shown in Figure 2, serves as the system's central processor, featuring a 1.5-GHz, quad-core CPU, sufficient RAM for simultaneous operations, and 40 GPIO pins to interface with sensors and relays across all 24

compartments.

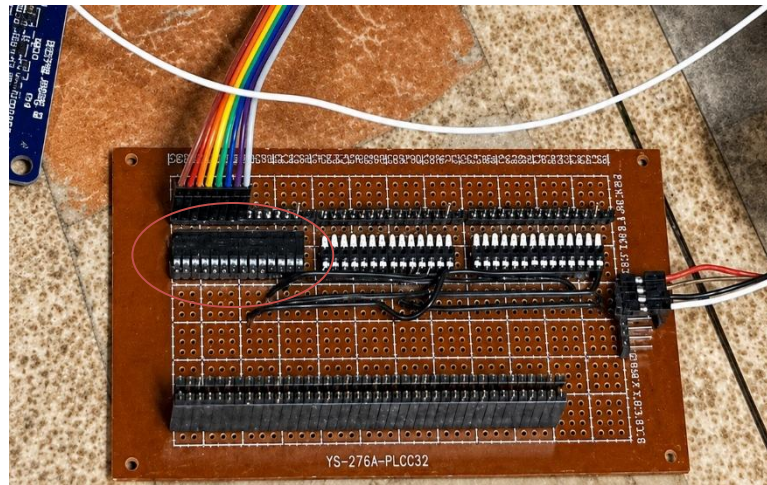


Figure 3. I/O expander (MCP23017)

The MCP23017, shown in Figure 3, is an I²C I/O expander chip. Three units are used in the system, each providing 16 GPIO pins, to independently control and monitor all 24 solenoid locks and door sensors via the Raspberry Pi.

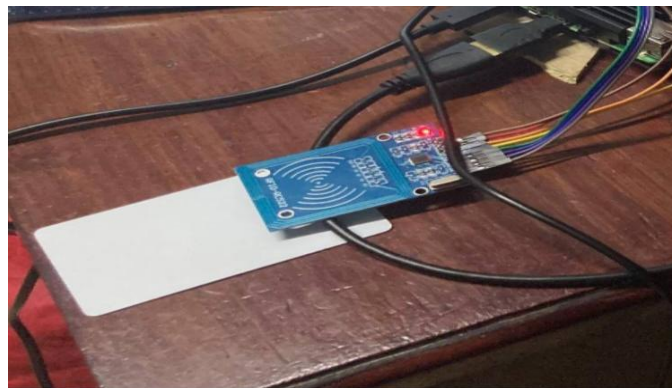


Figure 4. HF RFID reader (RC522)

The HF RFID reader (RC522), shown in Figure 4, is a 13.56 MHz module supporting ISO/IEC 14443A for contactless card read/write operations, installed on the kiosk panel to handle user registration, deposit, and retrieval authentication.



Figure 5. UHF RFID reader (915 MHz)

The UHF RFID reader (915 MHz), shown in Figure 5, is a long-range reader operating within the 902–928 MHz band capable of detecting tags from several meters away, mounted at the library entrance gate to automatically identify students carrying a registered dual-frequency card upon entry.



Figure 6. Solenoid lock (12v)

The 12-V solenoid lock, shown in Figure 6, is a 12-V DC electromechanical lock with a spring-loaded slanted tongue that allows doors to be pushed shut and can only be retracted via electrical impulse, providing sufficient holding force to secure each locker compartment against forced entry.



Figure 7. Magnetic door sensor (Reed Switch)

The magnetic door sensor (reed switch), shown in Figure 7, is a two-part magnetic switch with one piece fixed to the door frame and the other to the door itself, used to provide the Raspberry Pi with real-time open or closed status for each locker to prevent the solenoid from engaging on an improperly shut door.



Figure 8. USB webcam

The USB webcam, shown in Figure 8, is a 1080p wide-angle camera mounted on the kiosk that captures timestamped photo logs of each deposit and retrieval transaction, adding a visual security layer to the system's activity records.



Figure 9. Capacitive touchscreen monitor

The capacitive touchscreen monitor, shown in Figure 9, is a 10.1-in display connected to the Raspberry Pi via HDMI and USB that serves as the primary kiosk interface, offering more responsive and accurate touch input than resistive screens for smooth student interaction during registration and locker transactions.



Figure 10. Dual-frequency RFID card

The dual-frequency RFID card, shown in Figure 10, is a contactless smart card embedding two chips where one at 13.56 MHz (ISO/IEC 14443A) for short-range kiosk interaction via the RC522, and one at UHF (EPC Gen2/ISO 18000-6C) for long-range gate detection which issued to students as reusable digital claim cards for locker access and automatic entrance identification.

Design and Fabrication Process

This section covers the design and fabrication process of the LibLocker system, from planning and component selection to hardware assembly and software integration.

Design

During the design phase, the system architecture was defined to ensure seamless interaction among the user registration and token assignment module, HF RFID kiosk, UHF doorway detection system, Raspberry Pi controller, local database, camera module, and web-based monitoring dashboard. Hardware layout was planned to optimize close-range HF authentication at the kiosk, wide-range UHF detection at the library entrance, and secure locker control with camera coverage for transaction snapshots.

User interfaces were designed for the registration module, kiosk, and librarian dashboard to support clarity and efficient workflow. The kiosk guides students through authentication, locker assignment, and retrieval, while the dashboard presents real-time locker status, transaction logs, and verification images. System workflows covering offline operation, synchronization, unclaimed deposit handling, and error conditions were also mapped to ensure secure and reliable operation.

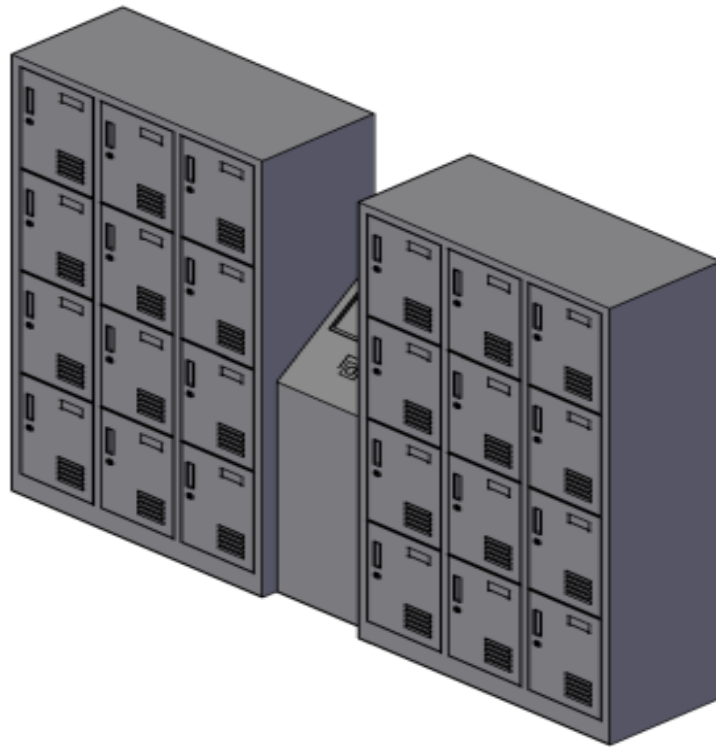


Figure 11. Three-dimensional design of the LibLocker

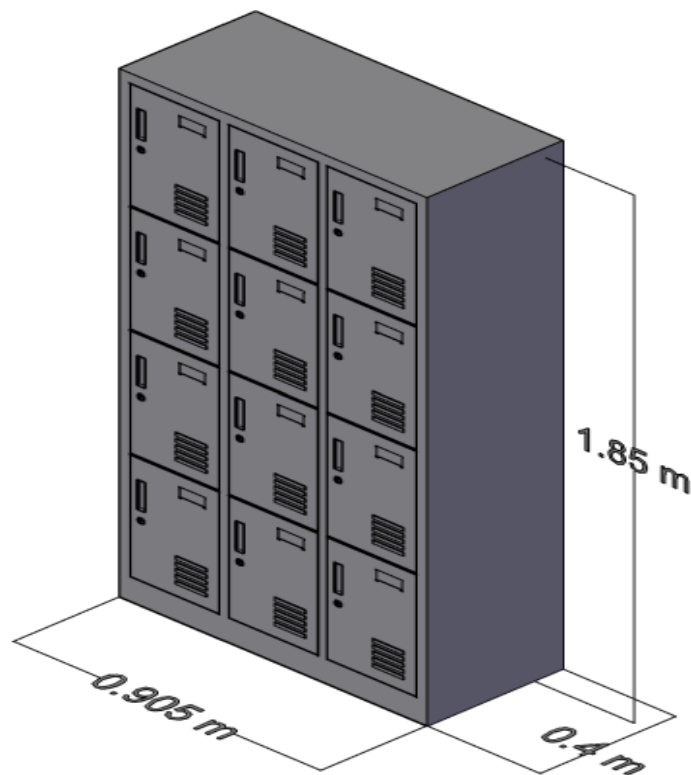


Figure 12. Side view of the design of the LibLocker



Figure 13. Front view of the design of the LibLocker

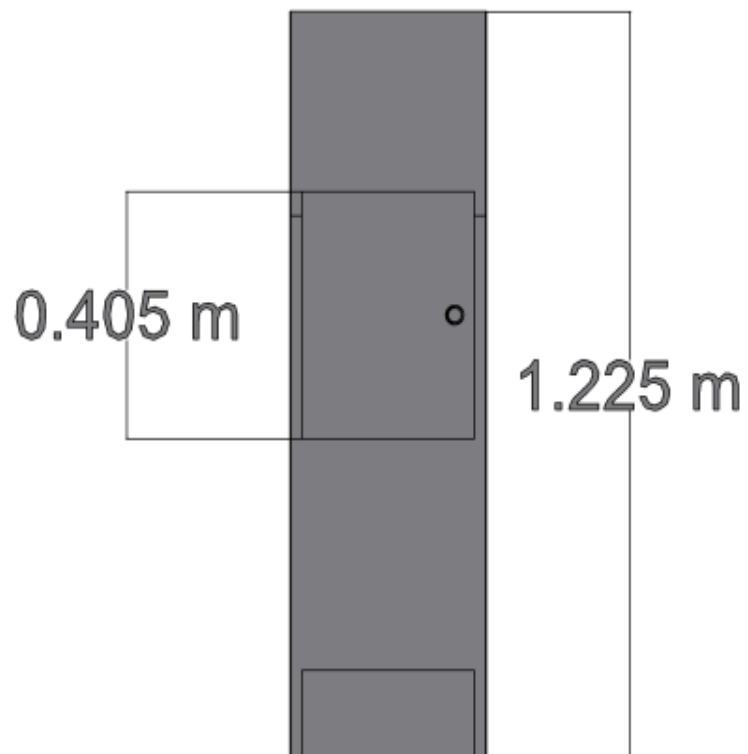


Figure 14. Rear view of the design of the LibLocker

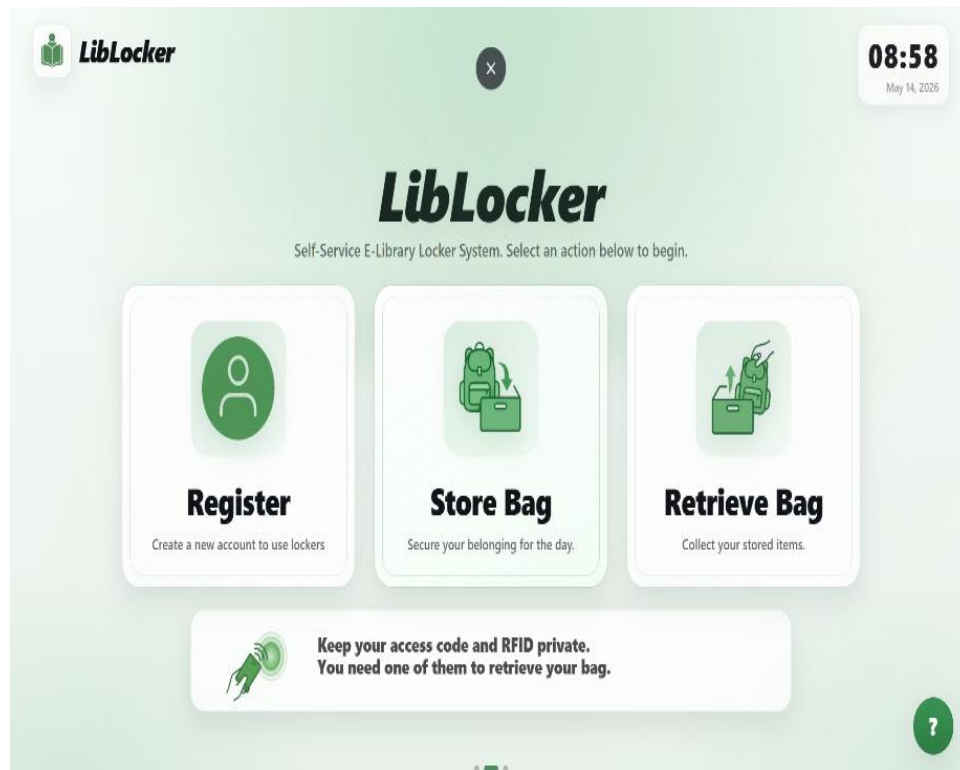


Figure 15. Kiosk main menu interface

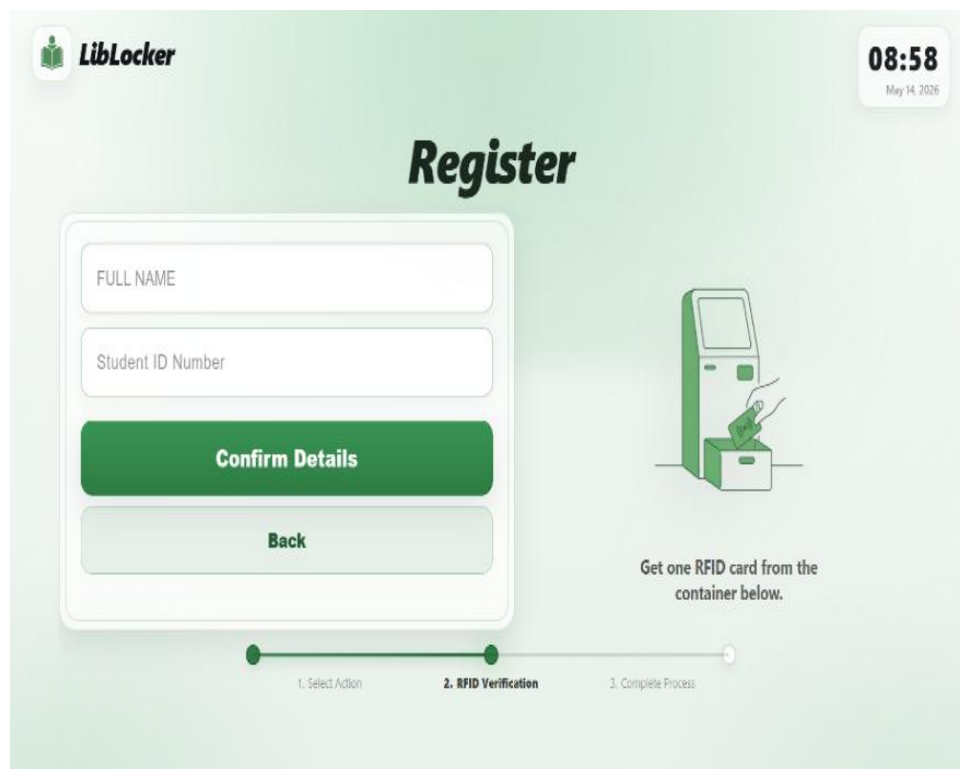


Figure 16. User registration interface

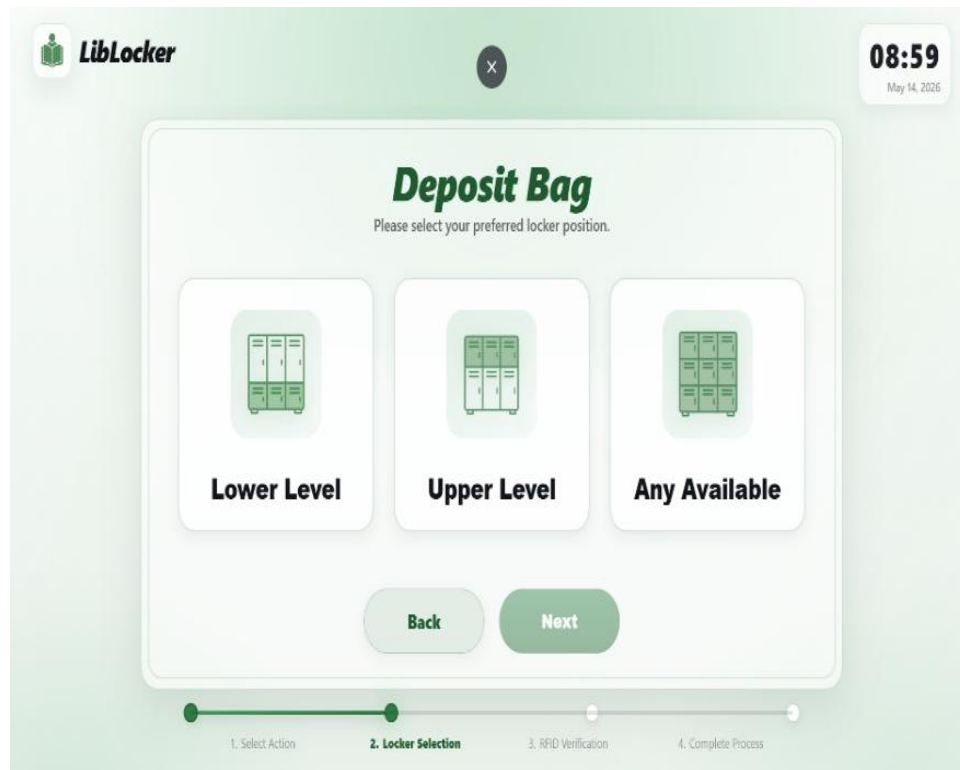


Figure 17. Locker selection interface

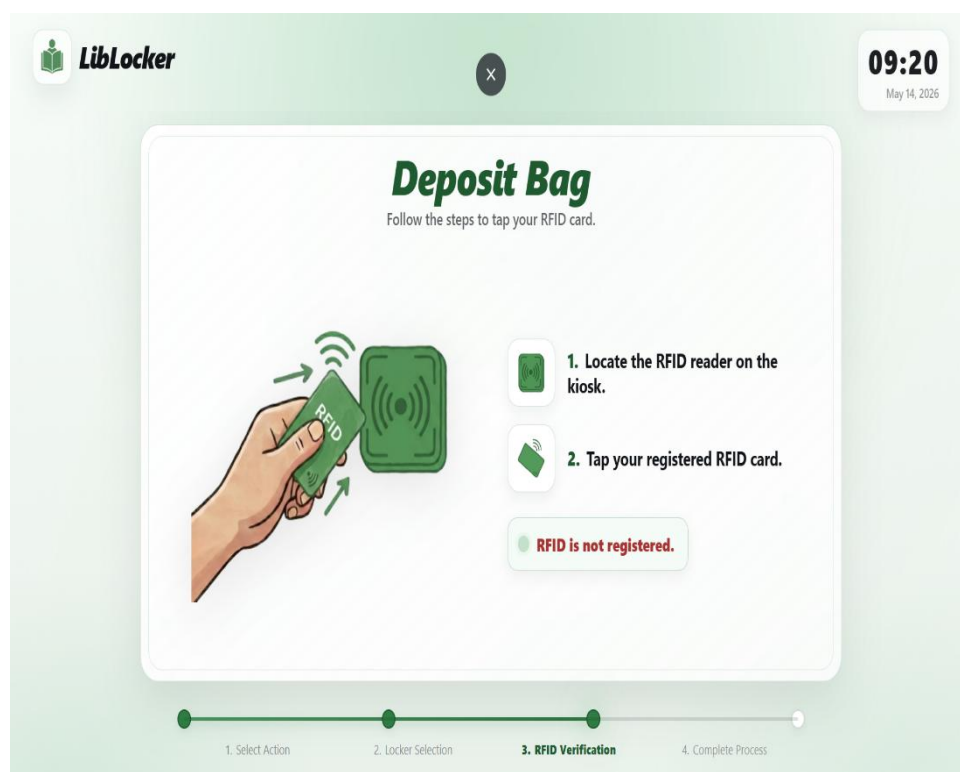


Figure 18. RFID verification prompt (Deposit)

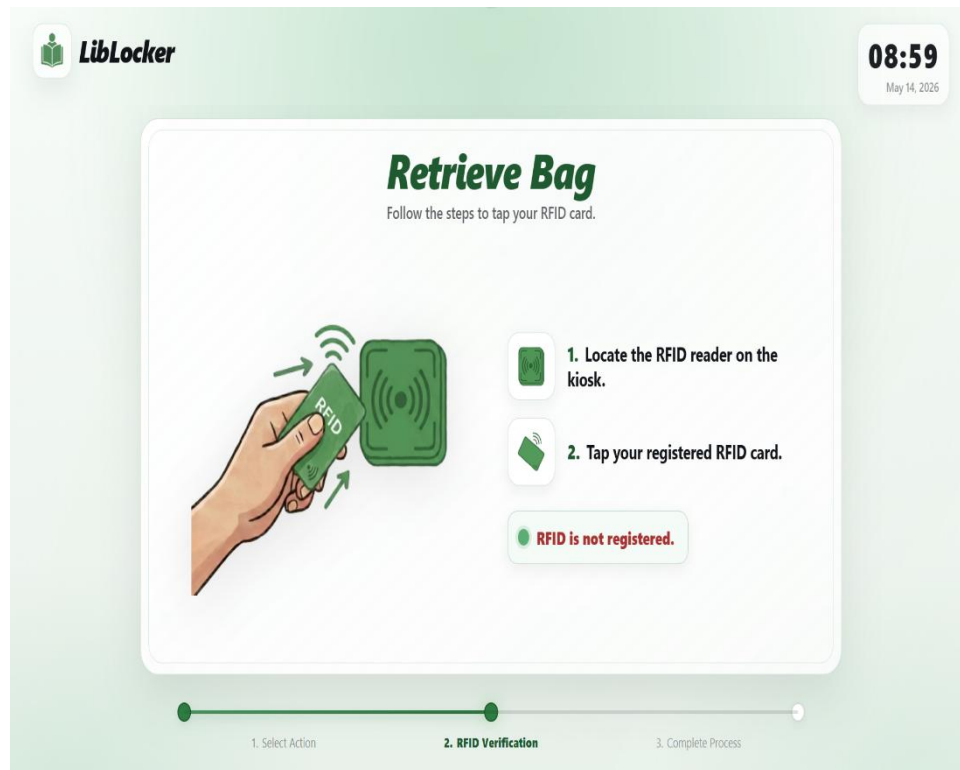


Figure 19. RFID verification prompt (Retrieval)

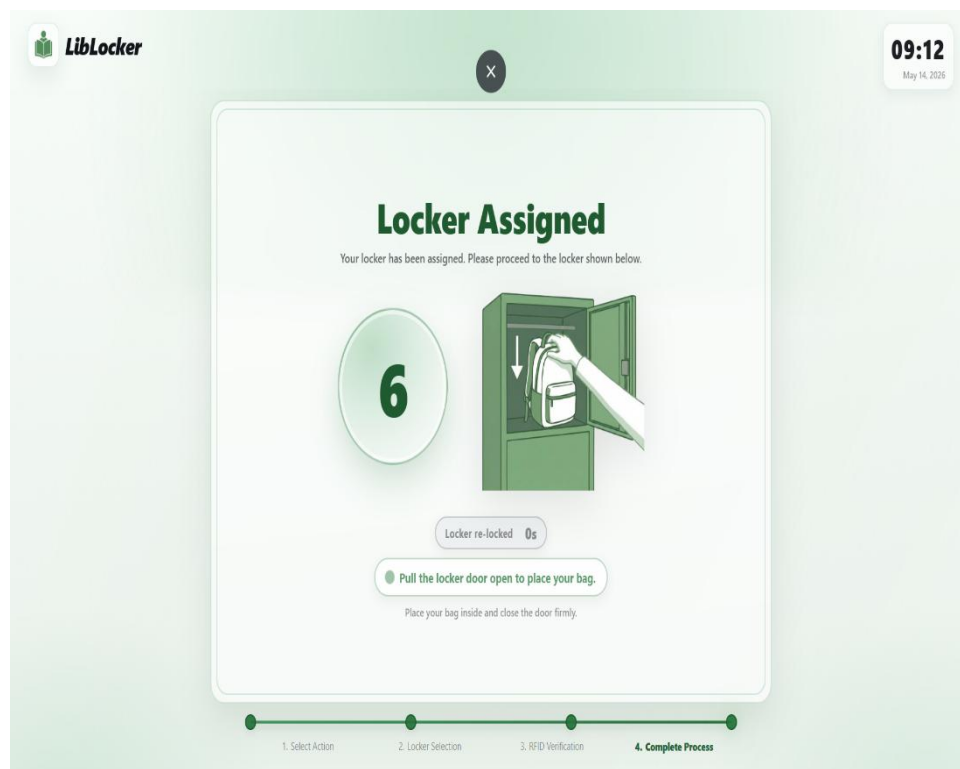


Figure 20. Deposit success screen



Figure 21. Retrieval success screen

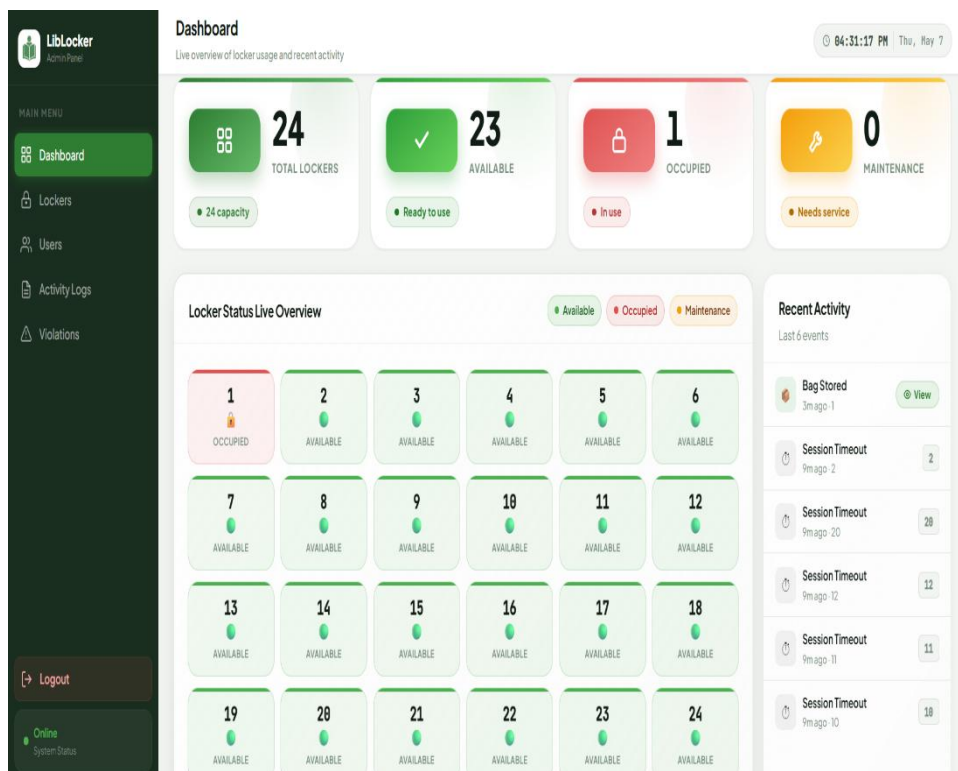


Figure 22. Dashboard overview

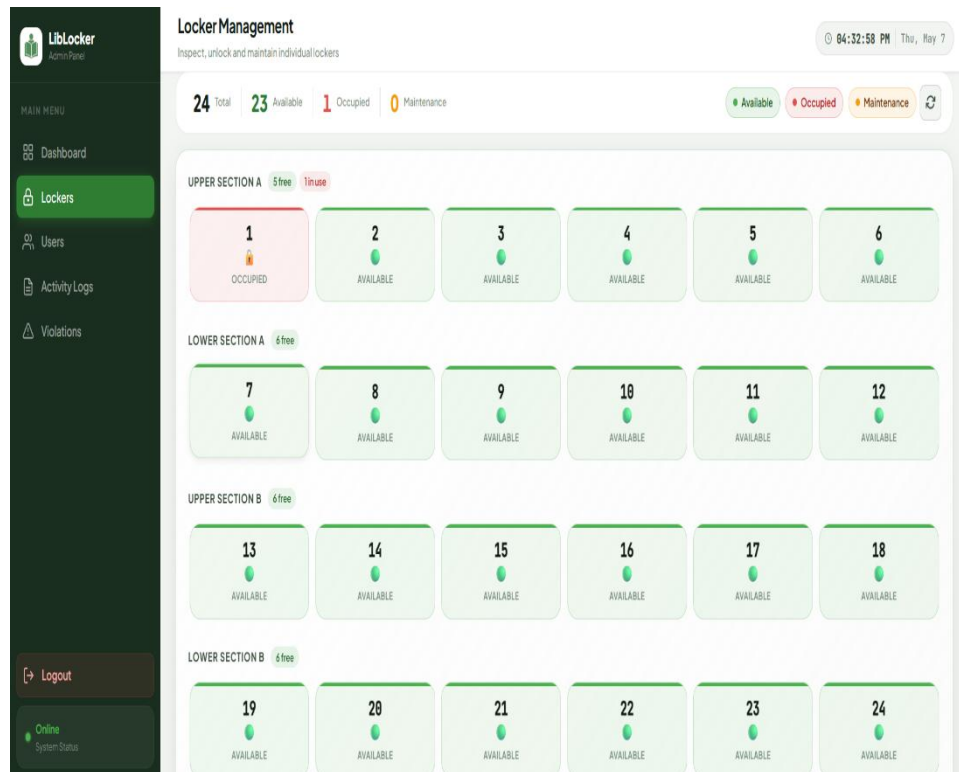


Figure 23. Locker status grid

Registered Users
All students with a registered RFID card

04:33:17 PM Thu, May 7

Registered Students
All users who have registered their RFID card

Search by name, student ID, RFID or UHF... Refresh

#	FULL NAME	STUDENT ID	RFID CARD	UHF TAG	REGISTERED	ACTIVE LOCKER	STATUS	ACTIONS
1	SEBASTIANTAGALOG	202210711	807388E39A4	—	May 4, 2026 10:54 PM	—	✓ Active	Manage
2	JOVAN NADALA	2020010769	80803830053F	—	May 4, 2026 10:53 PM	—	✓ Active	Manage
3	ADD	809755544556	804380C8392D	8179561242	May 4, 2026 10:32 PM	L-01	✓ Active	Manage
4	NEL	202210714	8083488639C7	—	May 4, 2026 10:31 PM	—	✓ Active	Manage
5	JONRL	202210706	—	—	May 4, 2026 10:19 PM	—	⚠ Pending	Manage
6	JOVAN P.	20221706	80A338C239A8	8179561258	May 4, 2026 10:18 PM	—	✓ Active	Manage
7	ROMEG	202110782	807348C839CF	8179561266	May 4, 2026 09:22 PM	—	✓ Active	Manage
8	GROME	202110753	80A460E1062E	8179561238	May 4, 2026 08:59 PM	—	✓ Active	Manage
9	ROMMMM	202110747	800365043958	—	May 4, 2026 08:58 PM	—	✓ Active	Manage

Figure 24. Active log view

Entry Violations
Real-time monitoring of UHF entry-validation violations

06:32:51 PM Mon, May 4

Type: All types Status: Unresolved only Refresh now

TIME	TYPE	STUDENT	UHF EPC	DESCRIPTION	STATUS	ACTION
2026-05-04 10:32	Entry without deposit	unknown	3593761798	UHF tag detected at entrance but no recent deposit session exists and the EPC is not registered to any user.	UNRESOLVED	Resolve
2026-05-04 10:32	Entry without deposit	unknown	0179561242	UHF tag detected at entrance but no recent deposit session exists and the EPC is not registered to any user.	UNRESOLVED	Resolve
2026-05-04 10:32	No entry after deposit	202110743 GEROME	—	Student 202110743 deposited a bag but did not enter the library within 300 seconds.	UNRESOLVED	Resolve
2026-05-04 10:30	Entry without deposit	unknown	3593761798	UHF tag detected at entrance but no recent deposit session exists and the EPC is not registered to any user.	UNRESOLVED	Resolve
2026-05-04 10:30	Entry without deposit	unknown	0179561242	UHF tag detected at entrance but no recent deposit session exists and the EPC is not registered to any user.	UNRESOLVED	Resolve

Recent Sessions LIVE
Status indicator from sessions.entered_library and sessions.violation_flag

CREATED	STUDENT	LOCKER	ACTION	ENTRY TIME	STATUS
2026-05-04 10:31	202110743 GEROME	Locker1	DEPOSIT	—	Violation

Figure 25. Violations page

Development

The development phase involved assembling the LibLocker prototype and integrating all hardware components, including the HF RFID reader, UHF doorway detection system, Raspberry Pi controller, solenoid locks, touchscreen kiosk, USB camera, and SD card module. Hardware wiring, mounting, and calibration were completed to ensure stable communication, accurate UHF detection, and reliable locker control.

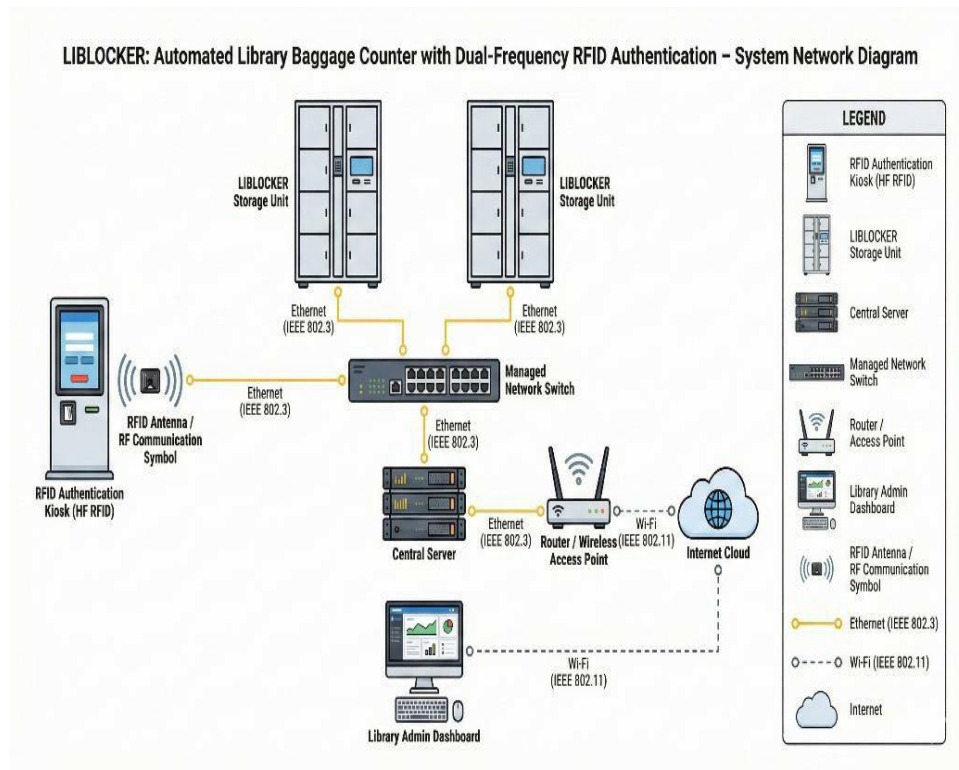


Figure 26. System network diagram of LibLocker

System software and firmware were developed to support user registration, RFID card encoding, dual-frequency RFID authentication, HF-UHF identity matching, automated locker assignment and retrieval, and camera-assisted transaction logging. The Raspberry Pi managed device communication, local data storage, and offline–online synchronization with the dashboard. A web-based dashboard was implemented for real-time monitoring, log management, and unclaimed deposit handling, and the system was iteratively tested and refined to ensure reliable operation under normal library conditions.

Principles of Operation

The operation of the LibLocker system begins with user registration, where student information is recorded and a dual-frequency RFID card is assigned and linked to the user's identity in the system database. For baggage deposit, the user taps the RFID card on the HF RFID kiosk to initiate authentication. Upon successful verification,

the Raspberry Pi local gateway assigns an available locker, activates the corresponding solenoid lock, and triggers the webcam to capture a snapshot of the transaction. All data are secured and stored in the local database. As the user passes through the library entrance or exit, the UHF doorway gate detects the RFID card and verifies that the movement corresponds to a valid locker transaction. The system's HF-UHF identity-matching process ensures that only authorized deposits and retrievals are permitted. During retrieval, the user taps the RFID card on the kiosk to unlock the assigned locker. The system records the transaction, captures a retrieval snapshot, updates the database, and displays locker status, records, and images on the web dashboard for librarian monitoring. Data are synchronized with the cloud server for backup and analytics, enabling efficient management of unclaimed items.

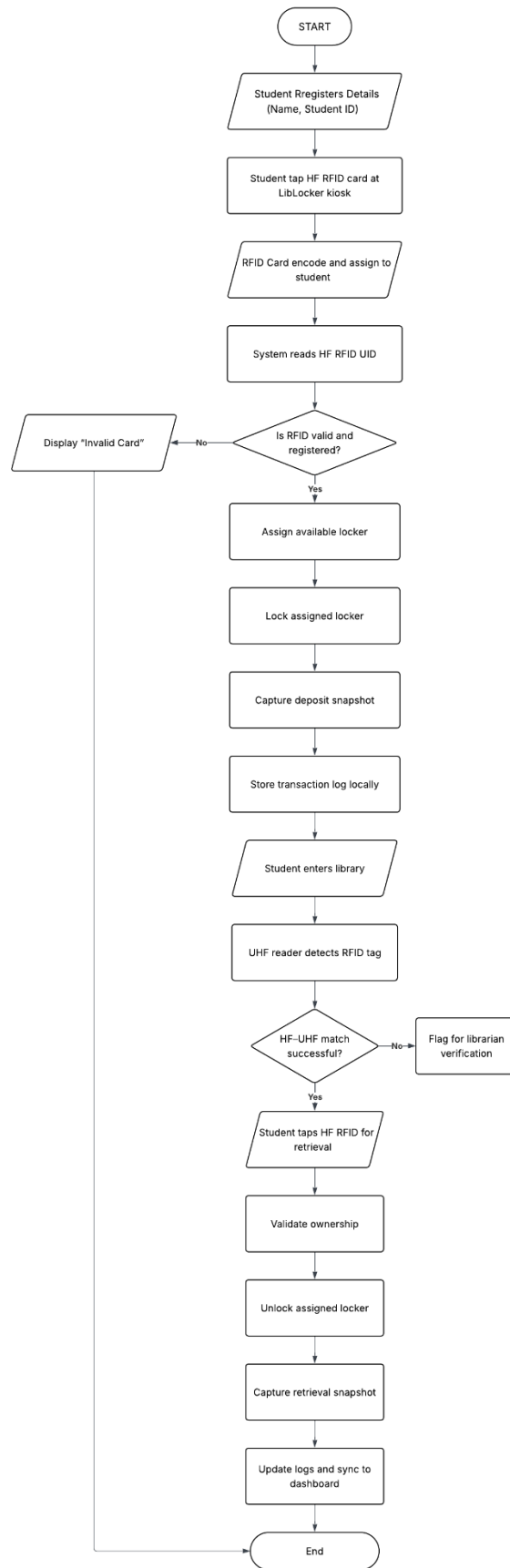


Figure 27. System operational flowchart

Figure 27 illustrates the main operational steps of the LibLocker system. The process begins when the student taps their HF RFID card at the kiosk. The card is validated, and if approved, the system assigns a locker, locks it, and captures a deposit snapshot. As the student enters the library, the UHF gate detects their RFID card and verifies that it matches the earlier HF transaction. A mismatch triggers staff verification to prevent unauthorized access. For retrieval, the student taps their HF card again; once confirmed, the system unlocks the assigned locker and records a retrieval snapshot. This ensures a secure, monitored deposit and retrieval process.

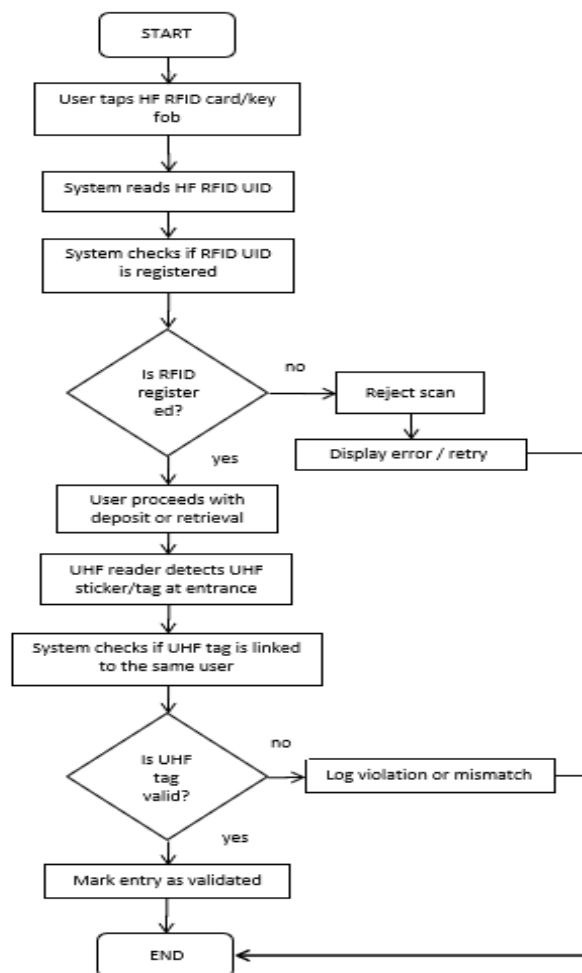


Figure 28. RFID authentication sequence

As shown in Figure 28, the RFID authentication sequence serves as the

primary identity verification method in LibLocker. The system uses an HF RFID reader at the kiosk for registration, deposit, and retrieval, and a UHF RFID reader at the entrance to detect whether a student enters the library after depositing a bag. This separation ensures close-range accuracy at the kiosk while enabling long-range entry validation at the door, allowing the system to distinguish between users who only deposited a bag and those who actually entered, supporting locker misuse prevention

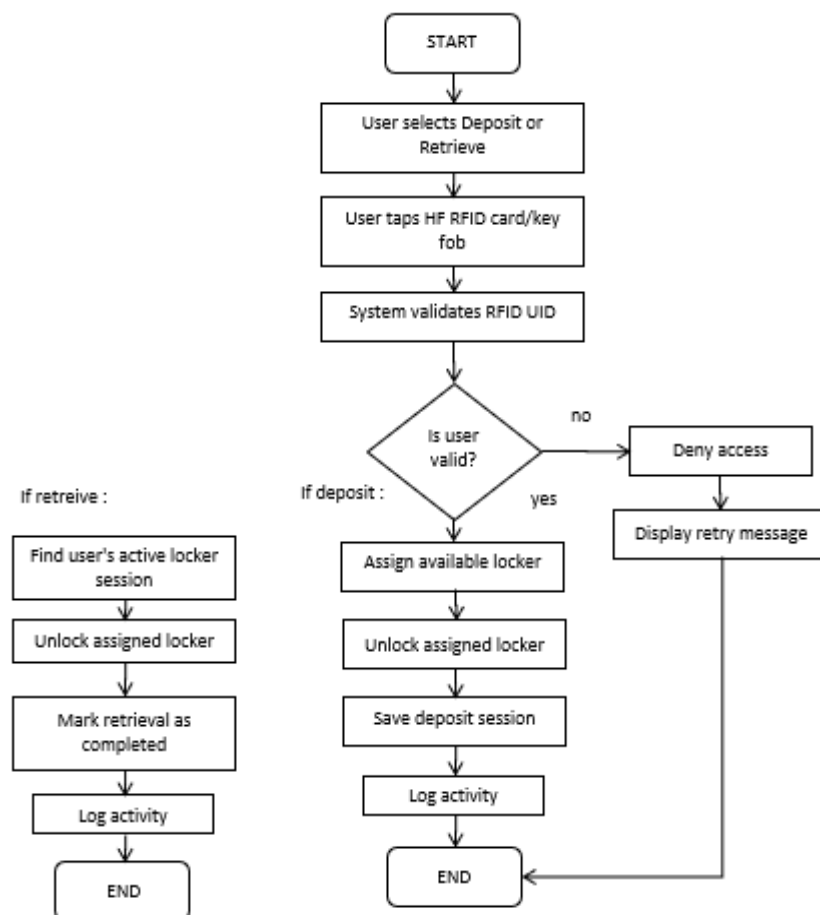


Figure 29. Locker access mechanism

As shown in Figure 29, the locker access mechanism ensures lockers cannot be opened without proper authentication. The system automatically assigns lockers during deposit and verifies the user's session during retrieval, limiting access to the assigned locker only. Transaction logging records all activities for monitoring and

accountability, confirming that the mechanism effectively supports both security and system control.

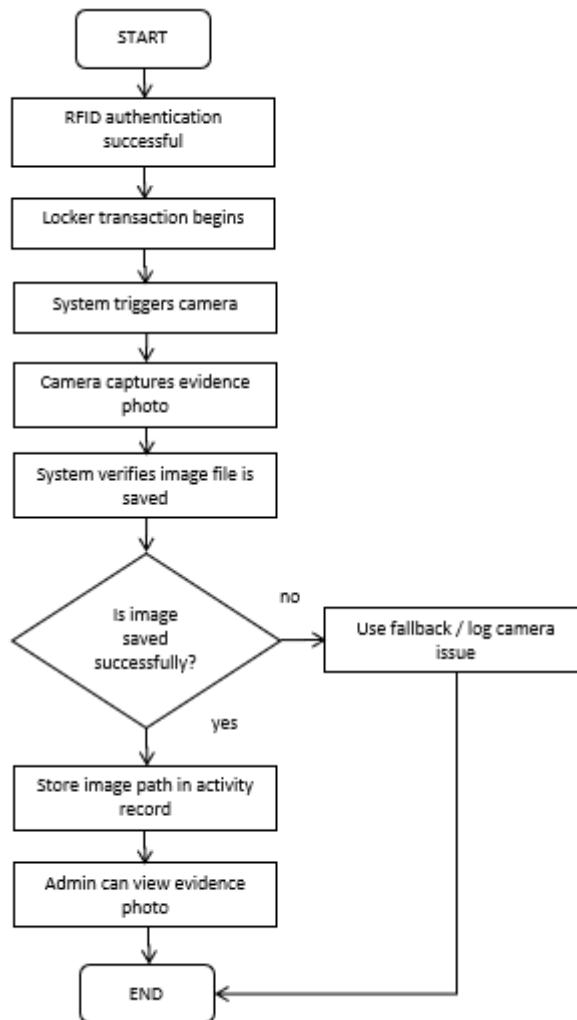


Figure 30. Camera-assisted verification

As shown in Figure 30, the camera-assisted verification feature captures photos during locker transactions and links them to corresponding activity logs, providing visual evidence for dispute resolution and accountability. This allows admins to verify both user actions and visual records, confirming the feature effectively supports the system's security and monitoring objectives.

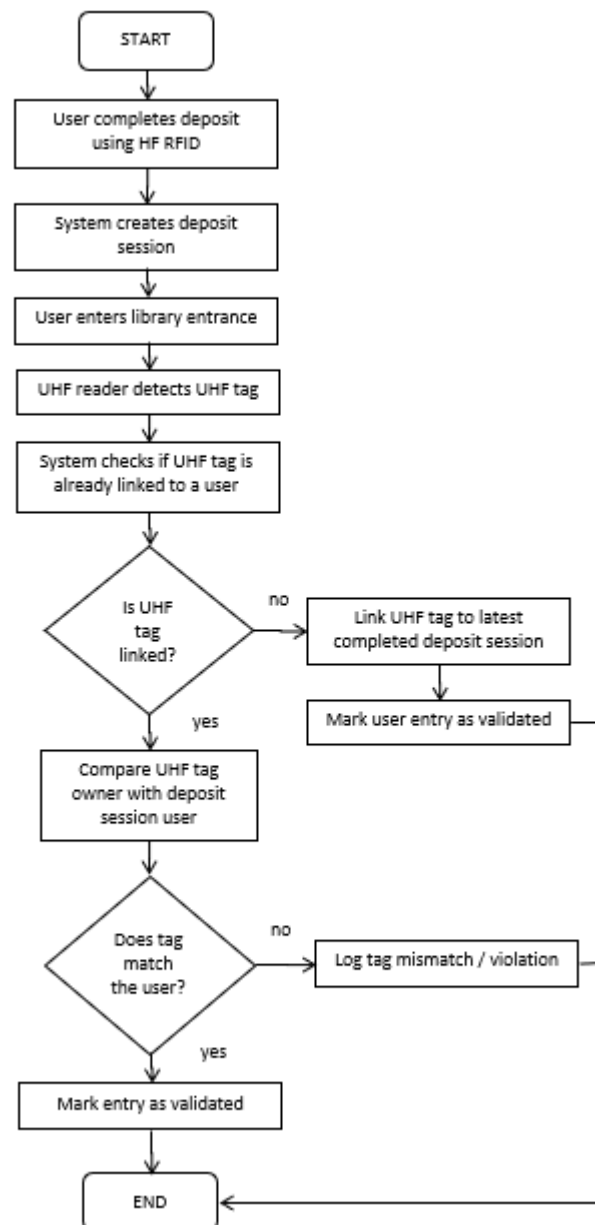


Figure 31. HF-UHF identity matching

As shown in Figure 31, the HF-UHF identity-matching logic links HF RFID at the kiosk with UHF RFID at the entrance to confirm that the student who deposits a bag is the same one who enters the library, with UHF association handled automatically without a separate registration step. Matched tags are marked as valid entries while unknown or mismatched tags are flagged as violations, strengthening monitoring and preventing locker misuse.

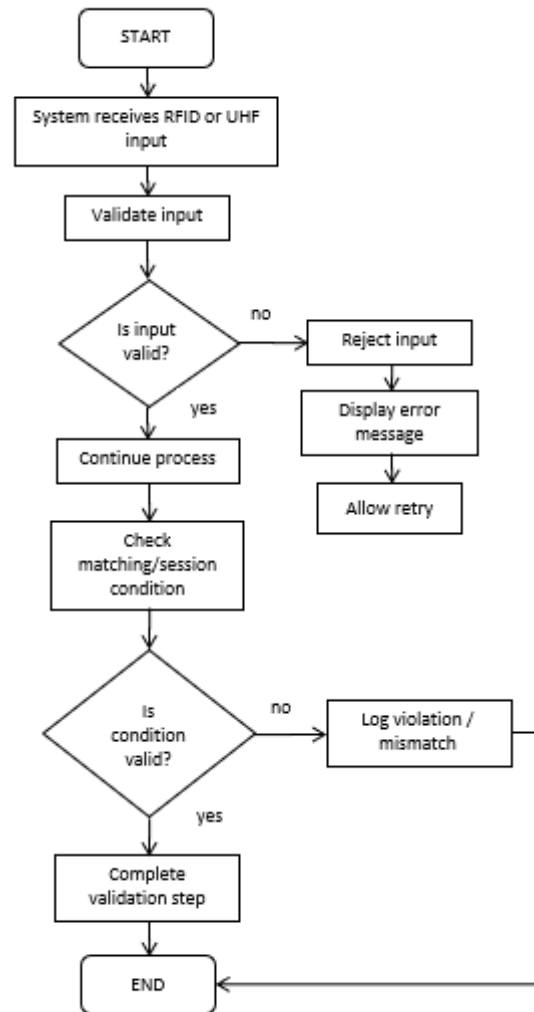


Figure 32. Error handling logic

As shown in Figure 32, the error handling logic ensures system reliability by rejecting invalid RFID scans and unauthorized access attempts while prompting users to correct their actions. Violations and mismatches are logged for admin review, confirming that the mechanism effectively supports both system security and user guidance.

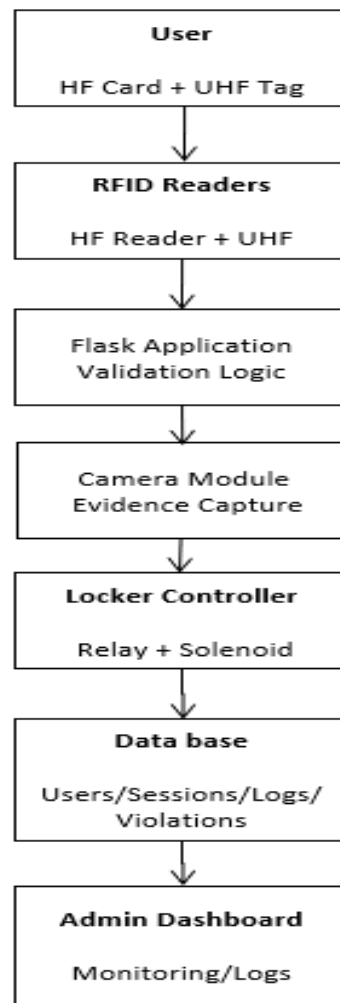


Figure 33. System integration flow

As shown in Figure 33, the system integration flow demonstrates how RFID readers, the Flask application, locker controller, camera, and database work together in sequence during transactions. Validation confirmed correct component interaction and complete data flow, with user details, locker activity, entry status, and evidence photos accurately recorded and displayed on the dashboard, verifying that the architecture effectively supports real-time locker management, RFID authentication, and monitoring.

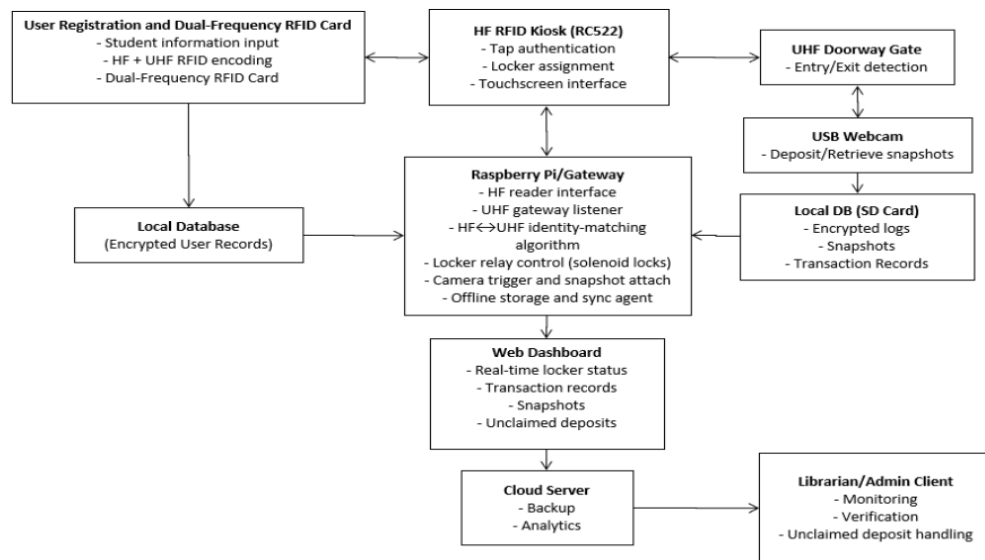


Figure 34. Process block diagram

As shown in Figure 34, the block diagram illustrates how LibLocker's components work together, with the HF RFID kiosk handling authentication and locker assignment, the UHF doorway gate detecting library entry for identity matching, and a USB webcam capturing transaction snapshots. All devices connect to the Raspberry Pi local gateway, which manages RFID reading, locker control, identity matching, and local storage, while the web dashboard provides real-time monitoring and the cloud server handles backup and analytics. Librarians use the dashboard to track transactions, resolve mismatches, and manage unclaimed deposits.

Testing and Evaluation

This section presents the testing and evaluation conducted to assess the LibLocker system's functionality, accuracy, and usability, covering component-level testing, ISO/IEC 25010-based evaluation, and user acceptability measurement through TAM.

Testing

The testing phase focused on evaluating the functionality, accuracy, and

stability of the LibLocker system. HF RFID tap authentication, RFID card issuance verification, and UHF doorway detection were tested for read accuracy, response time, and reliability. Solenoid lock operations were repeatedly tested to confirm secure locking and correct locker release, while the camera module was evaluated for clear snapshot capture during deposit and retrieval.

The HF-UHF identity-matching algorithm was validated using correct matches, mismatches, and unclaimed deposit scenarios to ensure reliable authentication. The dashboard was tested for real-time status updates, accurate transaction logging, and stable offline–online synchronization. A pilot test involving selected students and library staff assessed system usability, authentication flow, and retrieval accuracy under normal library operating conditions.

Evaluation

System evaluation was conducted based on ISO/IEC 25010 to assess LibLocker’s functional suitability, reliability, usability, performance efficiency, and security. On the other hand, librarians and selected students evaluated the user registration process, RFID card usage, kiosk authentication, and dashboard interface for clarity, accuracy, and ease of use. Lastly, user acceptability was measured using TAM, focusing on perceived usefulness, convenience, and overall satisfaction. Evaluation results were used to refine RFID sensitivity, dashboard layout, registration workflow, and HF-UHF identity-matching logic.

Data Analysis

Collected data from system testing, including RFID read rates, locker release accuracy, snapshot clarity, and dashboard update times, were analyzed by computing averages, accuracy percentages, and error frequencies. User evaluation scores based on ISO/IEC 25010 and TAM were summarized to determine overall system quality and acceptability. Results were interpreted to verify whether LibLocker met its performance requirements and to identify areas requiring additional optimization.

Table 1. Five-point Likert scale used to interpret survey results

NUMERICAL RATING	DESCRIPTIVE RATING
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

Table 1 was used to evaluate users' response to the surveys based on ISO/IEC 25010 and TAM distributed through survey forms.

User evaluation data based on ISO/IEC 25010 quality characteristics and TAM were analyzed using weighted mean, standard deviation, and percentage. The weighted mean was computed using the formula:

$$\text{Weighted mean} = \frac{\sum f(x)}{n}$$

From the given formula, f represents the frequency of responses, x represents the scale value, and n represents the total number of respondents.

RESULTS AND DISCUSSION

This chapter presents the results of the testing and evaluation of the LibLocker prototype to determine whether it met the study's objectives in terms of security, accuracy, efficiency, reliability, hardware and software functionality, and user acceptance.

Testing and Evaluation Results

Comprehensive testing confirmed that LibLocker's dual-frequency HF-UHF authentication successfully verified registered users while consistently denying unauthorized access, achieving 100 percent retrieval accuracy. The database accurately stored transaction records, locker assignments, timestamps, and images, maintaining stable offline–online synchronization without data loss. The user registration module correctly recorded student information, rejected duplicate and incomplete entries, and securely encoded RFID cards. Performance testing yielded an average transaction time of 6.8 s, within the 10-s target, with 95 percent system uptime. ISO/IEC 25010- and TAM-based evaluations obtained overall means of 4.63 and 4.63, respectively, both interpreted as "Strongly Agree", confirming high usability, reliability, security, and user satisfaction. These results validate LibLocker as a reliable and efficient automated baggage management solution for the CCAT e-Library.

Design Validation and Documentation

This section presents the validation of the system design and documentation completeness conducted prior to development, confirming that all components and diagrams were logically sound and accurately represented before implementation began.

Table 2. System design validation

COMPONENT/ PROCESS	DESIGN DESCRIPTION	VALIDATION METHOD	SUCCESS INDICATOR	RESULT	REMARKS (Passed/ Failed)
RFID Authentication Sequence	HF tap, then UHF detection, then validation	Diagram walkthrough	Correct sequence flow	Authentication flow worked correctly and sequentially.	Passed
Locker Access Mechanism	Authentication, then unlock, then log	Logic checking	Correct control logic	Locker unlocked only after valid verification.	Passed
Camera- Assisted Verification	Capture, then verify, then store	Flow validation	Correct trigger point	Camera triggered at the correct verification stage	Passed
HF-UHF Identity Matching	Compare HF ID and UHF tag	Logic analysis	Matching rule corrects	HF and UHF identities matched successfully.	Passed
Error Handling Logic	Invalid RFID/ mismatch	Scenario analysis	All cases handled	Invalid inputs were properly rejected.	Passed
System Integration Flow	RFID to camera to locker to database	Workflow tracing	Modules interact correctly	All modules communicated and functioned correctly.	Passed

Table 2 shows that all critical system design components passed validation prior to development. The RFID authentication sequence, locker access mechanism, camera-assisted verification, HF-UHF identity matching, error handling logic, and system integration flow were each verified through diagram walkthroughs and logic analysis. Every component produced the expected behavior, confirming that the system design was logically sound and ready for implementation.

Table 3. Documentation completeness quality

COMPONENT/ PROCESS	DESIGN DESCRIPTION	VALIDATION METHOD	SUCCESS INDICATOR	RESULT	REMARKS (Passed/ Failed)
Block Diagram	System components	Visual inspection	Complete architecture	All system components were properly labeled and connected.	Passed

Table 3. Continued...

COMPONENT/ PROCESS	DESIGN DESCRIPTION	VALIDATION METHOD	SUCCESS INDICATOR	RESULT	REMARKS (Passed/ Failed)
Data Flow Diagram (DFD)	Data movement	Flow verification	Complete data paths	Data flowed correctly between all system modules.	Passed
Process Flowchart	Step-by-step workflow	Sequence checking	Logical process flow	The workflow sequence was complete and logically arranged.	Passed
RFID Authentication Flow	HF-UHF sequence	Content validation	Clear authentication logic	Authentication steps were clearly defined and organized.	Passed
Locker Operation Flow	Access and release logic	Flow validation	Clear mechanism	Locker access and release logic were accurately represented.	Passed
Camera Verification Flow	Capture process	Diagram review	Accurate process	Camera verification process was correctly included.	Passed
Identity Matching Logic	HF-UHF comparison	Diagram analysis	Matching clarity	HF-UHF matching logic was clearly illustrated.	Passed
Block Diagram	System components	Visual inspection	Complete architecture	All system components were properly labeled and connected.	Passed

Table 3 shows that all required documentation was complete and accurate before development commenced. The block diagram, data flow diagram, process flowchart, RFID authentication flow, locker operation flow, camera verification flow, and identity matching logic were all validated through visual inspection, flow verification, and diagram review. Each document was found to correctly represent its corresponding system component, providing a reliable reference throughout the development process.

Hardware and Software Testing

This section presents the results of hardware and software testing conducted on the LibLocker system, evaluating each component's functionality and reliability through repeated trials across key operations including RFID authentication, locker control, database recording, synchronization, camera capture, kiosk interaction, and dashboard monitoring.

Table 4. User verification through the RFID module

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Scan RFID card during deposit	RFID successfully detected user	UID 04A2B3C1 detected; matched User #1 (Jonel)	UID 04D5E6F2 detected; matched User #2 (Jovan)	UID 04F1A6B5 detected; matched User #3 (Gerome)	Passed
Scan RFID card during retrieval	RFID successfully recognized user	UID 04A2B3C1 detected; matched User #1 (Jonel)	UID 04D5E6F2 detected; matched User #2 (Jovan)	UID 04F1A6B5 detected; matched User #3 (Gerome)	Passed

Table 4 shows that the RFID modules successfully detected and identified the correct user in every trial, both during deposit and retrieval operations. Each scan returned the expected UID and matched it to the corresponding registered user without error. This confirms that the HF RFID reader consistently performed accurate card recognition, which is the foundational step for all subsequent locker operations.

Table 5. Lock mechanism

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Locker opening during user deposit	Assigned locker opens after successful authentication	Locker 1 opened during deposit	Locker 4 Opened during deposit	Locker 8 Opened during deposit	Passed

Table 5. Continued....

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Locker opening during user retrieval	Correct locker opens after successful authentication	Locker 1 opened during retrieval	Locker 4 opened during retrieval	Locker 8 opened during retrieval	Passed
Attempt unauthorized unlock with expired session	No locker should open	No relay activated; lockers remained sealed	Access denied; no locker opened	Session timeout enforced; no locker opened	Passed

Table 5 shows that the solenoid-based locking and unlocking mechanism responded correctly to authentication results across all three trials. The assigned locker opened upon successful RFID verification during deposit, and the same locker unlocked upon authenticated retrieval. Critically, when an unauthorized unlock was attempted using an expired session, no relay was activated and all lockers remained sealed. This confirms that the locking mechanism is tightly coupled to the authentication logic, preventing access without valid session credentials.

Table 6. Database

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Save user information	User information saved successfully	User #1 information saved successfully	User #2 information saved successfully	User #3 information saved successfully	Passed
Save locker number and timestamp	Logs recorded correctly	Record saved with timestamp 2026-05-08 09:14:21, locker_no=1, user_id=1, action=deposit	Record saved with timestamp 2026-05-08 09:18:42, locker_no=4, user_id=2, action=deposit	Record saved with timestamp 2026-05-08 09:21:15, locker_no=8, user_id=3, action=deposit	Passed

Table 6 shows that the database correctly stored user information, locker assignments, timestamps, and transaction types across all three trials. Each record was saved with precise timestamps and accurate locker and user identifiers,

confirming that the database layer reliably captured all transactional data generated during system operation. This accurate record-keeping is essential for audit trails and locker status tracking.

Table 7. Synchronization logic

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Make deposit while offline; restore connection	Local log syncs once online	3 offline transactions synced in 2.1 s after reconnection	2 offline transactions synced in 1.8 s after reconnection	4 offline transactions synced in 2.4 s after reconnection	Passed
Real-time SSE update on dashboard	Dashboard updates within 3 s of event	Dashboard updated within 1.2 s of deposit	Dashboard updated within 1.4 s of retrieval	Dashboard updated within 1.1 s of force-unlock	Passed

Table 7 shows that the synchronization logic successfully handled offline queuing and real-time dashboard updates. Offline transactions synced within 1.8 to 2.4 s upon reconnection, and the SSE stream updated the dashboard within 1.1 to 1.4 s per event, both well within the 3-s target, with no data lost during offline periods.

Table 8. Camera snapshot function

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Deposit snapshot	Snapshot captured successfully	Saved as deposit_001.jpg and linked to TX ID 1	Saved as deposit_002.jpg and linked to TX ID 2	Saved as deposit_003.jpg and linked to TX ID 3	Passed
Retrieval snapshot	Snapshot linked correctly	Saved as retrieve_001.jpg; linked to TX ID 1	Saved as retrieve_002.jpg; linked to TX ID 2	Saved as retrieve_003.jpg; linked to TX ID 3	Passed

Table 8 shows that the camera snapshot function captured and correctly linked images to their corresponding transaction IDs in all trials. Deposit and retrieval snapshots were saved with accurate filenames and linked to the appropriate transaction records, confirming that the camera module integrates reliably with the

transaction logging system and provides a visual audit trail for every locker interaction.

Table 9. Kiosk interface

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Deposit snapshot	Snapshot captured successfully	Saved as deposit_001.jpg; linked to TX ID 1	Saved as deposit_002.jpg; linked to TX ID 2	Saved as deposit_003.jpg; linked to TX ID 3	Passed
Retrieval snapshot	Snapshot linked correctly	Saved as retrieve_001.jpg; linked to TX ID 1	Saved as retrieve_002.jpg; linked to TX ID 2	Saved as retrieve_003.jpg; linked to TX ID 3	Passed

Table 9 shows that deposit and retrieval snapshots were correctly captured, named, and linked to their transaction IDs across all three trials, with both procedures passing, confirming reliable camera snapshot operation.

Table 10. Monitoring dashboard

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
View locker status	All 24 lockers displayed with current status	Grid showed Lockers 1, 3, 5 occupied, 21 available	Grid showed Lockers 2, 7, 10, 14 occupied, 20 available	Grid showed Lockers 1, 3, 5, 7, 8 occupied, 19 available	Passed
Access transaction history	Log displays recent transactions in order	Activity log displayed 12 most recent transactions	Activity log displayed 15 most recent transactions	Activity log displayed 18 most recent transactions	Passed

Table 10 shows that the monitoring dashboard accurately reflected real-time locker status and transaction history throughout the test. The locker status grid correctly displayed occupied and available units as activity changed between trials, and the activity log grew incrementally with each new transaction. These results confirm that administrators have an accurate, up-to-date view of system activity, supporting effective oversight of the baggage counter.

Table 11. Student information entry

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Enter student name and number	Student data saved successfully	Jonel Andamon (2022-12345) saved as User #1	Jovan Nadala (2022-12346) saved as User #2	Gerome Voluntad (2022-12347) saved as User #3	Passed

Table 11 shows that the student information entry procedure successfully stored complete and accurate records for all three test users. Each student's name and ID number were correctly saved and assigned a unique user identifier in the system, confirming that the registration form reliably captures and persists new user data.

Table 12. Data validation

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Submit incomplete entry	Incomplete entry rejected	Error: "Name is required"; record not saved	Error: "Student number is required"; record not saved	Error: "Contact information is required"; record not saved	Passed
Submit duplicate student number	Duplicate entry rejected	Error: "Student number 2022-12345 already exists"	Error: "Student number 2022-12346 already exists"	Error: "Student number 2022-12347 already exists"	Passed
Submit valid student record	Valid entry accepted and saved	Valid record saved (04/28/2026 04:08:45 PM)	Valid record saved (04/29/2026 05:22:11 PM)	Valid record saved (04/30/2026 06:37:39 PM)	Passed

Table 12 shows that the data validation mechanism correctly enforced input integrity across all test scenarios. Incomplete entries were rejected with field-specific error messages, duplicate student numbers were flagged and blocked, and valid

records were accepted and stored with accurate timestamps. These validation controls prevent erroneous or conflicting data from entering the database, maintaining the integrity of the user registry throughout the system's operation.

Table 13. HF RFID assignment

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Assign HF RFID to registered student	HF UID linked to user successfully	UID 04A2B3C1 linked to User #1 (Jonel)	UID 04D5E6F2 linked to User #2 (Jovan)	UID 04F1A6B5 linked to User #3 (Gerome)	Passed

Table 13 shows that HF RFID cards were successfully assigned to each registered student, with each UID uniquely linked to the corresponding user profile in the database. This one-to-one mapping ensures that every HF card tap during a kiosk transaction can be resolved to a specific registered user, enabling reliable identity verification at the point of locker access.

Table 14. UHF RFID assignment

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Assign UHF RFID to registered student	UHF EPC linked to user successfully	EPC 0179561230 linked to User #1 (Jonel Andamon)	EPC 0179561226 linked to User #2 (Jovan Nadala)	EPC 0179561262 linked to User #3 (Gerome Voluntad)	Passed

Table 14 shows that UHF RFID tags were similarly linked to each registered student, establishing the second layer of the dual-frequency identity framework. Each tag's EPC was correctly associated with the corresponding user record, allowing the UHF reader at the library entrance to passively identify students without requiring manual interaction.

Table 15. Dual-frequency matching

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Verify HF and UHF IDs	IDs matched correctly	Match confirmed: both IDs retrieved User #1 (Jonel)	Match confirmed: both IDs retrieved User #2 (Jovan)	Match confirmed: both IDs retrieved User #3 (Gerome)	Passed
Repeat identity matching (deposit)	Matching process successful	UID 04A2B3C1 matched with EPC 0179561230 (Locker 1)	UID 04D5E6F2 matched with EPC 0179561226 (Locker 4)	UID 04F1A6B5 matched with EPC 0179561262 (Locker 8)	Passed
Table Repeat identity matching (retrieval)	HF-UHF records matched correctly	UID 04A2B3C1 verified for Locker 1 retrieval	UID 04D5E6F2 verified for Locker 4 retrieval	UID 04F1A6B5 verified for Locker 8 retrieval	Passed

Table 15 shows that the HF-UHF identity matching logic correctly paired and verified identifiers for all three users during both deposit and retrieval operations. The system confirmed that both the HF UID and UHF EPC resolved to the same registered user in every trial, ensuring that only the student who initiated the deposit could complete the retrieval. This dual-frequency linkage forms the core security mechanism of LibLocker, preventing locker access by any individual whose credentials do not match across both frequencies.

Table 16. Secured record storage

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Save student data	Student records saved correctly	Record saved (04/28/2026 08:14:22 AM)	Record saved (04/28/2026 09:32:41 AM)	Record saved (04/28/2026 10:48:09 AM)	Passed

Table 16. Continued...

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Save RFID identifiers	RFID identifiers saved successfully	UID stored (04/28/2026 11:06:17 AM)	UID stored (04/29/2026 01:21:53 PM)	UID stored (04/30/2026 03:37:28 PM)	Passed
Retrieve saved records	Stored data retrieved correctly	Retrieved (04/28/2026 04:09:35 PM)	Retrieved (04/29/2026 05:26:14 PM)	Retrieved (04/30/2026 06:41:57 PM)	Passed

Table 16 shows that all student data and RFID identifiers were saved to the database with accurate timestamps across all three trials. Records were retrievable upon request and returned the correct data, confirming that the storage module maintains data integrity and provides dependable access for subsequent system operations.

Table 17. RFID card encoding

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Encode RFID card	RFID encoded successfully	RFID encoded successfully	RFID encoded successfully	RFID encoded successfully	Passed
Scan encoded card	Encoded data recognized	UID recognized successfully upon scan	UID recognized successfully upon scan	UID recognized successfully upon scan	Passed

Table 17 shows that the RFID card encoding process was completed successfully in every trial. Cards were encoded without error, and the encoded UIDs were recognized correctly upon re-scanning, indicating that the encoding routine reliably writes and verifies data on the RFID card hardware.

Table 18. RFID card linking

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Scan RFID card	User identified successfully	User #1 identified via UID scan	User #2 identified via UID scan	User #3 identified via UID scan	Passed
Verify assigned user profile	Correct user profile displayed	Correct profile displayed – User #1	Correct profile displayed – User #2	Correct profile displayed – User #3	Passed

Table 18 shows that each encoded RFID card was correctly linked to its assigned user profile. Upon scanning, the system immediately identified the correct user and displayed the corresponding profile, confirming that the card-to-user association established during registration persists accurately and supports consistent identity resolution throughout the locker workflow.

Table 19. Reusability of RFID cards

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Deactivate RFID card	RFID deactivated successfully	UID 04A2B3C1 deactivated (04/28/2026 08:33:12 AM)	UID 04D5E6F2 deactivated (04/29/2026 09:48:37 AM)	UID 04F1A6B5 deactivated (04/30/2026 11:02:19 AM)	Passed
Reassign RFID to new user	RFID reassigned correctly; no link to old user	UID 04A2B3C1 reassigned to User #11; no longer linked to User #1	UID 04D5E6F2 reassigned to User #12; no longer linked to User #2	UID 04F1A6B5 reassigned to User #13; no longer linked to User #3	Passed

Table 19 shows that RFID cards were successfully deactivated and reassigned to new users without retaining any link to the previous holder, confirming that the

system supports a full card lifecycle and allows a fixed card pool to be reused across student batches without security risk.

Table 20. Data overwriting

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Re-encode used RFID card	Old data overwritten; new info saved	Database row updated; old reference removed	Database row updated; old reference removed	Database row updated; old reference removed	Passed
Scan updated RFID card	Updated information recognized	Updated RFID data recognized	Updated RFID data recognized	Updated RFID data recognized	Passed

Table 20 shows that re-encoding a previously used RFID card correctly overwrote the old data and replaced it with updated user information. Upon scanning, the card returned only the newly assigned user's data, with no trace of prior records. This data overwriting capability ensures that card reuse does not compromise identity accuracy and that the system remains reliable even as card assignments change over time.

Table 21. Unauthorized access detection

PROCEDURE	EXPECTED RESULT	ACTUAL RESULTS			REMARKS (Passed/ Failed)
		TRIAL 1	TRIAL 2	TRIAL 3	
Scan unregistered RFID card	Unregistered RFID rejected	Access denied	Access denied	Access denied	Passed
Scan invalid RFID card	Invalid RFID denied access	Invalid tag rejected	Invalid tag rejected	Invalid tag rejected	Passed
Attempt unauthorized retrieval	rejected unauthorized transaction	Unauthorized access blocked	Unauthorized access blocked	Unauthorized access blocked	Passed

Table 21 shows that the system blocked all three types of unauthorized access

during testing: unregistered cards, invalid cards, and retrieval attempt without a valid session. Layered access control of this kind is what keeps random or expired cards from being used to open lockers.

Performance and Security Testing

This section presents the results of performance and security testing conducted to evaluate the LibLocker system's encoding accuracy, RFID pairing reliability, locker release precision, transaction speed, and overall system uptime under controlled test conditions.

¹
Table 22. Encoding accuracy

TRIAL NO.	TAG UID	USER NO. LINKED	VERIFICATION RESULT	REMARKS (Passed/Failed)
1	04A2B3C1	1	Tag re-read retrieves correct user	Passed
2	04D5E6F2	2	Tag re-read retrieves correct user	Passed
3	04F1A6B5	3	Tag re-read retrieves correct user	Passed
4	04C3E8B9	4	Tag re-read retrieves correct user	Passed
5	04B2F1A8	5	Tag re-read retrieves correct user	Passed
6	04E9C2A1	6	Tag re-read retrieves correct user	Passed
7	04A1B8E5	7	Tag re-read retrieves correct user	Passed
8	04F8C3A2	8	Tag re-read retrieves correct user	Passed

Table 22. Continued...

TRIAL NO.	TAG UID	USER NO. LINKED	VERIFICATION RESULT	REMARKS (Passed/Failed)
9	04D2E5B7	9	Tag re-read retrieves correct user	Passed
10	0489B2C4	10	Tag re-read retrieves correct user	Passed

Table 22 shows that across the 10 encoding trials, every tag returned the correct user upon re-scanning. The 100 percent encoding accuracy exceeded the 95 percent target, which suggests that the encoding process was consistent enough that students would not need to come back for re-encoding under normal use.

Table 23. RFID pairing and authentication accuracy

TRIAL NO.	HF RFID	UHF RFID	EXPECTED RESULT	ACTUAL RESULT	REMARKS (Passed/Failed)
1	04A2B3C1	0179561230	Valid match	Valid match	Passed
2	04D5E6F2	0179561226	Valid match	Valid match	Passed
3	04F1A6B5	0197561262	Valid match	Valid match	Passed
4	04A2B3C1	0197561226	Invalid match	Rejected	Passed
5	04C3E8B9	0197561255	Valid match	Valid match	Passed

Table 23 shows that HF-UHF identity matched correctly, accepting all valid pairings and rejecting the deliberately mismatched pair in Trial 4, where a UHF tag did not correspond to the HF card used for deposit. This is the key scenario the dual-frequency check was designed to catch. Across all five trials, there were zero unauthorized accesses, consistent with the design intent.

Table 24. Locker unlocking mechanism for retrieval transaction

TRIAL NO.	USER NO.	LOCKER NO.	EXPECTED RESULT	ACTUAL RESULT	REMARKS (Passed/Failed)
1	1	1	Correct release	Correct release	Passed

Table 24. Continued...

TRIAL NO.	USER NO.	LOCKER NO.	EXPECTED RESULT	ACTUAL RESULT	REMARKS (Passed/Failed)
2	2	2	Correct release	Correct release	Passed
3	3	3	Correct release	Correct release	Passed
4	4	4	Correct release	Correct release	Passed
5	5	5	Correct release	Correct release	Passed

Table 24 shows that the locker unlocking mechanism for the retrieval process functioned properly as correct locker was activated for each user in all five trials. Every retrieval transaction matched the expected locker to the authenticated user, achieving 100 percent retrieval accuracy. This confirms that the system's locker assignment and authentication logic are tightly integrated, ensuring that students retrieve only their own designated locker without error.

Table 25. Transaction time and measurement

TRIAL NO.	ACTION	DURATION (s)	REMARKS (Passed/Failed)
1	Deposit	6.00	Passed
2	Retrieval	7.00	Passed
3	Deposit	7.00	Passed
4	Retrieval	6.00	Passed
5	Deposit	8.00	Passed
AVERAGE		6.80	PASSED

Table 25 shows that all transactions were completed faster than the 10-s target, averaging 6.8 s across five trials. The 6 to 8 s range reflects minor variations in user interaction speed, all within the acceptable limit. Per Nielsen's System Response Time guidelines, responses under 10 s sustain user attention and smooth interaction flow, confirming that the system can prevent queue build-up and efficiently serve multiple users during peak hours.

Table 26. System reliability

METRIC	OBSERVATION	REMARKS
System Uptime	9.5 h/10 h	95 % uptime
RFID Detection	Stable	Passed
Camera Capture	Stable	Passed
Data Sync	Stable	Passed

Table 26 reports 95 percent uptime across the 10-h test window, with RFID detection, camera capture, and data synchronization remaining stable throughout. The brief downtime was caused by a Wi-Fi reconnection event, which the offline-online sync logic handled automatically. For a prototype, 95 percent sustained uptime indicates that the system can run continuously without constant supervision under the tested conditions.

User Acceptability Evaluation

This section presents the user acceptability evaluation results based on ISO/IEC 25010 and the Technology Acceptance Model (TAM), measuring respondents' assessment of the system's quality, usability, and overall acceptability.

Table 27. Summary of evaluation based on ISO/IEC 25010

CRITERIA	MEAN	INTERPRETATION
Functional Suitability	4.68	Strongly Agree
Performance Efficiency	4.64	Strongly Agree
Reliability	4.51	Strongly Agree
Security	4.64	Strongly Agree
Usability	4.68	Strongly Agree
Compatibility	4.62	Strongly Agree
OVERALL MEAN	4.63	STRONGLY AGREE

Table 27 presents the ISO/IEC 25010-based evaluation results, with an overall mean of 4.63, interpreted as “Strongly Agree”. Functional Suitability and Usability

received the highest scores of 4.68, indicating that the respondents found the system fully capable of meeting its intended purpose and easy to interact with. Reliability received the lowest score at 4.51, which is consistent with the brief network-related downtime recorded during performance testing. Performance Efficiency and Security both scored 4.64, reflecting strong confidence in the system's speed and access control measures. Compatibility scored 4.62, confirming that respondents found the system compatible with the existing library environment. Overall, the evaluation based on ISO/IEC 25010 validate LibLocker as a high-quality system across all six evaluated quality dimensions.

Table 28. Summary of evaluation based on TAM

CRITERIA	MEAN	INTERPRETATION
Perceived Usefulness	4.65	Strongly Agree
Perceived Ease of Use	4.63	Strongly Agree
Behavioral Intention	4.61	Strongly Agree
User Satisfaction	4.62	Strongly Agree
OVERALL MEAN	4.63	STRONGLY AGREE

Table 28 presents the TAM-based evaluation results, also yielding an overall mean of 4.63 interpreted as "Strongly Agree". Perceived Usefulness received the highest score at 4.65, indicating that respondents clearly recognized LibLocker's value in improving upon the manual baggage handling process. Perceived Ease of Use scored 4.63, reflecting that users found the system intuitive and accessible without extensive training. Behavioral Intention and User Satisfaction scored 4.61 and 4.62, respectively, demonstrating that respondents would willingly adopt the system and were satisfied with its performance during the evaluation period. Taken together, the TAM-based evaluation results confirm strong user acceptance and a high likelihood of successful adoption if LibLocker were permanently deployed.

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the summary of the study, the conclusions drawn based on the results of the testing and evaluation, and the recommendations formulated for future improvement of the automated LibLocker system.

Summary

This study focused on the design and development of LibLocker, an RFID-based smart locker management system intended to improve the security, efficiency, and accessibility of locker usage in academic environments. The system integrated dual-frequency RFID technology, a Raspberry Pi microcontroller, a software-based kiosk interface, and an administrative dashboard to automate the locker management process. The development of the system involved hardware integration, software development, and system testing to ensure that all components functioned properly. The hardware components included HF and UHF RFID readers, a Raspberry Pi 4 Model B controller, MCP23017 I/O expanders, solenoid locks, magnetic reed switches, a USB webcam, and a capacitive touchscreen monitor. The software components consisted of the user kiosk interface, administrative dashboard, secured database, and HF-UHF identity-matching algorithm.

Several testing procedures were conducted to evaluate the performance and reliability of the system. These included hardware communication testing, kiosk and dashboard functionality testing, RFID authentication testing, user registration validation, data logging verification, and transaction performance testing. The results showed that the system was able to accurately identify authorized users, open the correct locker, and record transaction data in the database with 100 percent accuracy across all tested scenarios. User evaluation based on ISO/IEC 25010 and TAM both yielded an overall mean of 4.63, both interpreted as "Strongly Agree". These results

indicate that the LibLocker system provides a reliable, secure, and efficient alternative to traditional manual locker management methods.

Conclusions

Based on the results and findings of the study, different conclusions were made. First, the LibLocker system successfully integrated dual-frequency RFID technology with an automated locker management system, providing secure and automated access while addressing the limitations of traditional manual baggage handling. Another conclusion is that the hardware and software components functioned cohesively, enabling accurate RFID authentication, proper locker operation, and seamless coordination among system modules. Additionally, the system reliably recorded and managed locker transactions through its secured database and monitoring dashboard, ensuring data accuracy, traceability, and real-time visibility of system activity. Likewise, the system demonstrated efficient transaction processing, with deposit and retrieval operations completed within an average of 6.8 s, well within the 10-s target. This contributed to reduced waiting time and improved user experience. The system also exhibited stable operation throughout testing, with consistent RFID detection, successful offline–online data synchronization, and reliable camera snapshot capture, indicating strong system reliability. Lastly, the evaluation results based on ISO/IEC 25010 and TAM revealed that the users found the system was easy to use, convenient, and effective, reflecting a high level of user acceptability. Overall, the LibLocker system achieved the objectives of the study and demonstrated its potential as an efficient, secure, and reliable locker management solution suitable for implementation in academic environments.

Recommendations

Based on the findings and limitations of the study, the following recommendations are proposed:

1. Enhance system scalability and integration with centralized campus databases to improve data management and coordination among departments.
2. Improve hardware durability and optimize system performance to ensure reliability and efficiency for long-term operation.
3. Develop a mobile or web notification feature that can inform users and administrators about locker transactions, unclaimed baggage, and system activity in real time.
4. Continuously refine the user interface and overall user experience to make the system more accessible to all users, including students with limited mobility or visual impairments
5. Conduct further testing and evaluation in larger environments and with more respondents to assess the system's effectiveness, reliability, and adaptability under actual operational conditions.
6. Implement the LibLocker system permanently in the e-library to provide a more organized, secure, and efficient baggage management process compared to the traditional method.

REFERENCES

- Aguila, J. I., & Balba, N. (2025). SmartLibrary: Comprehensive RFID-based library system for attendance tracking and inventory management. *Asia Pacific Journal of Management and Sustainable Development*, 13(1), 127–132. doi:10.70979/mzac6221
- Arfan, M. (2024). Designing seamless user interactions in RFID-based smart libraries. New York, NY: Springer.
- Bharatiraja, C., Sridhar, R., Vidhya, P., & Sundaram, K. (2023). IoT-based centralized smart locker system with RFID and cloud monitoring. *International Journal of Advanced Computer Science and Applications*, 14(7), 112–120.
- Bhat, R., Manohar, S., & Acharya, B. (2025). Secure dormitory locker access using RFID and IoT. *Journal of Information Security and Applications*, 60, 102115.
- Bhat, S. A., Ananya, K., Chaithra, B., & Upadhyaya, A. (2025). RFID-based locker security, student and visitor monitoring system in hostel. In *Advances in Engineering Research* (pp. 136–148). Paris, France: Atlantis Press. doi:10.2991/978-94-6463-762-5_13
- Costa, F., Genovesi, S., Borgese, M., Michel, A., Dicandia, F. A., & Manara, G. (2021). A review of RFID sensors: *The new frontier of Internet of Things. Sensors*, 21(9), 3138. doi:10.3390/s21093138
- De La Salle University Manila. (2018). Book monitoring system with real-time inventory using RFID technology (2018). Retrieved from departments/electronics-computer-engineering/thesis-archive/book-monitoring-system-with-real-time-inventory-using-rfid-technology-2018/
- Duroc, Y. (2022). From identification to sensing: RFID is one of the key technologies in the IoT field. *Sensors*, 22(19), 7523. doi:10.3390/s22197523
- Dwivedi, Y. K., Kapoor, K. K., Williams, M. D., & Williams, J. (2012). RFID systems in libraries: An empirical examination of factors affecting system use and user satisfaction. *International Journal of Information Management*, 33(2), 367–377. doi:10.1016/j.ijinfomgt.2012.10.008
- Dwivedi, Y. K., Singh, M., & Tiwari, S. (2012). Analyzing the impact of RFID on library circulation and inventory management. *Library Hi Tech*, 30(2), 259–274.
- International Organization for Standardization. (2011). ISO/IEC 25010:2011-Systems and software engineering-Systems and software Quality Requirements and Evaluation (SQuaRE)-System and software quality models. ISO.
- IJRASET. (n.d.). RFID based automated library management system. Retrieved from <https://www.ijraset.com/research-paper/rfid-based-automated-library-management-system>
- Jain, A., Kumar, R., & Mehta, S. (2024). The role of RFID in digital libraries: Smart automation and services. *International Journal for Multidisciplinary Research*. Retrieved from <https://www.ijfmr.com>

- Liu, Y., Wang, H., Chen, X., & Zhang, L. (2021). Enhancing human-object interaction recognition via RFID and vision sensor fusion. *IEEE Transactions on Multimedia*, 23, 17234–17245.
- Munoz-Ausecha, C., Ruiz-Rosero, J., & Ramirez-Gonzalez, G. (2021). RFID applications and security: A review. *Computation*, 9(6), 69. doi:10.3390/computation9060069
- Nielsen, J. (1993). Usability engineering. Morgan Kaufmann Publishers. (See also: Nielsen, J. (1993). Response Time Limits. Nielsen Norman Group. <https://www.nngroup.com/articles/response-times-3-important-limits/>)
- Nirudi, Y. (2025). Evolution of user satisfaction with RFID in libraries: Case study of NITHM Library. SSRN Electronic Journal. Retrieved from <https://papers.ssrn.com>
- Roberts, C. M. (2006). Radio frequency identification (RFID). *Computers & Security*, 25(1), 18–26. <https://doi.org/10.1016/j.cose.2005.12.003>
- Sureshkumar, T., Abul Hussain, A., & Radha Jeyalakshmi, A. (2025). Enhancing library efficiency and user experience through RFID and kiosk integration. *International Journal of Engineering Research & Technology*. Retrieved from <https://www.ijert.org>
- Vedanth, V. S., Rohan, M. S., Sai, S. V., Ramaswamy, T., & Sreenivasulu, Y. (2023). RFID-based automated library management system. *International Journal for Research in Applied Science and Engineering Technology*, 11(1), 885–890. doi:10.22214/ijraset.2023.48607
- Verma, R., & Garg, P. (2020). RFID technology in library applications. *International Journal of Computing and Technology*, 7(7). Retrieved from <https://www.ijcat.org>
- Wang, X., Wang, X., Yan, Y., Liu, J., & Zhao, Z. (2022). RF access: Barrier-free access control systems with UHF RFID. *Applied Sciences*, 12(22), 11592.
- Want, R. (2006). An introduction to RFID technology. *IEEE Pervasive Computing*, 5(1), 25–33. <https://doi.org/10.1109/MPRV.2006.2>
- Yusuf, M., Akintunde, A., Habeeb, S., & Quadri, A. (2023). Radio frequency identification (RFID) based library management system. *International Journal of Technology and Systems*, 8(2). Retrieved from <https://iprjb.org>

APPENDICES

APPENDIX TABLES

Appendix Table 1. Summary of responses for evaluation based on ISO/IEC 25010 in terms of functional suitability and performance efficiency

RESPONDENT NO.	FUNCTIONAL SUITABILITY					PERFORMANCE EFFICIENCY				
	FS1	FS2	FS3	FS4	MEAN	PE1	PE2	PE3	PE4	MEAN
1	5	5	5	5	5	5	5	5	5	5
2	5	5	5	5	5	4	4	4	5	4.25
3	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	4	5	5	4.75
6	4	4	4	5	4.25	5	4	4	5	4.5
7	4	4	4	4	4	4	5	4	4	4.25
8	5	5	5	5	5	5	5	5	5	5
9	5	5	5	5	5	5	5	5	5	5
10	5	5	5	5	5	5	5	5	5	5
11	5	5	5	5	5	5	5	5	4	4.75
12	5	5	5	5	5	5	5	5	5	5
13	5	5	5	5	5	5	5	5	5	5
14	5	5	5	5	5	5	5	5	5	5
15	5	5	5	5	5	5	5	5	5	5
16	4	3	5	4	4	5	5	5	5	5
17	5	5	5	5	5	4	4	4	5	4.25
18	5	5	5	5	5	5	5	4	5	4.75
19	4	3	5	4	4	5	4	4	4	4.25
20	3	4	4	4	3.75	4	4	4	4	4
21	5	5	4	5	4.75	4	5	5	4	4.5
22	5	5	5	5	5	5	5	5	4	4.75
23	5	5	5	5	5	5	5	5	5	5
24	5	5	5	5	5	5	5	5	5	5
25	4	4	4	3	3.75	4	3	3	4	3.5
26	4	4	4	4	4	4	4	4	4	4
27	5	5	5	5	5	5	5	5	5	5
28	5	5	5	5	5	5	5	5	5	5
29	5	5	5	5	5	5	5	5	5	5
30	4	4	4	4	4	4	4	4	3	3.75
GRAND MEAN	4.67	4.63	4.73	4.7	4.68	4.7	4.63	4.6	4.63	4.64

Appendix Table 2. Summary of responses for evaluation based on ISO/IEC 25010 in terms of reliability and security

RESPONDENT NO.	RELIABILITY					SECURITY				
	R1	R2	R3	R4	MEAN	S1	S2	S3	S4	MEAN
1	4	4	5	5	4.5	5	5	5	5	5
2	4	5	4	5	4.5	4	4	4	5	4.25
3	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	4	4
5	5	4	5	5	4.75	5	5	4	5	4.75
6	4	4	5	5	4.5	4	3	4	3	3.5
7	4	4	4	4	4	4	4	4	4	4
8	5	5	5	5	5	5	5	5	5	5
9	5	5	5	5	5	5	5	5	5	5
10	4	4	4	4	4	5	5	5	5	5
11	5	4	4	4	4.25	5	5	5	5	5
12	5	5	5	5	5	5	5	5	5	5
13	5	5	5	5	5	5	5	5	5	5
14	5	5	5	5	5	5	5	5	5	5
15	5	5	5	5	5	5	5	5	5	5
16	5	4	5	5	4.75	4	5	4	5	4.5
17	5	5	4	4	4.5	5	5	5	4	4.75
18	5	5	5	5	5	5	5	5	5	5
19	4	4	4	4	4	5	5	4	5	4.75
20	4	4	4	4	4	4	4	4	4	4
21	5	4	4	4	4.25	4	4	4	5	4.25
22	4	4	5	4	4.25	5	4	5	4	4.5
23	5	5	5	5	5	5	5	5	5	5
24	4	5	5	4	4.5	5	5	5	5	5
25	4	3	4	3	3.5	4	4	5	4	4.25
26	4	4	4	4	4	4	4	4	4	4
27	5	4	5	5	4.75	5	5	5	4	4.75
28	5	5	5	5	5	5	5	5	5	5
29	5	5	5	5	5	5	5	5	5	5
30	4	3	3	3	3.25	4	4	4	4	4
GRAND MEAN	4.57	4.4	4.6	4.57	4.51	4.67	4.63	4.63	4.63	4.64

Appendix Table 3. Summary of responses for evaluation based on ISO/IEC 25010 in terms of usability and compatibility

RESPONDENT NO.	USABILITY					COMPATIBILITY				
	U1	U2	U3	U4	MEAN	C1	C2	C3	C4	MEAN
1	5	5	4	5	4.75	4	5	5	5	4.75
2	4	5	5	5	4.75	4	5	4	5	4.5
3	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	4	4.75	5	5	5	4	4.75
6	5	5	5	4	4.75	4	4	5	5	4.5
7	4	4	4	4	4	5	4	4	4	4.25
8	5	5	5	5	5	5	5	5	5	5
9	5	5	5	5	5	5	5	5	5	5
10	5	5	5	5	5	5	4	4	4	4.25
11	5	5	5	5	5	5	5	4	4	4.5
12	5	5	5	5	5	5	5	5	5	5
13	5	5	5	5	5	5	5	5	5	5
14	5	5	5	5	5	5	5	5	5	5
15	5	5	5	5	5	5	5	5	5	5
16	5	4	4	5	4.5	5	4	4	4	4.25
17	5	4	4	5	4.5	5	5	4	5	4.75
18	5	5	5	5	5	5	5	5	5	5
19	5	5	5	5	5	4	4	5	4	4.25
20	4	4	4	4	4	4	4	4	4	4
21	4	4	3	5	4	5	5	5	5	5
22	5	5	4	4	4.5	5	5	5	5	5
23	5	5	5	5	5	5	4	5	5	4.75
24	5	5	5	5	5	5	5	5	5	5
25	4	4	4	5	4.25	4	4	4	4	4
26	4	4	4	4	4	4	4	4	4	4
27	5	4	5	4	4.5	4	5	4	5	4.5
28	5	5	5	5	5	5	5	5	5	5
29	5	5	5	5	5	5	5	5	5	5
30	4	4	4	5	4.25	4	4	3	3	3.5
GRAND MEAN	4.73	4.67	4.6	4.73	4.68	4.67	4.63	4.57	4.6	4.62

Appendix Table 4. Summary of responses for evaluation based on TAM in terms of perceived usefulness and perceived ease of use

RESPONDENT NO.	PERCEIVED USEFULNESS					PERCEIVED EASE OF USE				
	PU1	PU2	PU3	PU4	MEAN	PE1	PE2	PE3	PE4	MEAN
1	5	5	5	4	4.75	5	4	5	4	4.5
2	5	4	4	4	4.25	4	4	5	5	4.5
3	5	5	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4	4	4
5	5	4	4	5	4.5	5	5	5	5	5
6	4	5	5	4	4.5	4	4	4	5	4.25
7	4	4	4	4	4	4	4	4	4	4
8	5	5	5	5	5	5	5	5	5	5
9	5	5	5	5	5	5	5	5	5	5
10	5	5	5	5	5	5	5	4	5	4.75
11	5	5	5	4	4.75	5	5	4	5	4.75
12	5	5	5	5	5	5	5	5	5	5
13	5	5	5	4	4.75	5	5	5	5	5
14	5	5	5	5	5	5	5	5	5	5
15	5	5	5	5	5	5	5	5	5	5
16	5	4	5	4	4.5	5	5	5	5	5
17	5	5	4	5	4.75	4	5	4	4	4.25
18	5	5	5	5	5	5	5	5	5	5
19	4	5	4	5	4.5	4	5	4	5	4.5
20	4	4	4	4	4	5	4	4	4	4.25
21	5	5	5	5	5	5	5	4	4	4.5
22	5	5	5	4	4.75	5	4	5	4	4.5
23	5	5	5	5	5	5	5	5	5	5
24	5	5	5	5	5	5	5	5	5	5
25	4	4	4	3	3.75	4	4	3	4	3.75
26	4	4	4	4	4	4	4	4	4	4
27	5	5	5	5	5	4	4	5	5	4.5
28	5	5	5	5	5	5	5	5	5	5
29	5	5	5	5	5	5	5	5	5	5
30	4	4	4	3	3.75	4	4	4	4	4
GRAND MEAN	4.73	4.7	4.67	4.5	4.65	4.67	4.63	4.57	4.67	4.63

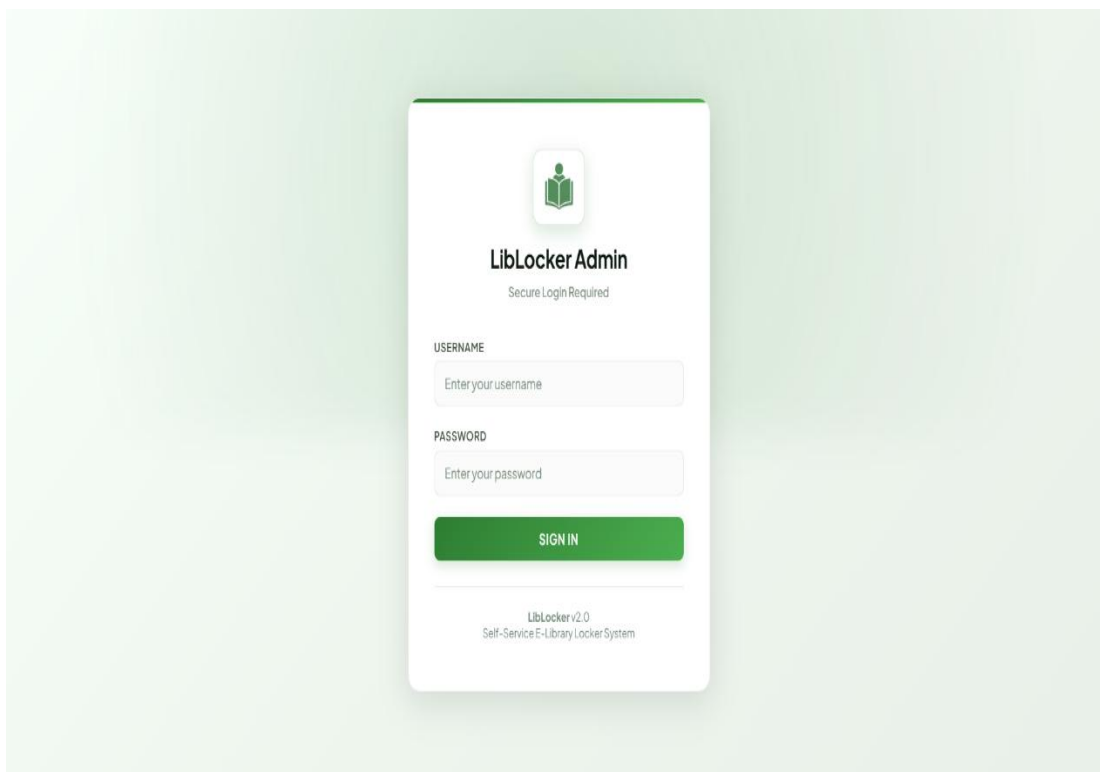
Appendix Table 5. Summary of responses for evaluation based on TAM in terms of behavioral intention and user satisfaction

RESPONDENT NO.	BEHAVIORAL INTENTION				USER SATISFACTION			
	BI1	BI2	BI3	MEAN	US1	US2	US3	MEAN
1	5	5	5	5	5	5	5	5
2	4	4	4	4	5	4	4	4.33
3	5	5	5	5	5	5	5	5
4	4	4	4	4	4	4	4	4
5	4	4	4	4	5	5	5	5
6	4	4	4	4	4	4	4	4
7	4	4	4	4	4	4	4	4
8	5	5	5	5	5	5	5	5
9	5	5	5	5	5	5	5	5
10	5	5	4	4.67	5	5	4	4.67
11	5	5	4	4.67	5	5	5	5
12	5	5	5	5	5	5	5	5
13	5	5	5	5	5	5	5	5
14	5	5	5	5	5	5	3	4.33
15	5	5	5	5	5	5	5	5
16	5	5	5	5	5	5	5	5
17	5	5	5	5	5	4	5	4.67
18	5	5	5	5	5	5	5	5
19	5	5	4	4.67	4	4	4	4
20	4	4	4	4	4	4	4	4
21	5	4	5	4.67	4	4	5	4.33
22	3	5	5	4.33	4	4	4	4
23	5	5	5	5	5	5	5	5
24	5	5	5	5	5	5	5	5
25	3	4	4	3.67	4	4	4	4
26	4	4	4	4	4	4	4	4
27	5	5	4	4.67	5	5	5	5
28	5	5	5	5	5	5	5	5
29	5	5	5	5	5	5	5	5
30	4	4	4	4	4	4	5	4.33
GRAND MEAN	4.6	4.67	4.57	4.61	4.67	4.6	4.6	4.62

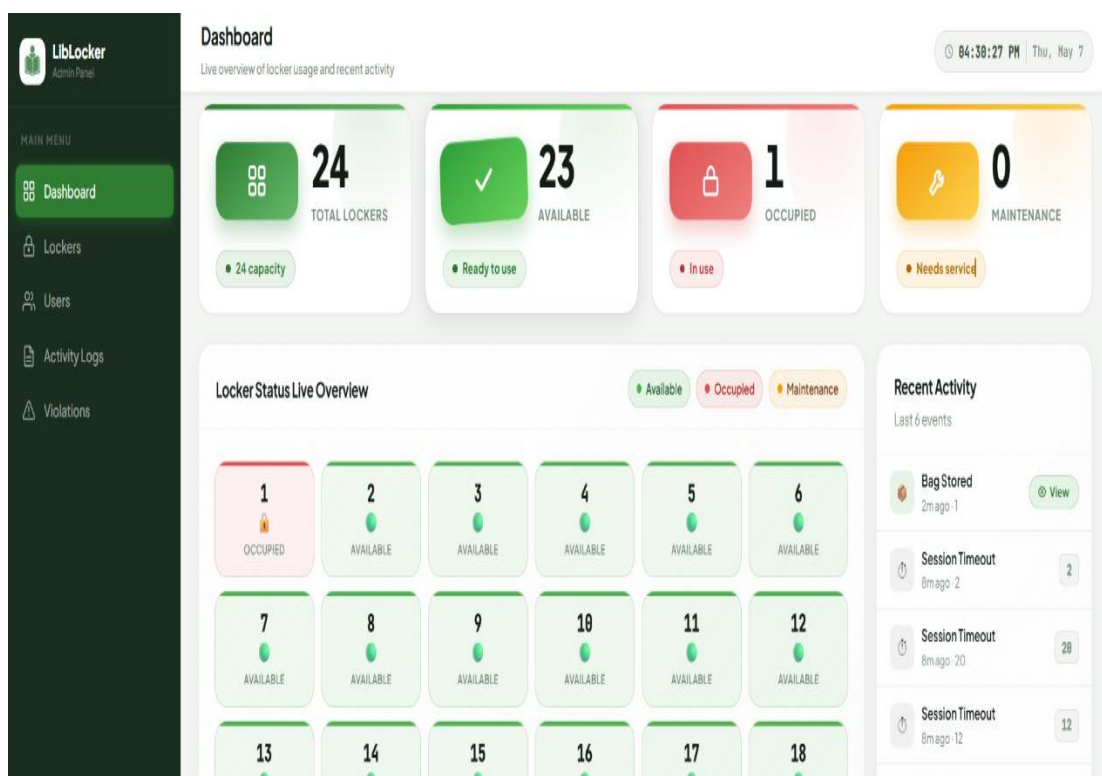
Appendix Table 6. Cost analysis

QTY	UNIT	DESCRIPTION	UNIT PRICE (P)	TOTAL PRICE (P)
1	pc	Raspberry Pi 4 Model B	4,000	4,000
1	pc	HF RFID Reader Module	300	300
1	pc	UHF 915 MHz RFID Reader	7,000	7,000
1	pc	5V 3A Power Supply	500	500
1	pc	Micro SD Card (64GB)	350	350
24	pcs	Solenoid Lock (12V)	300	7,200
24	pcs	Magnetic Reed Switch	60	1,440
3	pcs	8-Channel Relay Module	300	900
3	pcs	I/O Expander Chip	300	900
1	pc	Industrial Power Supply	1000	1,000
1	pc	10.1" Capacitive Touchscreen Monitor	5,500	5,500
1	pc	USB Webcam	600	600
1	pc	HDMI Cable, Wiring & Adapters	500	500
30	pcs	Reusable Dual-Frequency cards	25	750
2	pcs	Locker	7,000	14,000
1	pc	wooden Kiosk Casing	4,000	4,000
TOTAL				48,940

APPENDIX FIGURES



Appendix Figure 1. Admin login interface



Appendix Figure 2. Admin main interface

LibLocker
Admin Panel

MAIN MENU

- Dashboard
- Lockers
- Users**
- Activity Logs
- Violations

Logout

Online
System Status

Registered Users

All students with a registered RFID card

04:33:17 PM Thu, May 7

Registered Students

All users who have registered their RFID card

11 users

Search by name, student ID, RFID or UHF...

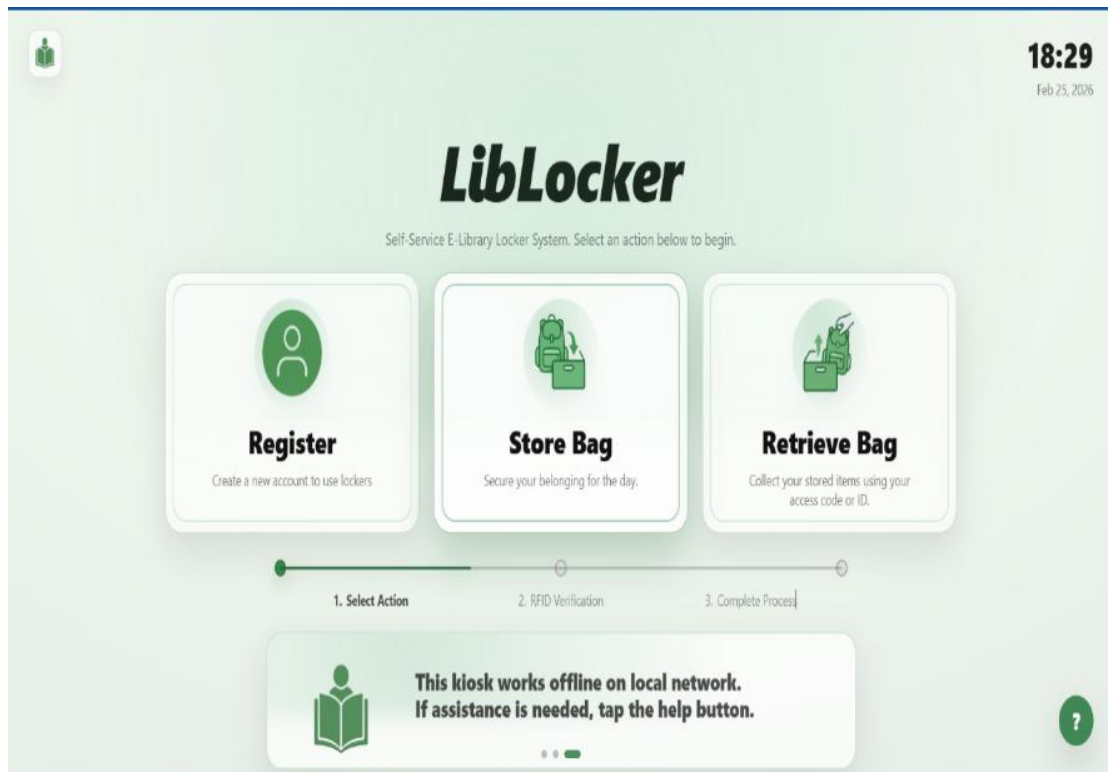
Refresh

#	FULL NAME	STUDENT ID	RFID CARD	UHF TAG	REGISTERED	ACTIVE LOCKER	STATUS	ACTIONS
1	SEBASTIANTAGALOG	202210711	007308E39AA	—	May 4, 2026 10:54 PM	—	✓ Active	Manage
2	JOVAN NADALA	2020018769	008C0830053F	—	May 4, 2026 10:53 PM	—	✓ Active	Manage
3	ADD	009755544556	009438C083920	0179561242	May 4, 2026 10:32 PM	L-01	✓ Active	Manage
4	NEL	202210714	0003486A39C7	—	May 4, 2026 10:31 PM	—	✓ Active	Manage
5	JONRL	202210706	—	—	May 4, 2026 10:19 PM	—	⚠ Pending	Manage
6	JOVAN P.	20221786	00A330C23968	0179561250	May 4, 2026 10:16 PM	—	✓ Active	Manage
7	ROMEG	202110782	007348CE39CF	0179561266	May 4, 2026 09:22 PM	—	✓ Active	Manage
8	GROME	202110753	00A460E1062E	0179561230	May 4, 2026 08:59 PM	—	✓ Active	Manage
9	ROMMMM	202110747	000365043950	—	May 4, 2026 08:58 PM	—	✓ Active	Manage

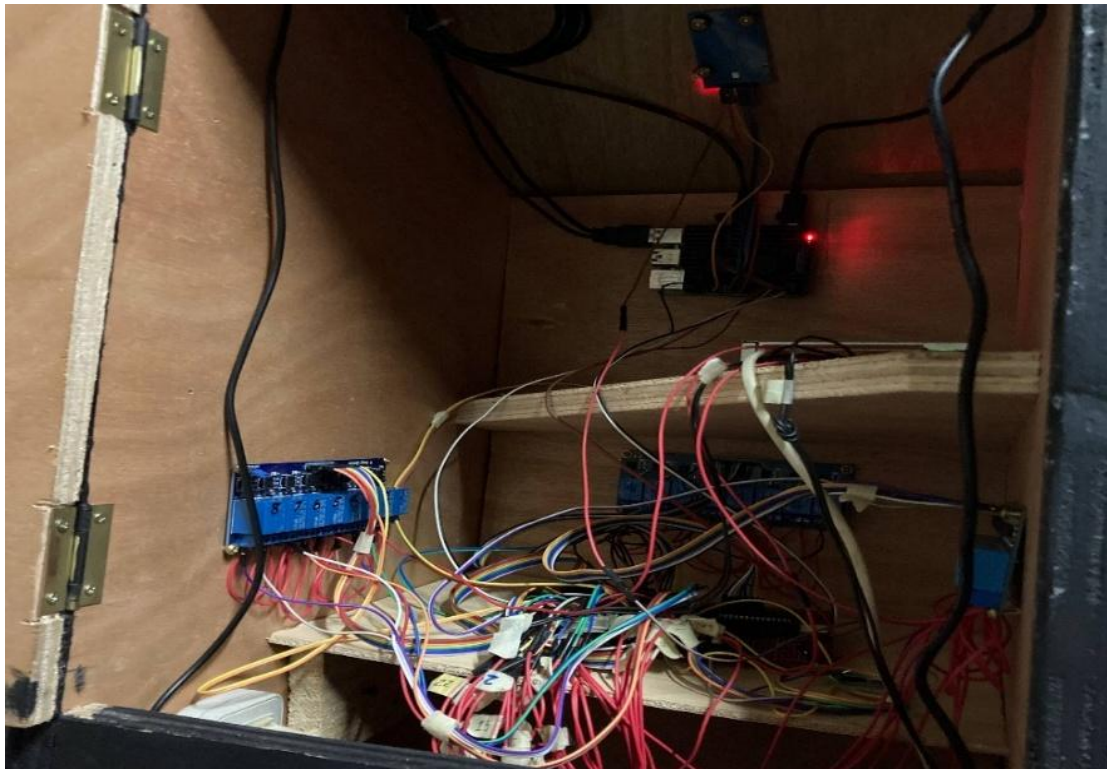
Appendix Figure 3. User transaction saved in admin dashboard



Appendix Figure 4. Landing page of kiosk interface



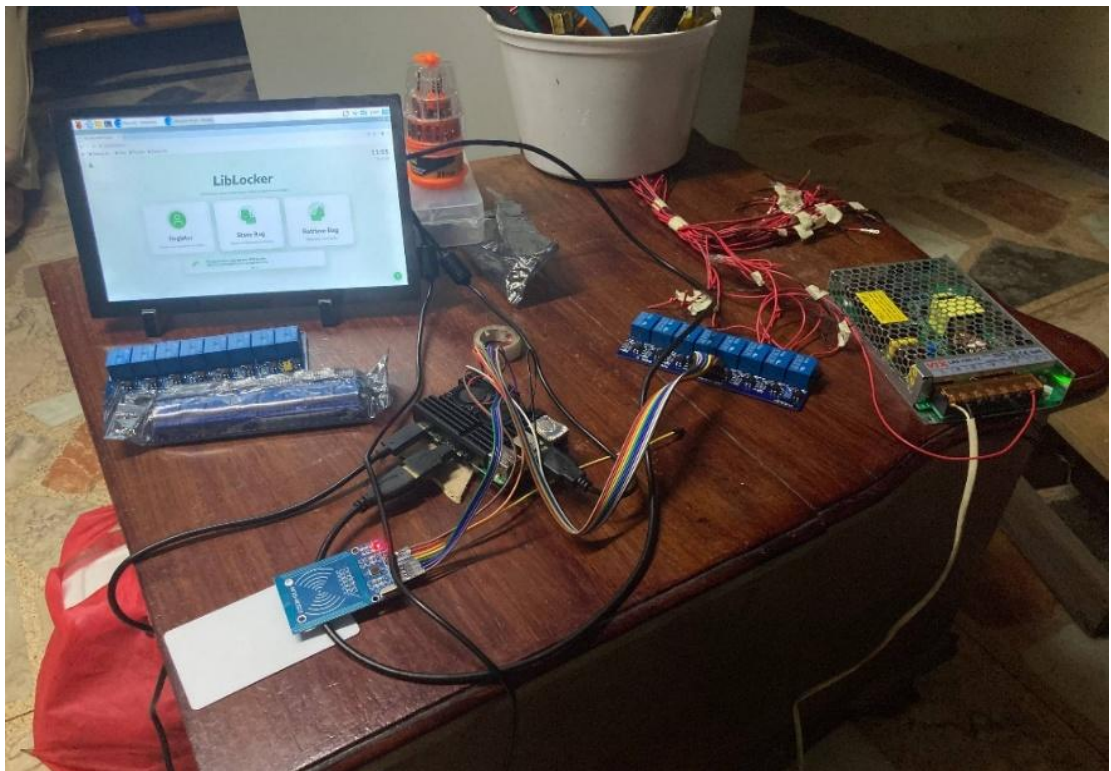
Appendix Figure 5. Kiosk main interface



Appendix Figure 6. Components wiring of kiosk



Appendix Figure 7. Wirings of solenoid, reed switch, and locker



Appendix Figure 8. Components testing



Appendix Figure 9. Kiosk interface testing



Appendix Figure 10. Kiosk and locker integration



Appendix Figure 11. Kiosk casing (Front view) with UHF RFID reader (915 MHz)



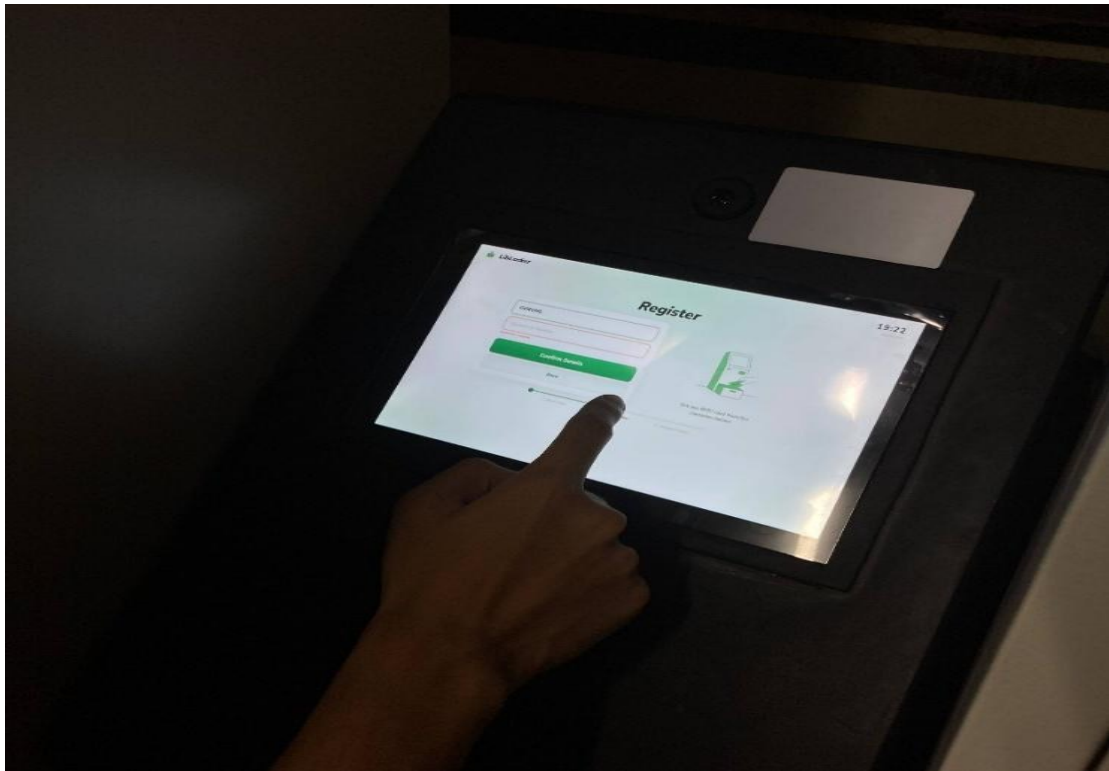
Appendix Figure 12. Kiosk casing (Side view)



Appendix Figure 13. Final setup of prototype

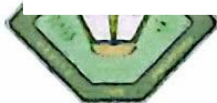


Appendix Figure 14. Deployed at e-library



Appendix Figure 15. Prototype testing

COMMUNICATION LETTER WITH LIBRARY STAFF



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU CCAT Campus
Rosario, Cavite
(046) 437-9507 / (046) 437-6659
cvsuroosario@cvsu.edu.ph
www.cvsu-rosario.edu.ph



DEPARTMENT OF ENGINEERING

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

January 14, 2026

DAN MARIE F. NAVARRETE, RL, MLIS

Campus Librarian

This Campus

Ma'am:

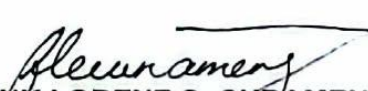
We, the undersigned 4th Year Bachelor of Science in Computer Engineering students of Cavite State University-CCAT Campus, respectfully write this letter to formally inform your good office of our intention to implement our approved design project/thesis entitled **"Development of LibLocker: An Automated Library Baggage Counter with Dual Frequency RFID Authentication for Cavite State University-CCAT Campus e-Library."**

The primary objective of this project, entitled "Development of LibLocker: An Automated Library Baggage Counter with Dual Frequency RFID Authentication for Cavite State University-CCAT Campus e-Library," is to design and develop an IoT-based system that enhances library security and operational efficiency. The proposed system aims to automate baggage counting and access monitoring through dual-frequency RFID authentication, reducing manual intervention, improving accuracy in monitoring entries and exits, and strengthening asset and user security within the e-Library. In this regard, we respectfully request permission to implement, test, and evaluate the proposed system within your office, with all activities conducted under proper coordination and in strict compliance with university policies, data privacy regulations, and ethical standards.

We sincerely hope for your kind consideration and approval of this request. Should you require further details or clarification regarding our project, we would be more than willing to discuss them with you at your convenience. Thank you very much for your time and support.

Respectfully yours,


JONEEL B. ANDAMON
Researcher, BSCpE 401 B


FRANK LORENZ C. CURAMENG
Researcher, BSCpE 401 B

Madala
JOVAN P. NADALA
Researcher, BSCpE 401 B

Voluntad
GEROME Q. VOLUNTAD
Researcher, BSCpE 401 B

Noted:

Sunny
SUNNY MAY S. RONAS, MENG
Research Adviser

Yna
PRINCESS YNA D. JUYAD, REE, RME
Technical Critic

APPROVED:
Navarrete
DAN MARIE F. NAVARRETE, RL, MLIS
Campus Librarian

SOURCE CODE

Index.html

```

<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>LibLocker</title>
  <link rel="stylesheet" href="/assets/css/styles.css" />
  <link rel="stylesheet" href="/assets/css/kiosk_redesign.css" />
</head>
<body class="page page-landing">
  <main class="landing-screen" role="button" aria-label="Touch the screen to start"
  tabindex="0">
    <section class="landing-center" aria-label="LibLocker start screen">
      <div class="landing-logo" aria-hidden="true">
        
      </div>
      <h1 class="landing-title">LibLocker</h1>
      <p class="landing-subtitle">
        <span data-i18n-key="landingTouchPrefix">Touch the screen</span>
        <span data-i18n-key="landingTouchSuffix">to start</span>
      </p>
      <div class="landing-indicator" aria-hidden="true">
        <span></span>
        <span></span>
        <span></span>
      </div>
    </section>

    <div class="landing-wave landing-wave-left" aria-hidden="true"></div>

```



```

</div>

<div class="clock">
  <div id="time" class="clock-time">00:00</div>
  <div id="date" class="clock-date">---</div>
</div>

</header>

<main class="kiosk-home-main">
  <section class="kiosk-home-hero">
    <h1 class="kiosk-home-title">LibLocker</h1>
    <p class="kiosk-home-subtitle" data-i18n-key="subtitle">
      Self-Service E-Library Locker System &nbsp;—&nbsp; Select an action below to
begin.
    </p>
  </section>
  <section class="kiosk-actions" aria-label="Main actions">
    <a href="/pages/register.html" class="kiosk-action-card" data-action="register">
      <div class="kiosk-card-inner">
        <div class="kiosk-card-icon">
          
        </div>
        <div class="kiosk-card-title" data-i18n-key="registerTitle">Register</div>
        <div class="kiosk-card-desc" data-i18n-key="registerDesc">Create a new
account to use lockers</div>
      </div>
    </a>
    <a href="/pages/store_select.html" class="kiosk-action-card kiosk-action-card--
featured" data-action="store">
      <div class="kiosk-card-inner">
        <div class="kiosk-card-icon">
          
        </div>

```

```

    <div class="kiosk-card-title" data-i18n-key="storeTitle">Store Bag</div>

    <div class="kiosk-card-desc" data-i18n-key="storeDesc">Secure your
    belongings for the day.</div>

    </div>

</a>

<a href="/pages/retrieve.html" class="kiosk-action-card" data-action="retrieve">

    <div class="kiosk-card-inner">

        <div class="kiosk-card-icon">

        </div>

        <div class="kiosk-card-title" data-i18n-key="retrieveTitle">Retrieve Bag</div>

        <div class="kiosk-card-desc" data-i18n-key="retrieveDescHome">Collect your
        stored items.</div>

    </div>

</a>

</section>

<section class="kiosk-hint rfid-carousel" aria-label="RFID instruction"
tabindex="0">

    <div class="rfid-carousel-track">

        <div class="rfid-slide is-active" role="group" aria-label="1 of 3">

            <div class="rfid-icon"></div>

            <div class="rfid-text">

                <div class="rfid-title" data-i18n-key="helpRfidLine1">Choose an action, then
                tap your RFID card.</div>

                <div class="rfid-title" data-i18n-key="helpRfidLine2">Wait for confirmation
                before leaving the kiosk.</div>

            </div>

        </div>

        <div class="rfid-slide" role="group" aria-label="2 of 3" aria-hidden="true">

            <div class="rfid-icon"></div>

            <div class="rfid-text">

```

```

    <div class="rfid-title" data-i18n-key="helpPrivacyLine1">Keep your access
code and RFID private.</div>

```

```

    <div class="rfid-title" data-i18n-key="helpPrivacyLine2">You need it to
retrieve your bag.</div>

```

```

    </div>

```

```

</div>

```

```

<div class="rfid-slide" role="group" aria-label="3 of 3" aria-hidden="true">

```

```

    <div class="rfid-icon"></div>

```

```

    <div class="rfid-text">

```

```

        <div class="rfid-title" data-i18n-key="helpLanguageLine1">Need assistance
or language help?</div>

```

```

        <div class="rfid-title" data-i18n-key="helpLanguageLine2">Tap the ? button at
the bottom right.</div>

```

```

        </div>

```

```

    </div>

```

```

</div>

```

```

<div class="rfid-dots" aria-label="Slide indicators">

```

```

    <button class="rfid-dot is-active" type="button" aria-label="Go to slide
1"></button>

```

```

    <button class="rfid-dot" type="button" aria-label="Go to slide 2"></button>

```

```

    <button class="rfid-dot" type="button" aria-label="Go to slide 3"></button>

```

```

</div>

```

```

</section>

```

```

</main>

```

```

<div class="help-btn" role="button" tabindex="0" aria-
label="Help"><span>?</span></div>

```

```

<div class="lang-pill" aria-label="Language selector">

```

```

    <span class="lang-label" data-i18n-key="languageLabel">Language:</span>

```

```

    <select id="langSelect" aria-label="Select language">

```

```

        <option value="en" selected data-i18n-key="optionEnglish">English</option>

```

```

        <option value="fil" data-i18n-key="optionFilipino">Filipino</option>

```

```

    </select>

```

```

</div>

<script src="/assets/js/app.js"></script>

<script src="/assets/js/kiosk_enhancements.js"></script>

</body>

</html>

```

ADMIN

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>LibLocker Admin Login</title>

<style>

body{

  font-family: Arial, sans-serif;

  background:#f2f5f2;

  display:flex;

  justify-content:center;

  align-items:center;

  height:100vh;

}

.login-form{

  width:320px;

  padding:30px;

  background:#fff;

  border-radius:12px;

  display:flex;

  flex-direction:column;

  gap:15px;

  box-shadow:0 4px 10px rgba(0,0,0,0.1);

```

```

}
.login-form h2{
    text-align:center;
    color:#2e7d32;
}
.login-form input{
    padding:10px;
    border:1px solid #ccc;
    border-radius:8px;
}
.login-form button{
    padding:10px;
    border:none;
    border-radius:8px;
    background:#2e7d32;
    color:white;
    cursor:pointer;
}
.error-box{
    color:red;
    font-size:14px;
    text-align:center;
}
</style>
</head>
<body>
<form method="POST" class="login-form">
    <h2>LibLocker Admin Login</h2>
    {% if error %}
    <div class="error-box">{{ error }}</div>
    {% endif %}

```

```

<input
  type="text"
  name="username"
  placeholder="Username"
  required
>
<input
  type="password"
  name="password"
  placeholder="Password"
  required
>
<button type="submit">Sign In</button>
</form>
</body>
</html>

```

CSS

```

:root{--green:#2f7d44;--green-2:#4fa56a;--text:#1f2a1f;--muted:#66736a;--
card:rgba(255,255,255,0.85);--stroke:rgba(47,125,68,0.25);--shadow:0 16px 40px
rgba(0,0,0,0.10)}

*{box-sizing:border-box}

html,body{height:100%;overflow:hidden}

body{margin:0;min-height:100vh;font-family:ui-sans-serif,system-ui,-apple-
system,Segoe UI,Roboto,Arial,sans-serif;color:var(--text);background:radial-
gradient(1200px 600px at 50% 20%,#f8fffa 0%,#eef6ee 55%,#e9f0e9 100%)}

.bg-glow{position:fixed;inset:-120px -160px auto -
160px;height:520px;background:radial-gradient(circle at 50%
50%,rgba(79,165,106,0.18) 0%,rgba(79,165,106,0) 70%);filter:blur(18px);pointer-
events:none}

.topbar{position:relative;width:100%;padding:20px 36px 0;display:flex;justify-
content:space-between;align-items:flex-start}

.brand{display:flex;gap:12px;align-items:center}

```

```

.brand-icon{width:62px;height:62px;border-
radius:16px;background:rgba(255,255,255,0.72);border:1px solid
rgba(47,125,68,0.15);box-shadow:0 10px 26px rgba(0,0,0,0.06);display:grid;place-
items:center}

.brand-logo-img{width:46px;height:46px;object-fit:contain}

.brand-wordmark{font-weight:900;font-style:italic;letter-spacing:-0.6px;font-
size:34px;color:#111;line-height:1;user-select:none}

.clock{text-align:right;color:#111;padding:9px 16px 11px;border-
radius:18px;background:rgba(255,255,255,0.54);border:1px solid
rgba(47,125,68,0.12);backdrop-filter:blur(8px)}

.clock-time{font-weight:800;font-size:38px;letter-spacing:0.5px;font-variant-
numeric:tabular-nums}

.clock-date{margin-top:2px;font-size:14px;color:rgba(0,0,0,0.55)}

.wrap{width:min(1200px,92vw);margin:0 auto;padding:10px 0 70px;min-
height:calc(100vh - 80px)}

.hero{text-align:center;margin-top:8px}

.title{margin:6px 0;font-size:clamp(44px,6vw,74px);font-weight:900;letter-spacing:-
1px;color:#111;font-style:italic}

.subtitle{margin:0;color:rgba(0,0,0,0.55);font-size:19px}

/* ACTION CARDS */

.actions{margin:22px auto 0;width:min(1100px,92vw);display:grid;grid-template-
columns:repeat(3,1fr);gap:20px}

.action-card{position:relative;border-radius:24px;background:linear-
gradient(135deg,rgba(255,255,255,0.85),rgba(255,255,255,0.65));border:1px solid
rgba(47,125,68,0.15);backdrop-filter:blur(10px);box-shadow:0 8px 32px
rgba(0,0,0,0.06);overflow:hidden;transition:all .3s cubic-bezier(0.4,0,0.2,1);text-
decoration:none;color:inherit;display:block}

.action-card::after{content:"";position:absolute;inset:10px;border-
radius:18px;border:2px solid rgba(79,165,106,0.15);pointer-events:none;transition:all
.3s}

.action-card:hover{transform:translateY(-8px);box-shadow:0 24px 56px
rgba(47,125,68,0.12)}

.card-inner{padding:28px 24px 26px;text-align:center}

.card-icon{width:104px;height:104px;margin:0 auto 16px;border-
radius:20px;background:linear-
gradient(135deg,rgba(79,165,106,0.15),rgba(79,165,106,0.05));display:flex;align-
items:center;justify-content:center;transition:all .3s}

```

```

.card-icon-img{width:82px;height:82px;object-fit:contain;filter:drop-shadow(0 10px
18px rgba(47,125,68,0.10));transition:transform .3s}

.action-card:hover .card-icon{transform:scale(1.1) translateY(-2px)}

.action-card:hover .card-icon-img{transform:scale(1.08) rotate(5deg)}

.card-title{font-size:30px;font-weight:800;color:#111;margin-bottom:8px}

.card-desc{font-size:14px;color:rgba(0,0,0,0.55);line-height:1.35;max-
width:260px;margin:0 auto}

/* HOME */

.page-home .wrap{width:min(1380px,96vw);min-height:calc(100vh -
96px);display:flex;flex-direction:column;justify-content:center;gap:24px;padding:4px 0
54px}

.page-home .actions{width:min(1280px,96vw);gap:28px;margin-top:0}

.page-home .action-card{min-height:340px;border-radius:28px}

.page-home .card-inner{padding:36px 28px 32px;display:flex;flex-
direction:column;align-items:center;justify-content:center}

.page-home .card-icon{width:148px;height:148px;margin-bottom:20px}

.page-home .card-icon svg,.page-home .card-icon-img{width:112px;height:112px}

.page-home .card-title{font-size:clamp(36px,2.6vw,42px);margin-bottom:12px}

.page-home .card-desc{font-size:17px;line-height:1.5;max-width:320px}

/* STEPPER */

.stepper{margin:22px auto 0;width:min(780px,92vw);text-align:center}

.stepper-line{position:relative;height:22px;margin:0 auto 10px}

.stepper-
line::after{content:"";position:absolute;left:0;right:0;top:10px;height:3px;border-
radius:99px;background:rgba(0,0,0,0.18)}

.stepper-progress{position:absolute;left:0;top:10px;height:3px;border-
radius:99px;background:rgba(47,125,68,0.85);transition:width .5s cubic-
bezier(0.4,0,0.2,1)}

.stepper-dot{position:absolute;top:3px;width:20px;height:20px;border-
radius:50%;background:#d9ded9;border:2px solid
rgba(0,0,0,0.22);transform:translateX(-50%);transition:all .3s}

.stepper-dot--active{background:rgba(47,125,68,0.95);border-
color:rgba(47,125,68,0.55)}

.stepper-dot--done{background:rgba(47,125,68,0.95);border-
color:rgba(47,125,68,0.55)}

```

```

.stepper-labels{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;font-size:15px;color:rgba(0,0,0,0.45)}

.stepper-label{display:flex;justify-content:center;gap:6px;align-items:center}

.stepper-label--active{color:rgba(0,0,0,0.75);font-weight:800}

.stepper-3 .stepper-labels,.stepper-4 .stepper-labels{font-size:15px}

.stepper-3--step1 .stepper-progress{width:0%}.stepper-3--step2 .stepper-progress{width:50%}.stepper-3--step3 .stepper-progress{width:100%}

.stepper-4--step1 .stepper-progress{width:0%}.stepper-4--step2 .stepper-progress{width:33.333%}.stepper-4--step3 .stepper-progress{width:66.666%}.stepper-4--step4 .stepper-progress{width:100%}

/* RFID CAROUSEL */

.rfid-hint{margin:22px auto 0;width:min(820px,92vw);display:flex;align-items:center;gap:18px;padding:16px 18px;border-radius:20px;background:rgba(255,255,255,0.75);border:2px solid rgba(79,165,106,0.22);box-shadow:0 18px 44px rgba(0,0,0,0.10);position:relative;overflow:hidden}

.rfid-hint.rfid-carousel{display:block;cursor:pointer;user-select:none;padding:10px 14px 20px;border-radius:18px}

.rfid-carousel-track{position:relative;min-height:72px}

.rfid-slide{position:absolute;inset:0;display:flex;align-items:center;gap:12px;opacity:0;visibility:hidden;transform:translateY(6px);transition:opacity .42s ease,transform .42s ease,visibility .42s linear}

.rfid-slide.is-active{opacity:1;visibility:visible;transform:translateY(0)}

.rfid-dots{position:absolute;left:50%;bottom:6px;transform:translateX(-50%);display:flex;gap:8px}

.rfid-dot{width:8px;height:8px;border:none;border-radius:999px;padding:0;background:rgba(0,0,0,0.25);cursor:pointer;transition:width .2s,background-color .2s}

.rfid-dot.is-active{width:22px;background:rgba(47,125,68,0.85)}

.rfid-title{font-size:18px;font-weight:800;color:rgba(0,0,0,0.70);line-height:1.15}

.rfid-icon-img{width:90px;height:66px;object-fit:contain}

/* PILL BUTTONS */

.pill-btn{display:inline-flex;align-items:center;justify-content:center;height:60px;min-width:168px;padding:0 34px;border-radius:999px;font-weight:800;font-size:20px;cursor:pointer;border:1px solid rgba(47,125,68,0.25);box-shadow:0 16px 34px rgba(0,0,0,0.10);transition:transform .12s ease}

```

```

.pill-btn--solid{background:linear-
gradient(180deg,rgba(47,125,68,0.95),rgba(47,125,68,0.80));color:#fff}

.pill-btn--ghost{background:rgba(47,125,68,0.14);color:rgba(47,125,68,0.95)}

.pill-btn:active{transform:translateY(1px)}

.pill-btn[disabled]{opacity:0.55;cursor:not-allowed}

/* REGISTER */

.page-register .grid{margin-top:4px;display:grid;grid-template-columns:1.1fr
0.9fr;gap:32px;align-items:center}

.page-register .form-card{border-
radius:24px;background:rgba(255,255,255,0.55);border:2px solid
rgba(79,165,106,0.22);box-shadow:0 18px 48px rgba(0,0,0,0.10);padding:16px}

.page-register .input-box{margin-bottom:14px}

.page-register .input-box input{width:100%;height:70px;padding:0 22px;font-
size:24px;border-radius:16px;border:2px solid
rgba(47,125,68,0.40);background:rgba(255,255,255,0.72);outline:none;transition:all
.3s}

.page-register .input-box input:focus{border-color:rgba(47,125,68,0.75);box-
shadow:0 0 0 3px rgba(76,175,80,0.1);transform:translateY(-2px)}

.page-register .input-box input.input-invalid{border-
color:rgba(160,30,30,0.80);animation:shake .5s}

.page-register .input-error{display:none;font-size:14px;font-
weight:700;color:rgba(160,30,30,0.92)}

.page-register .input-error:not(:empty){display:block;min-height:18px;margin-top:6px}

.page-register .btn-primary{width:100%;height:72px;border:1px solid
rgba(33,101,53,0.40);border-radius:16px;background:linear-
gradient(180deg,rgba(58,145,83,0.98),rgba(47,125,68,0.90));color:#fff;font-
size:25px;font-weight:800;cursor:pointer;margin-top:6px;transition:transform
.12s,filter .12s}

.page-register .link-back{width:100%;height:70px;border-radius:16px;border:1px solid
rgba(47,125,68,0.34);background:linear-
gradient(180deg,rgba(47,125,68,0.24),rgba(47,125,68,0.18));color:rgba(34,98,52,0.9
8);font-size:24px;font-weight:800;margin-top:10px;cursor:pointer;transition:transform
.12s}

.page-register .scan-icon-
wrap{position:relative;width:220px;height:220px;display:grid;place-items:center}

.page-register .scan-rfid-icon{width:160px;height:auto;position:relative;z-index:3}

```

```

.page-register .scan-pulse{position:absolute;width:160px;height:160px;border-
radius:50%;border:4px solid rgba(79,165,106,0.30);opacity:0;animation:register-
scan-pulse 2s ease-out infinite}

.page-register .scan-pulse--b{animation-delay:.65s}.page-register .scan-pulse--
c{animation-delay:1.3s}

.page-register .scan-title{font-size:28px;font-
weight:800;color:rgba(0,0,0,0.76);margin-top:14px}

.page-register .scan-subtext{font-size:19px;font-
weight:650;color:rgba(0,0,0,0.55);margin-top:10px}

.page-register .register-toast{font-size:16px;font-
weight:700;opacity:0;transform:translateY(4px);transition:opacity .18s,transform
.18s;overflow:hidden}

.page-register .register-toast.is-show{opacity:1;transform:translateY(0)}

.page-register .register-toast.is-error{color:rgba(160,30,30,0.95)}

.page-register .register-toast.is-success{color:rgba(33,111,55,0.95)}

.page-register .right-caption{font-size:22px;font-
weight:700;color:rgba(0,0,0,0.65);line-height:1.3;text-align:center}

.page-register .kiosk-illustration{width:min(340px,30vw)}

/* LOCKER SELECT */

.page-locker .locker-card{width:min(1180px,92vw);margin:0 auto;border-
radius:26px;background:rgba(255,255,255,0.62);border:2px solid
rgba(79,165,106,0.24);box-shadow:0 22px 60px rgba(47,125,68,0.18)}

.page-locker .locker-card-inner{padding:32px 36px 28px;display:flex;flex-
direction:column;align-items:center;justify-content:center}

.page-locker .locker-options{width:100%;max-width:1000px;display:grid;grid-
template-columns:repeat(3,1fr);gap:28px;margin:20px 0 28px}

.page-locker .locker-option{border:2px solid
rgba(47,125,68,0.20);background:rgba(255,255,255,0.72);border-
radius:24px;cursor:pointer;box-shadow:0 10px 28px
rgba(0,0,0,0.07);transition:transform .2s,border-color .2s,box-shadow .2s;min-
height:220px;position:relative;overflow:hidden}

.page-locker .locker-option:hover{transform:translateY(-6px);border-
color:rgba(47,125,68,0.40)}

.page-locker .locker-option--active{border-width:3px;border-
color:rgba(47,125,68,0.70);background:rgba(255,255,255,0.90);transform:translateY(
-4px)}

```

```

.page-locker .locker-option-inner{padding:32px 24px 28px;text-align:center;display:flex;flex-direction:column;align-items:center;justify-content:center;gap:16px}

.page-locker .locker-option-icon{width:140px;height:140px;border-radius:24px;background:linear-gradient(145deg,rgba(79,165,106,0.15),rgba(79,165,106,0.04));display:grid;place-items:center;transition:transform .2s}

.page-locker .locker-option:hover .locker-option-icon,.page-locker .locker-option--active .locker-option-icon{transform:scale(1.05)}

.page-locker .locker-option-icon img{width:96px;height:96px;object-fit:contain}

.page-locker .locker-option-title{font-size:30px;font-weight:900;color:#111}

.page-locker .locker-option--active .locker-option-title{color:rgba(30,90,45,0.95)}

.page-locker .locker-actions{display:flex;justify-content:center;gap:24px;margin-top:28px}

/* RETRIEVE */

.retrieve-card{width:min(1180px,92vw);margin:0 auto;border-radius:26px;background:rgba(255,255,255,0.62);border:2px solid rgba(79,165,106,0.24);box-shadow:0 22px 60px rgba(47,125,68,0.18)}

.retrieve-card-inner{padding:28px 36px 24px}

.retrieve-content{display:grid;grid-template-columns:1fr 1fr;gap:32px;align-items:center;padding:8px 0}

.retrieve-illustration img{max-height:380px;max-width:100%;filter:drop-shadow(0 14px 32px rgba(0,0,0,0.10))}

.retrieve-steps{display:flex;flex-direction:column;gap:20px;padding-left:10px}

.step-row{display:flex;align-items:center;gap:20px}

.step-icon{width:68px;height:68px;border-radius:16px;background:rgba(255,255,255,0.85);border:1px solid rgba(47,125,68,0.18);box-shadow:0 10px 24px rgba(0,0,0,0.07);display:grid;place-items:center;flex-shrink:0}

.step-icon img{width:40px;height:40px;object-fit:contain}

.step-text{font-size:21px;color:rgba(0,0,0,0.70);font-weight:700;line-height:1.3}

.retrieve-status{margin-top:8px;border-radius:14px;padding:14px 20px 14px 42px;font-size:19px;font-weight:800;color:rgba(47,125,68,0.95);background:rgba(255,255,255,0.80);border:1.5px solid rgba(47,125,68,0.20);position:relative}

.retrieve-status::before{content:"";position:absolute;left:14px;top:50%;transform:translateY(-

```

```

50%);width:18px;height:18px;border-
radius:50%;background:#4caf50;animation:pulse-dot 2s infinite}

/* ASSIGNED */

.page-assigned .assigned-card{width:min(1180px,92vw);margin:10px auto 0;border-
radius:26px;background:rgba(255,255,255,0.62);border:2px solid
rgba(79,165,106,0.24);box-shadow:0 22px 60px rgba(47,125,68,0.18)}

.page-assigned .assigned-card-inner{padding:28px 36px 22px;display:flex;flex-
direction:column;justify-content:center;align-items:center;gap:16px;text-
align:center;min-height:400px}

.page-assigned .assigned-heading{font-size:clamp(40px,4.4vw,58px);font-
weight:950;color:rgba(79,165,106,0.78);margin:0 0 6px}

.page-assigned .assigned-subtitle{margin:0;font-size:17px;font-
weight:650;color:rgba(0,0,0,0.55)}

.page-assigned .assigned-content{display:grid;grid-template-columns:max-content
max-content;justify-content:center;align-items:center;column-gap:32px}

.page-assigned .locker-badge{width:200px;height:200px;border-
radius:50%;background:radial-gradient(circle at 30% 25%,rgba(79,165,106,0.30)
0%,rgba(79,165,106,0.04) 70%);border:2.5px solid rgba(47,125,68,0.28);box-
shadow:0 18px 44px rgba(47,125,68,0.15);display:grid;place-items:center}

.page-assigned .locker-number{font-size:88px;font-
weight:950;color:rgba(47,125,68,0.95);line-height:1}

.page-assigned .assigned-illustration{width:min(360px,32vw);object-
fit:contain;filter:drop-shadow(0 16px 36px rgba(0,0,0,0.10))}

.page-assigned .assigned-note{font-size:19px;font-
weight:750;color:rgba(0,0,0,0.62);max-width:560px;text-align:center}

/* COMPLETE */

.complete-card{width:min(1180px,92vw);margin:0 auto;border-
radius:26px;background:rgba(255,255,255,0.62);border:2px solid
rgba(79,165,106,0.24);box-shadow:0 22px 60px rgba(47,125,68,0.18)}

.complete-card-inner{padding:28px 36px 24px;display:flex;flex-
direction:column;align-items:center;justify-content:center;gap:18px;text-align:center}

.complete-title{font-size:clamp(42px,4.5vw,58px);font-
weight:900;color:rgba(47,125,68,0.85);letter-spacing:-0.5px;margin-bottom:6px}

.complete-subtitle{font-size:19px;font-weight:600;color:rgba(0,0,0,0.58);max-
width:640px}

.complete-illustration img{max-height:400px;max-width:min(480px,44vw);object-
fit:contain}

.complete-actions{display:flex;justify-content:center;margin-top:10px}

```

```
/* REGISTER COMPLETE */
```

```
.page-register-complete{display:flex;flex-direction:column}

.page-register-complete .register-complete-wrap{flex:1;display:flex;align-
items:center;justify-content:center;width:min(1000px,92vw);margin:0 auto;padding-
bottom:80px}

.page-register-complete .register-complete-card{width:100%;max-
width:780px;background:rgba(255,255,255,0.9);backdrop-filter:blur(16px);border-
radius:32px;border:1px solid rgba(255,255,255,0.8);box-shadow:0 24px 48px
rgba(0,0,0,0.04);animation:slideUpFade .7s cubic-bezier(0.16,1,0.3,1) forwards}

.page-register-complete .register-complete-card-inner{padding:52px 48px;text-
align:center;display:flex;flex-direction:column;align-items:center}

.page-register-complete .register-complete-badge{width:124px;height:124px;margin-
bottom:28px;border-radius:50%;background:linear-
gradient(135deg,#4CAF50,#2E7D32);box-shadow:0 16px 32px
rgba(76,175,80,0.25);display:grid;place-items:center;animation:scaleBounce .6s
cubic-bezier(0.34,1.56,0.64,1) .2s both}

.success-svg{width:64px;height:64px}

.success-check{stroke:#fff;stroke-width:5;stroke-linecap:round;stroke-
linejoin:round;stroke-dasharray:48;stroke-dashoffset:48;animation:drawCheck .6s
cubic-bezier(0.65,0,0.45,1) .6s forwards}

.page-register-complete .register-complete-title{margin:0 0 16px;font-size:54px;font-
weight:800;color:#1a1a1a;letter-spacing:-0.5px;animation:slideUpFade .6s ease .3s
both}

.page-register-complete .register-complete-subtitle{margin:0;font-size:25px;font-
weight:500;color:#555;line-height:1.5;animation:slideUpFade .6s ease .4s both}

.redirect-indicator{margin-top:44px;display:flex;align-items:center;justify-
content:center;font-size:20px;font-
weight:600;color:rgba(47,125,68,0.8);background:rgba(47,125,68,0.08);padding:14px
28px;border-radius:99px;animation:slideUpFade .6s ease .5s both}

.redirect-dots span{animation:loadingDots 1.4s infinite;opacity:0;margin-left:2px}

.redirect-dots span:nth-child(1){animation-delay:0s}.redirect-dots span:nth-
child(2){animation-delay:.2s}.redirect-dots span:nth-child(3){animation-delay:.4s}
```

```
/* LANDING */
```

```
.page-landing{margin:0;width:100vw;min-height:100vh;display:flex;align-
items:center;justify-content:center;overflow:hidden;cursor:pointer;background:radial-
gradient(980px 500px at 50% 54%,rgba(116,194,139,0.12) 0%,transparent
72%),linear-gradient(180deg,#f8fbf8 0%,#f0f6f1 58%,#ebf3ee 100%)}

.page-landing .landing-center{position:relative;z-
index:8;width:min(900px,92vw);margin:-86px auto 0;text-align:center}
```

```

.page-landing .landing-
logo{width:clamp(140px,17vw,220px);height:clamp(140px,17vw,220px);margin:0
auto 8px;border-radius:50%;background:radial-gradient(circle at 50%
42%,rgba(120,215,150,0.24) 0%,transparent 82%);display:grid;place-
items:center;box-shadow:0 22px 48px rgba(79,165,106,0.14)}

.page-landing .landing-logo
img{width:clamp(90px,11vw,150px);height:clamp(90px,11vw,150px);object-
fit:contain;filter:drop-shadow(0 14px 24px rgba(47,125,68,0.28))}

.page-landing .landing-title{margin:0;font-size:clamp(72px,9vw,124px);font-
weight:900;font-style:italic;letter-spacing:-1.5px;line-height:0.94;color:#115f3a}

.page-landing .landing-subtitle{margin:clamp(112px,15vh,148px) auto 0;font-
size:clamp(32px,4.3vw,62px);font-
weight:500;color:rgba(36,40,45,0.92);animation:landingSubtitleFloat 2.8s ease-in-out
infinite}

.page-landing .landing-subtitle span{color:#0b8550;font-weight:800}

.page-landing .landing-indicator{width:min(420px,76vw);margin:24px auto
0;display:flex;align-items:center;justify-content:center;gap:16px}

.page-landing .landing-indicator span:nth-child(2){width:44px;height:8px;border-
radius:999px;background:rgba(66,182,114,0.92);animation:landingIndicatorBounce
1.9s ease-in-out infinite}

/* HELP & LANG */

.help-btn{position:fixed;right:30px;bottom:26px;width:58px;height:58px;border-
radius:999px;background:rgba(47,125,68,0.90);color:#fff;display:grid;place-
items:center;box-shadow:0 18px 40px rgba(0,0,0,0.14);cursor:pointer;z-index:20}

.help-btn span{font-size:26px;font-weight:900}

.lang-pill{position:fixed;right:100px;bottom:28px;display:flex;align-
items:center;gap:10px;background:rgba(255,255,255,0.80);border:1px solid
rgba(47,125,68,0.18);border-radius:999px;padding:12px 20px;box-shadow:0 18px
44px rgba(0,0,0,0.10);opacity:0;visibility:hidden:pointer-
events:none;transform:translateX(24px) scale(0.94);transition:opacity .24s,transform
.24s,visibility .24s step-end;z-index:19}

.lang-pill.lang-pill--open{opacity:1;visibility:visible;pointer-
events:auto;transform:translateX(0) scale(1)}

.lang-pill select{border:none;outline:none;background:transparent;font-
weight:700;font-size:17px;color:rgba(0,0,0,0.65)}

/* UHF SPINNER */

.uhf-spinner{width:76px;height:76px;border-radius:50%;border:6px solid
rgba(47,125,68,0.15);border-top-color:rgba(47,125,68,0.85);animation:uhf-spin .85s
linear infinite;margin:0 auto 28px}

```

```
/* KEYFRAMES */
```

```
@keyframes register-scan-  
pulse{0%{opacity:.85;transform:scale(0.72)}100%{opacity:0;transform:scale(1.5)}}
```

```
@keyframes pulse-dot{0%,100%{opacity:1}50%{opacity:.3}}
```

```
@keyframes uhf-spin{to{transform:rotate(360deg)}}
```

```
@keyframes shake{0%,100%{transform:translateX(0)}25%{transform:translateX(-  
10px)}75%{transform:translateX(10px)}}
```

```
@keyframes  
slideUpFade{0%{opacity:0;transform:translateY(24px)}100%{opacity:1;transform:tran  
slateY(0)}}
```

```
@keyframes  
scaleBounce{0%{opacity:0;transform:scale(0.5)}70%{opacity:1;transform:scale(1.05)}  
100%{opacity:1;transform:scale(1)}}
```

```
@keyframes drawCheck{100%{stroke-dashoffset:0}}
```

```
@keyframes loadingDots{0%{opacity:0}50%{opacity:1}100%{opacity:0}}
```

```
@keyframes  
landingSubtitleFloat{0%,100%{transform:translateY(0)}50%{transform:translateY(-  
7px)}}
```

```
@keyframes  
landingIndicatorBounce{0%,100%{transform:translateY(0)}50%{transform:translateY(  
-4px)}}
```

```
/* RESPONSIVE */
```

```
@media(max-width:980px){.actions{grid-template-  
columns:1fr;width:min(560px,92vw)}.retrieve-content{grid-template-  
columns:1fr}.page-register .grid{grid-template-columns:1fr}}
```

```
@media(max-width:1100px),(max-height:650px){body{zoom:.88}}
```

```
@media(max-width:1024px),(max-height:600px){body{zoom:.82}}
```

```
@media(hover:none){.action-card:hover,.locker-option:hover{transform:none}.action-  
card:active{transform:scale(0.98)}.locker-option:active{transform:scale(0.97)}.pill-  
btn:active,.page-register .btn-primary:active{transform:translateY(2px)}}
```

```
@media(prefers-reduced-motion:reduce){*,*::before,*::after{animation-  
duration:.01ms!important;animation-iteration-count:1!important;transition-  
duration:.01ms!important}}
```

```
@media(min-width:1240px) and (min-height:780px){.page-home  
.wrap{gap:26px}.page-home .action-card{min-height:340px}.page-home .card-  
icon{width:148px;height:148px}.page-home .card-title{font-  
size:clamp(36px,2.6vw,42px)}}
```

```
@media(max-height:820px) and (min-width:1024px){.topbar{padding-top:16px}.clock-time{font-size:30px}.page-home .wrap{gap:18px}.page-home .action-card{min-height:280px}.page-home .card-title{font-size:30px}}
```

SQL

```
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE users (
```

```
  id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
  full_name TEXT NOT NULL,
```

```
  student_id TEXT UNIQUE NOT NULL,
```

```
  rfid_uid TEXT UNIQUE NOT NULL,
```

```
  uhf_epc TEXT UNIQUE,
```

```
  violation_count INTEGER DEFAULT 0,
```

```
  access_status TEXT DEFAULT 'ACTIVE',
```

```
  created_at TEXT DEFAULT (datetime('now'))
```

```
);
```

```
CREATE TABLE lockers (
```

```
  id INTEGER PRIMARY KEY,
```

```
  label TEXT UNIQUE NOT NULL,
```

```
  locker_number INTEGER UNIQUE,
```

```
  locker_group TEXT,
```

```
  status TEXT DEFAULT 'available',
```

```
  current_user_id INTEGER,
```

```
  is_occupied INTEGER DEFAULT 0,
```

```
  FOREIGN KEY(current_user_id) REFERENCES users(id)
```

```
);
```

```
CREATE TABLE sessions (
```

```
  id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
  user_id INTEGER,
```

```
  locker_id INTEGER,
```

```
  action TEXT,
```

```
  status TEXT DEFAULT 'active',
```

```

    created_at TEXT DEFAULT (datetime('now')),
    completed_at TEXT,
    FOREIGN KEY(user_id) REFERENCES users(id),
    FOREIGN KEY(locker_id) REFERENCES lockers(id)
);

CREATE TABLE transactions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER,
    locker_id INTEGER,
    action TEXT,
    status TEXT DEFAULT 'DONE',
    created_at TEXT DEFAULT (datetime('now')),
    FOREIGN KEY(user_id) REFERENCES users(id),
    FOREIGN KEY(locker_id) REFERENCES lockers(id)
);

CREATE TABLE violations (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER,
    session_id INTEGER,
    violation_type TEXT,
    timestamp TEXT DEFAULT (datetime('now')),
    resolved INTEGER DEFAULT 0,
    FOREIGN KEY(student_id) REFERENCES users(id),
    FOREIGN KEY(session_id) REFERENCES sessions(id)
);

CREATE TABLE activity_logs (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    activity_type TEXT,
    user_id INTEGER,
    locker_id INTEGER,
    timestamp TEXT DEFAULT (datetime('now')),

```

```
FOREIGN KEY(user_id) REFERENCES users(id),  
FOREIGN KEY(locker_id) REFERENCES lockers(id)  
);
```

**RESEARCH INSTRUMENT
(SURVEY QUESTIONNAIRE)**



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CCAT Campus
 Rosario, Cavite
 ☎ (046) 437-9505 / ☎ (046) 437-6659
 cvsuroario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

EVALUATION INSTRUMENT FOR
**DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER
 WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY
 CCAT CAMPUS E-LIBRARY**
 (For Evaluators)

Dear participants,

This assessment aims to understand your perspectives on both technical and non-technical aspects of the project. Your participation in this survey indicates your willingness to share opinions and information essential for evaluating the project. Please rest assured that your identity will remain anonymous, and any personal information you provide will be kept confidential. The data gathered will be used strictly for educational purposes.

Name of Respondent: _____ Age: _____ Gender: _____
 Year and Section: _____

General Instruction: Evaluate the characteristics of the prototype using the rating scale below. Place a checkmark (✓) on the corresponding numerical rating.

ISO/IEC 25010

The ISO 25010 standard establishes a comprehensive model consisting of quality characteristics and associate sub-characteristics, which serves as criteria for the systematic evaluation of software product quality.

Rating Scale

NUMERICAL RATING	DESCRIPTION
5	Excellent
4	Very Satisfactory
3	Satisfactory
2	Unsatisfactory
1	Needs Improvement

INDICATOR	5	4	3	2	1
A. Functional Suitability					
The system correctly matches RFID tags (HF tap and UHF detection) during deposit and retrieval.					
The system performs all required functions (registration, deposit, retrieval, logging) properly.					
The system provides accurate and complete transaction records.					
The system correctly assigns and releases lockers based on user authentication.					
B. Performance Efficiency					
The system processes deposit transactions quickly					
The system processes retrieval transactions quickly.					
The system responds without noticeable delay during operation.					
The system helps reduce waiting time or queues in the library.					
C. Compatibility					
The system integrates well with RFID devices and the monitoring dashboard.					
The system works properly with network connectivity.					
The system continues to function even without network connectivity (offline mode).					
The system synchronizes data correctly between offline and online modes.					
D. Reliability					
The system operates without errors or crashes during use.					
The system performs well under different conditions (e.g., weak signal or offline mode).					
The system properly synchronizes data during offline-online operation.					
The system recovers correctly after interruptions or failures.					
E. Usability					
The system is easy to use.					
The kiosk interface is clear and easy to understand.					

The system instructions are easy to follow.					
The system is convenient for managing personal belongings in the library.					

F. Security

The system prevents unauthorized access to lockers.					
The dual-frequency RFID authentication (HF and UHF) ensures correct user identification.					
The system accurately validates user identity before allowing access.					
The camera snapshot feature improves security monitoring.					

Respondent's Signature



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CCAT Campus
 Rosario, Cavite
 ☎ (046) 437-9505 / ☎ (046) 437-6659
 cvsuroario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

EVALUATION INSTRUMENT FOR
**DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER
 WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY
 CCAT CAMPUS E-LIBRARY**
 (For Evaluators)

Dear participants,

This assessment aims to understand your perspectives on both technical and non-technical aspects of the project. Your participation in this survey indicates your willingness to share opinions and information essential for evaluating the project. Please rest assured that your identity will remain anonymous, and any personal information you provide will be kept confidential. The data gathered will be used strictly for educational purposes.

Name of Respondent: _____ Age: _____ Gender: _____
 Year and Section: _____

General Instruction: Evaluate the characteristics of the prototype using the rating scale below. Place a checkmark (✓) on the corresponding numerical rating.

TECHNOLOGY ACCEPTANCE MODEL

The Technology Acceptance Model establishes a comprehensive model consisting of quality characteristics and associate sub-characteristics, which serves as criteria for the systematic evaluation to accept and use technology.

Rating Scale

NUMERICAL RATING	DESCRIPTION
5	Excellent
4	Very Satisfactory
3	Satisfactory
2	Unsatisfactory
1	Needs Improvement

INDICATOR	5	4	3	2	1
A. Perceived Usefulness					
The system improves the efficiency of bag deposit and retrieval.					
The system provides accurate identification using RFID authentication.					
The system enhances the security of stored belongings.					
The system helps reduce waiting time in the library baggage counter.					
B. Perceived Ease-of-Use					
The system is easy to use.					
The kiosk interface is clear and easy to understand.					
The system can be used without assistance or prior training.					
Navigating the system (kiosk and dashboard) is simple and straightforward.					
C. Behavioral Intention (BI)					
I am willing to use this system regularly when visiting the library					
I would recommend this system to other students or users.					
I prefer this system over manual baggage counters					
D. User Satisfaction					
I am satisfied with the overall performance of the system.					
The system meets my expectations in terms of functionality and usability.					
Overall, I am satisfied with my experience using the system.					

Respondent's Signature

SAMPLE DATA SHEET



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CCAT Campus
 Rosario, Cavite
 ☎ (046) 437-9505 / 📠 (046) 437-6659
 cvsurosario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

EVALUATION INSTRUMENT FOR
**DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE
 COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE
 STATE UNIVERSITY CCAT CAMPUS E-LIBRARY**
 (For Evaluators)

Dear participants,

This assessment aims to understand your perspectives on both technical and non-technical aspects of the project. Your participation in this survey indicates your willingness to share opinions and information essential for evaluating the project. Please rest assured that your identity will remain anonymous, and any personal information you provide will be kept confidential. The data gathered will be used strictly for educational purposes.

Name of Respondent: Carlo A. Nacala Age: 25 Gender: Male
 Year and Section: BSE 402 Math

General Instruction: Evaluate the characteristics of the prototype using the rating scale below. Place a checkmark (✓) on the corresponding numerical rating.

ISO/IEC 25010

The ISO 25010 standard establishes a comprehensive model consisting of quality characteristics and associate sub-characteristics, which serves as criteria for the systematic evaluation of software product quality.

Rating Scale

NUMERICAL RATING	DESCRIPTION
5	Excellent
4	Very Satisfactory
3	Satisfactory
2	Unsatisfactory
1	Needs Improvement

INDICATOR	5	4	3	2	1
A. Functional Sultability					
Does the system accurately read HF RFID taps and UHF detection?	/				
Does the HF-UHF matching algorithm correctly verify user identity?	/				
Are locker assignments, snapshots, and logs recorded correctly?	/				
Does the dashboard display real-time locker status and transaction history?	/				
B. Performance Efficiency					
Does the system process RF reads and update the dashboard quickly?	/				
Does the kiosk respond promptly during deposit and retrieval operations?	/				
Does the system operate efficiently without unnecessary delays?	/				
C. Compatibility					
Does the system work reliably under different network conditions?	/				
Is it fully compatible with both HF and UHF RFID technologies?	/				
D. Reliability					
Do HF and UHF RFID readers perform consistently during repeated use?	/				
Do solenoid locks remain stable during multiple locking/unlocking cycles?	/				
Does the system function properly during weak network or offline conditions?	/				
E. Usability					
Is the kiosk interface easy for students to navigate?	/				
Is the dashboard clear and easy for librarians to interpret?	/				
Are system messages, alerts, and instructions easy to understand?	/				
F. Security					
Are RFID logs, snapshots, and user data securely stored?	/				
Does the system protect sensitive information in compliance with the Data Privacy Act (RA 10173)?	/				

G. Maintainability					
Can the system be updated or modified easily (software and database)?	/				
Are hardware components (RFID readers, locks, camera) easy to replace or repair?	/				
H. Portability					
Can the system be installed in different campus areas without major reconfiguration?	/				
Can the dashboard and logs be accessed across different devices (PC, laptop)?	/				



Respondent's Signature



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CCAT Campus
 Rosario, Cavite
 ☎ (046) 437-9505 / 📠 (046) 437-6659
 cvsurosalio@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

EVALUATION INSTRUMENT FOR
**DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE
 COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE
 STATE UNIVERSITY CCAT CAMPUS E-LIBRARY**
 (For Evaluators)

Dear participants,

This assessment aims to understand your perspectives on both technical and non-technical aspects of the project. Your participation in this survey indicates your willingness to share opinions and information essential for evaluating the project. Please rest assured that your identity will remain anonymous, and any personal information you provide will be kept confidential. The data gathered will be used strictly for educational purposes.

Name of Respondent: Carlo A. Nodala Age: 25 Gender: Male
 Year and Section: DSE 402 Math

General Instruction: Evaluate the characteristics of the prototype using the rating scale below. Place a checkmark (✓) on the corresponding numerical rating.

TECHNOLOGY ACCEPTANCE MODEL

The Technology Acceptance Model establishes a comprehensive model consisting of quality characteristics and associate sub-characteristics, which serves as criteria for the systematic evaluation to accept and use technology.

Rating Scale

NUMERICAL RATING	DESCRIPTION
5	Excellent
4	Very Satisfactory
3	Satisfactory
2	Unsatisfactory
1	Needs Improvement

INDICATOR	5	4	3	2	1
A. Perceived Usefulness					
The system improves the efficiency of bag deposit and retrieval.	/				
The system provides accurate identification using RFID authentication.	/				
The system enhances the security of stored belongings.	/				
The system helps reduce waiting time in the library baggage counter.	/				
B. Perceived Ease-of-Use					
The system is easy to use.	/				
The kiosk interface is clear and easy to understand.	/				
The system can be used without assistance or prior training.	/				
Navigating the system (kiosk and dashboard) is simple and straightforward.	/				
C. Behavioral Intention (BI)					
I am willing to use this system regularly when visiting the library	/				
I would recommend this system to other students or users.	/				
I prefer this system over manual baggage counters	/				
D. User Satisfaction					
I am satisfied with the overall performance of the system.	/				
The system meets my expectations in terms of functionality and usability.	/				
Overall, I am satisfied with my experience using the system.	/				


Respondent's Signature

GANTT CHART

LIBLOCKER — PROJECT WORK PLAN / GANTT CHART January 2026 – May 2026																						
		January 2026				February 2026				March 2026				April 2026				May 2026				
No.	Activity / Description	W1 01/05	W2 01/12	W3 01/19	W4 01/26	W5 02/02	W6 02/09	W7 02/16	W8 02/23	W9 03/02	W10 03/09	W11 03/16	W12 03/23	W13 03/30	W14 04/06	W15 04/13	W16 04/20	W17 04/27	W18 05/04	W19 05/11	W20 05/18	W21 05/25
PLANNING & LITERATURE REVIEW																						
1	Topic finalization and objective setting																					
2	Review of related literature																					
3	Conceptual / IPO framework development																					
SYSTEM DESIGN																						
4	Hardware architecture and component planning																					
5	Software, UI/UX, and database design																					
6	3D modeling of locker prototype and kiosk																					
SYSTEM DEVELOPMENT																						
7	Hardware assembly and locker fabrication																					
8	Raspberry PI setup and GPIO/I2C configuration																					
9	HF/UHF RFID module integration (RCS22 & 915 MHz)																					
10	Solenoid lock and reed switch wiring																					
11	Flask web app and kiosk UI development																					
12	Cloud database and web dashboard development																					
13	USB webcam and snapshot capture integration																					
14	Offline-online synchronization implementation																					
TESTING & DEBUGGING																						
15	HF/UHF RFID read accuracy and response testing																					
16	Solenoid lock and reed switch functional testing																					
17	HF-UHF identity matching algorithm validation																					
18	Dashboard real-time sync and logging testing																					
19	System debugging and refinement																					
PILOT TESTING & EVALUATION																						
20	Online survey with video demonstration																					
21	ISO/IEC 25010 quality standards evaluation (online)																					
22	TAM survey distribution and collection (online)																					
23	Data analysis and result interpretation																					
MANUSCRIPT WRITING																						
24	Chapters 1–3 writing																					
25	Chapters 4–5 writing (Results & Conclusion)																					
26	Appendices, references, and formatting																					
27	Proofreading and final revision																					
FINAL SUBMISSION																						
28	Final manuscript compilation and submission																					
LEGEND: Completed Upcoming / In Progress																						

CURRICULUM VITAE

JOVAN P. NADALA

Noveleta, Cavite | jovannadala@gmail.com | 09150926681

[linkedin.com/in/jovannadala](https://www.linkedin.com/in/jovannadala) | <https://github.com/jovan-nadala/MY-PORTFOLIO>



PROFESSIONAL SUMMARY

Motivated Computer Engineering Student with hands-on experience in web development and embedded systems development. Skilled in building user-centered applications using modern web technologies and IoT platforms. Passionate about creating efficient, scalable systems, and continuously improving technical expertise.

EDUCATION

Cavite State University - CCAT Campus

Bachelor of Science in Computer Engineering

Rosario, Cavite

Sept. 2022 - Present

Holy Cross Catholic School of Noveleta

Senior High School, STEM - With Honors

Noveleta, Cavite

2020 - 2022

TECHNICAL SKILLS

Programming: Python, C++

Web Development: HTML, CSS, JavaScript, Firebase

Tools: Figma, MIT App Inventor, Visual Studio Code

Soft Skills: Teamwork, Time Management, Communication, Adaptability, Problem Solving

EXPERIENCE

On-the-Job Training: Provincial Information and Communication Technology Office

Trece Martires City, Cavite

- Assisted in developing a requisition system for office transactions.
- Conducted network mapping across multiple offices.
- Performed PC setup and troubleshooting tasks.
- Presented system project to the technical and admin team, head of the office.

PROJECTS

Liblocker: Automated Library Baggage Counter (Design Project: 2026)

- Designed and developed an RFID-based baggage storage system with dual-frequency authentication for university e-library users.

Requisition Form Inventory System (2025)

- Developed a system for monitoring inventory and tracking office requisitions.

IoT-Enabled Department Resource Management System (2024)

- Built a real-time resource monitoring platform using IoT technologies.

Healthcare Appointment System (2024)

- Web platform for booking, rescheduling, and canceling medical appointments

CERTIFICATIONS

- Civil Service Professionals Eligibility - *Passed Civil Service Exam – Professional Level (2025)*
- Figma Certification - *UI/UX design and interface development (2025)*

SEMINARS AND TRAININGS

- ASEAN Data Science Explorers Enablement Session (2024)
- Gen AI to Z: A Career Summit in an AI-Driven World (2026)
- Arduino Day Philippines 2026 (2026)

REFERENCES

Engr. Kristian Daile Arellano

College Instructor - LPU-C

Contact: 09154998195/daile.arellano19@gmail.com

Princess Yna D. Juyad, REE, RME

College Instructor - CvSU-CCAT Campus

Contact: 09759697883/princessyna.juyad@cvsu.edu.ph

JONEL B. ANDAMON

Computer Engineering Student | Aspiring Web & Software Developer

Tanza, Cavite, Philippines • +63 994 810 8334 • andamonjonel06@gmail.com

Portfolio: dyoneeel.github.io/My-Portfolio • LinkedIn: linkedin.com/in/jonel-andamon



PROFESSIONAL SUMMARY

Fourth-year Computer Engineering student and hands-on builder of full-stack web and mobile applications using HTML, CSS, PHP, MySQL, JavaScript, and Flutter. Civil Service Eligible (Professional Level) with proven ability to design role-based systems, integrate relational databases, and deliver functional projects from concept to deployment. Seeking an internship or junior developer role where I can contribute to real-world software while expanding my expertise in modern web technologies.

TECHNICAL SKILLS

Languages: PHP, JavaScript, Python, C++, HTML5, CSS3, SQL,

Frameworks & Libraries: Node.js, Flutter, Bootstrap, React

Databases: MySQL, Firebase

Soft Skills: Team Collaboration, Problem Solving, Time Management, Fast Learner, Adaptability

EXPERIENCE

IT/Arduino Team Intern - James Polymers Manufacturing Corporation (August 2025) | *On-the-Job Training*

- Designed and built a prototype of a quality control system using Arduino Nano and Sentry2 vision camera for automated defect detection
- Calibrated vision parameters to accurately classify defective vs acceptable products

PROJECTS

TourGO - Car Rental Web Platform (2024) | *PHP, MySQL, JavaScript, HTML/CSS*

- Built a web-based casino with custom virtual currency, user wallets, and transaction tracking
- Developed backend logic for game outcomes and persistent data handling

RAWR Casino - Crypto-Themed Web Application (2024) | *PHP, MySQL, JavaScript, HTML/CSS*

- Built a web-based casino with custom virtual currency, user wallets, and transaction tracking
- Developed backend logic for game outcomes and persistent data handling

Escape - Multiplayer Mobile Game (2024) | *Flutter*

- Created an offline multiplayer board game with custom mechanics (wormholes and portals)
- Implemented turn-based logic and responsive UI for smooth gameplay

EDUCATION

Cavite State University - CCAT Campus (2022 – Present)

Bachelor of Science in Computer Engineering

Cavite Science Integrated School (2020 – 2022)

Senior High School, STEM Strand - Graduated with High Honors

ELIGIBILITY, CERTIFICATIONS & TRAINING

- **Civil Service Eligibility (Professional Level)** - Civil Service Commission (2025)
- **GEN AI TO Z: A Career Summit in an AI-Driven World** - Vibe Coders PH × UP EMC² Fraternity (2026)
- **ASEAN Data Science Explorers Enablement Session** - DICT IV-A & WITech Batangas (2024)

LANGUAGES

- Filipino
- English

GEROME Q. VOLUNTAD

B. Toledo St., Ligtong 4, Rosario,
Cavite 0942 330 4929
geromevoluntad1@gmail.com

**OBJECTIVE**

Motivated and adaptable individual seeking a position where I can utilize my strong communication skills, dedication, and eagerness to learn. Ready to contribute to a dynamic team and help achieve organizational goals, while continuously developing my personal and professional abilities in a supportive work environment.

PERSONAL INFORMATION

Date of Birth: September 20, 2003
Age: 22
Gender: Male
Citizenship: Filipino
Civil Status: Single
Languages: Filipino, English

EDUCATIONAL ATTAINMENT

Bachelor of Science in Computer Engineering
Cavite State University – CCAT
Campus Tejeros Convention,
Rosario, Cavite 4th Year

CERTIFICATIONS & ACHIEVEMENTS

National Certificate II (NC II) in Computer Systems Servicing
TESDA Certified

SKILLS

Proficient in Microsoft Office Suite (Word, Excel, PowerPoint)
Basic knowledge in C++, HTML, Java
Familiar with UI/UX design using Figma
Strong problem-solving and time management skills
Hardworking and goal-oriented

SUMMARY OF RECOMMENDATIONS



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU CCAT Campus
Rosario, Cavite
(046) 437-9507 / (046) 437-6659
cvsauroario@cvsu.edu.ph
www.cvsu-rosario.edu.ph

**RESEARCH, INNOVATION AND EXTENSION SERVICES
DEPARTMENT OF ENGINEERING**

SUMMARY OF RECOMMENDATIONS

Name: : JONEL B. ANDAMON
JOVAN P. NADALA
GEROME Q. VOLUNTAD

Degree//Specialization : **BS IN COMPUTER ENGINEERING**

Study Title: : **DEVELOPMENT OF LIBLOCKER: AN AUTOMATED
LIBRARY BAGGAGE COUNTER WITH DUAL
FREQUENCY RFID AUTHENTICATION FOR CAVITE
STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY**

Type of Paper : ☐ Title Defense ☐ Outline Defense ☒ Final Defense

Type of Study : ☐ Thesis ☒ Design Project ☐ Capstone
☐ EDP ☐ Case Study

Recommendations:

1.	Follow the format in objectives
2.	Fix the data presentation
3.	Remove the objective 7
4.	
5.	
6.	
7.	
8.	
9.	
10.	

Oral Review Committee	Signature	Date
Lead Panel: Erica Toledo Dimaunahan		05-28-26
Panel Member 1: Jhon Patrick S. De Mesa		05-28-26
Panel Member 2: Mart Amelson Diquit		05-28-26
Research Adviser: Sunny May S. Ronas		05-28-26
Technical Critic: Princess Yna D. Juyad		05-28-26

Noted by:

EMJAY A. CAMPO
Unit Research Coordinator, BSCPE

05-28-26
Date

SUNNY MAY S. RONAS, MEng
Department Chairperson

05-28-26
Date

COMPLIANCE OF RECOMMENDATIONS



Republic of the Philippines
CAVITE STATE UNIVERSITY
CvSU CCAT Campus
Rosario, Cavite
(046) 437-9507 / (046) 437-6659
cvsurosario@cvsu.edu.ph
www.cvsu-rosario.edu.ph

**RESEARCH, INNOVATION AND EXTENSION SERVICES
DEPARTMENT OF ENGINEERING**

COMPLIANCE TO RECOMMENDATIONS

Name: : JONEL B. ANDAMON
JOVAN P. NADALA
GEROME Q. VOLUNTAD

Degree//Specialization : BS IN COMPUTER ENGINEERING

Study Title: : DEVELOPMENT OF LIBLOCKER: AN AUTOMATED
LIBRARY BAGGAGE COUNTER WITH DUAL
FREQUENCY RFID AUTHENTICATION FOR CAVITE
STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY

Type of Paper : ☐ Title Defense ☐ Outline Defense ☒ Final Defense

Type of Study : ☐ Thesis ☒ Design Project ☐ Capstone
☐ EDP ☐ Case Study

No.	Nature of Revision	Previous	Revision(s) Made
1.	Follow the format in objectives	The objective is too long and redundant.	Objectives were rewritten and reformatted properly.
2.	Fix the data presentation	Testing results are presented with objectives attainment and summary separately.	Objectives attainment and summary were removed and similar tests were labeled.
3.	Remove the objective 7	Objective 7 is included.	Objective 7 was removed.
4.			
5.			
6.			

Oral Review Committee	Signature	Date
Lead Panel: Erica Toledo Dimaunahan		05-28-26
Panel Member 1: Jhon Patrick S. De Mesa		05-28-26
Panel Member 2: Mart Amelson Diquit		05-28-26
Research Adviser: Sunny May S. Ronas		05-28-26
Technical Critic: Princess Yna D. Juyad		05-28-26

Noted by:

EMJAY A. CAMPO

Unit Research Coordinator, BSCPE

SUNNY MAY S. RONAS, MEng
Date

Department Chairperson

Date

ROUTING SLIP (FOR UNDERGRADUATE)



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CvSU CCAT Campus
 Rosario, Cavite
 (046) 437-9507 / (046) 437-6659
 cvsurosario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

DEPARTMENT OF ENGINEERING

ROUTING SLIP (for Undergraduate)

Name of Student(s): ANDAMON, JONEL B., NADALA, JOVAN P., VOLUNTAD, GEROME Q.

Type of Study: ☐ Thesis ☐ Narrative ☐ EDP
☒ Design Project ☐ Case Study ☐ Teaching Portfolio

Title of Study: DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY

Type of Manuscript: ☐ Outline (if applicable) ☒ Final Manuscript

	DATE RECEIVED	DATE RELEASED	REMARKS	SIGNATURE
SUNNY MAY S. RONAS Adviser	5-28-24	5-28-24	OK	
PRINCESS YNA D. JUYAD Technical Critic	5-28-24	5-28-24	OK	
EMJAY A. CAMPO Department Research Coordinator, BSCPE	5-28-24	5-28-24	OK	
SUNNY MAY S. RONAS, MEng Department Chairperson	5-28-24	5-28-24	OK	
MICHAEL EDWARD T. ARMINTIA English Critic	06-01-24 06-15-24	06-12-24 06-15-24	for revision for printing	
ARIEL G. SANTOS, EdD Director for Instructions				
KARLO JOSE E. NABABLIT, PhD Director for Research and Extension (for thesis only)	6-14	6-14	approved	
LAURO B. PASCUA, EdD Campus Administrator	6-17-24	6-14-24	OK	

CERTIFICATION OF ENGLISH CRITIQUE



Republic of the Philippines
CAVITE STATE UNIVERSITY
(CvSU)
DON SEVERINO DE LAS ALAS CAMPUS
Indang, Cavite
☎ 415-0010 loc 270/200 ☎ 862-0806
www.cvsu.edu.ph



**CENTER FOR LANGUAGE LEARNING AND RESEARCH
(CvSU-CELLAR)**

15 June 2026

CERTIFICATION

This is to certify that the narrative report titled ***DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE UNIVERSITY-CCAT CAMPUS E-LIBRARY*** of **JONEL B. ANDAMON, JOVAN P. NADALA, and GEROME Q. VOLUNTAD** under **Bachelor of Science in Computer Engineering** from **Cavite State University-CCAT, Campus**, has been edited and proofread by the undersigned English Critic.

MICHAEL EDWARD T. ARMINTIA, REE
Accredited English Critic
Cavite State University-CCAT Campus

**MENTEE CONSENT/ATTESTATION FORM FOR UTILIZATION OF
INTELLECTUAL PROPERTY-RELATED WORK AND OUTPUT**



Republic of the Philippines
CAVITE STATE UNIVERSITY
 CvSU CCAT Campus
 Rosario, Cavite
 (046) 437-9507 / (046) 437-6659
 cvsurosario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

KNOWLEDGE MANAGEMENT CENTER

MENTEE CONSENT/ATTESTATION FORM FOR UTILIZATION OF INTELLECTUAL PROPERTY-RELATED WORK AND OUTPUT

Title of Work/Output:

DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER
WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE
UNIVERSITY-CCAT CAMPUS E-LIBRARY

I, JONEL B. ANDAMON, hereby consent to the use of the:

- | | |
|---|--|
| ☒ Thesis/Dissertation | ☒ Invention Patent |
| ☒ Research Work | ☐ Industrial Design |
| ☐ Creative Work | ☒ Utility Model |

mentioned above, or parts thereof, which is a product/output of the work I did with my mentor/mentor(s) as part of the requirements for earning my degree/diploma/certificate, which will be used for:

- | | |
|--|---|
| ☒ Publication | ☒ Patent/UM/Industrial Design Application |
| ☒ Paper or Poster Presentation | |
| ☒ Copyright Application | |

I give my consent/am attesting to this out of my free will, and with full understanding that:

- ✓ the mentor-mentee relationship is an important, dynamic academic relationship that contributes to scientific, technological and creative advances;
- ✓ the work we did is a collaborative work, and the contribution of each is as important as the contribution of the other; and
- ✓ my mentor has fully disclosed the possibility that our work would be utilized for IP-related outputs and/or application prior.

Signed this JUNE 15, 2026.

JONEL B. ANDAMON
 Printed Name and Signature of Mentee

- Additional consent/attestations on the next page -

Additional Attestations for Publication/Paper or Poster Presentation:

Since the original idea for the work is:

- ☐ Mine
- ☐ My Mentor's/Mentors'
- ☒ Ours

I hereby agree to be the:

- ☐ Corresponding Author
- ☒ 1st Author
- ☐ 2nd Author
- ☐ ____ Author

for our output.


JONEL B. ANDAMON

Printed Name and Signature of Mentee



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU CCAT Campus
 Rosario, Cavite
 (046) 437-9507 / (046) 437-6659
 cvsurosalario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

KNOWLEDGE MANAGEMENT CENTER

**MENTEE CONSENT/ATTESTATION FORM FOR UTILIZATION OF
 INTELLECTUAL PROPERTY-RELATED WORK AND OUTPUT**

Title of Work/Output:

DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER
WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE
UNIVERSITY-CCAT CAMPUS E-LIBRARY

I, JOVAN P. NADALA, hereby consent to the use of the:

- | | |
|---|--|
| ☒ Thesis/Dissertation | ☒ Invention Patent |
| ☒ Research Work | ☐ Industrial Design |
| ☐ Creative Work | ☒ Utility Model |

mentioned above, or parts thereof, which is a product/output of the work I did with my mentor/mentor(s) as part of the requirements for earning my degree/diploma/certificate, which will be used for:

- | | |
|--|---|
| ☒ Publication | ☒ Patent/UM/Industrial Design Application |
| ☒ Paper or Poster Presentation | |
| ☒ Copyright Application | |

I give my consent/am attesting to this out of my free will, and with full understanding that:

- ✓ the mentor-mentee relationship is an important, dynamic academic relationship that contributes to scientific, technological and creative advances;
- ✓ the work we did is a collaborative work, and the contribution of each is as important as the contribution of the other; and
- ✓ my mentor has fully disclosed the possibility that our work would be utilized for IP-related outputs and/or application prior.

Signed this JUNE 15, 2026.

JOVAN P. NADALA

Printed Name and Signature of Mentee

- Additional consent/attestations on the next page -

Additional Attestations for Publication/Paper or Poster Presentation:

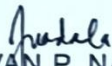
Since the original idea for the work is:

- ☐ Mine
- ☐ My Mentor's/Mentors'
- ☒ Ours

I hereby agree to be the:

- ☐ Corresponding Author
- ☐ 1st Author
- ☒ 2nd Author
- ☐ ____ Author

for our output.


JOVAN P. NADALA

Printed Name and Signature of Mentee



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU CCAT Campus
 Rosario, Cavite
 (046) 437-9507 / (046) 437-6659
 cvsurosario@cvsu.edu.ph
 www.cvsu-rosario.edu.ph

KNOWLEDGE MANAGEMENT CENTER

**MENTEE CONSENT/ATTESTATION FORM FOR UTILIZATION OF
 INTELLECTUAL PROPERTY-RELATED WORK AND OUTPUT**

Title of Work/Output:

DEVELOPMENT OF LIBLOCKER: AN AUTOMATED LIBRARY BAGGAGE COUNTER
WITH DUAL FREQUENCY RFID AUTHENTICATION FOR CAVITE STATE
UNIVERSITY-CCAT CAMPUS E-LIBRARY

I, GEROME Q. VOLUNTAD, hereby consent to the use of the:

- | | |
|---|--|
| ☒ Thesis/Dissertation | ☒ Invention Patent |
| ☒ Research Work | ☐ Industrial Design |
| ☐ Creative Work | ☒ Utility Model |

mentioned above, or parts thereof, which is a product/output of the work I did with my mentor/mentor(s) as part of the requirements for earning my degree/diploma/certificate, which will be used for:

- | | |
|--|---|
| ☒ Publication | ☒ Patent/UM/Industrial Design Application |
| ☒ Paper or Poster Presentation | |
| ☒ Copyright Application | |

I give my consent/am attesting to this out of my free will, and with full understanding that:

- ✓ the mentor-mentee relationship is an important, dynamic academic relationship that contributes to scientific, technological and creative advances;
- ✓ the work we did is a collaborative work, and the contribution of each is as important as the contribution of the other; and
- ✓ my mentor has fully disclosed the possibility that our work would be utilized for IP-related outputs and/or application prior.

Signed this JUNE 15, 2026.

GEROME Q. VOLUNTAD

Printed Name and Signature of Mentee

- Additional consent/attestations on the next page -

Additional Attestations for Publication/Paper or Poster Presentation:

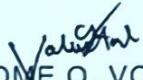
Since the original idea for the work is:

- ☐ Mine
- ☐ My Mentor's/Mentors'
- ☒ Ours

I hereby agree to be the:

- ☐ Corresponding Author
- ☐ 1st Author
- ☐ 2nd Author
- ☒ 3rd Author

for our output.


GEROME Q. VOLUNTAD

Printed Name and Signature of Mentee